Alliance CrossData

Page Information

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Release Notes

New SMTP email client

A full SMTP client application has been added to CrossData/400 to support sending CSV and other files using email. The older email API based on the IBM QtmmSendMail API is retained for compatibility. However, all new applications that you create should use the new API. The new API program is API200. Sample code can be found in the source file QCLSRC in the product library. Note that you must first configure the email parameters before using the API. On the main menu you will find a new menu option for email.

The new email API communicates directly with your mail server. It is fully compatible with Lotus Notes, Microsoft Exchange, and all Internet mail servers. The new email API does not use QSNADS or Mail Server Framework. The IBM mail services have proven unreliable and do not provide adequate feedback on failed mail delivery. The new API lets you specify From and Reply-to addresses that are different. You can also attach any AS/400 database or IFS file as an attachment. Please refer to the mail documentation later in this manual for information on using this API.

Escaping quote characters in data

When Excel saves a file in CSV format and it finds the quote character (usually “”) in a character string, it changes it to 2 quote characters. So a field like this:

```
Toys "R" Us
```

Becomes

```
Toys ""R"" Us
```

This capability has been added to CrossData/400 in the CVFILEX2 API and other places. See the sample code in source file QCLSRC, member SAMPEX2, for a demonstration of how to use this new API.

Inbound data conversion

You can now convert CSV and Tab-delimited files to an AS/400 externally described database file. CrossData/400 will use the external definition for a file to map the CSV or Tab-delimited data to the AS/400 file. Interactive panels and APIs (Application Program Interfaces) now support this direct import of data. The Programmer’s Toolkit has also been updated to let you use CrossData/400 functions directly.

Query support

CrossData/400 now provides direct support for IBM Query. CrossData/400 can run any query you have created and direct the output to a CSV, Tab-delimited, or ASCII fixed length field file. The original query can specify interactive or printed output. CrossData/400 will re-direct the output to a file and perform data conversion for you. Interactive menus and APIs (Application Program Interfaces) are provided for the direct query support.

New Commands

Commands are being added to the API support in CrossData/400. Commands make it easier to embed support for CrossData/400 in your CL programs and provide a convenient user interface for more experience users. Full help support is provided with the commands. The available commands are documented in the API section below. New commands are being added frequently and all existing high level APIs will have corresponding command support.

Application Concepts

Why data conversion

AS/400 character data is stored in the EBCDIC character set and numeric data is stored in various packed formats in the DB2/400 relational database system. This is a flexible and efficient means of storing data, but is incompatible with the data formats on Web sites, PCs, workstations, LAN servers, UNIX hosts, and almost all non-IBM systems. In order to provide cross platform integration it is necessary to convert the data from packed, EBCDIC formats to a format that can be used by the receiving system. This usually means converting the data to ASCII formats readable by the receiving system. Many applications can read CSV (Comma Separated Values) files in ASCII format, or fixed length ASCII files. UNIX applications can often handle tab-delimited ASCII files. Alliance CrossData/400 provides the ability to convert any AS/400 file into one of these formats without the need for complex programming for each database file.

CSV (Comma Separated Values) format

The CSV format is defined by Microsoft as a way to interface data directly into Microsoft Excel and other Microsoft applications. In Windows 95 and Windows NT a file with the extension of .CSV is recognized as an Excel file. Clicking on this file will launch Excel and open the file in spreadsheet mode. CSV data is characterized by fields separated by the comma (“,”) character. Character fields are surrounded in double quote marks and numeric fields have leading signs. A Carriage Return and Line Feed terminate the record. A CSV file with customer data might look like this:

```
“Acme Corp”, “John Smith”, “212-555-9999”, “1000.21” (CR/LF) “Bromm
```

In this example the numeric value represents the year-to-date sales. A CSV file in this format can be directly imported into Excel. Alliance CrossData/400 can convert any standard externally or internally described file to CSV format.

ASC (Fixed length ASCII) files

Many database systems are able to read ASCII files where the fields are of a fixed length and there are no field separators. Some examples of these types of systems are database systems by Oracle, Sybase, and Microsoft, contact management systems from Symantec (ACT!) and Telemagic, and many UNIX applications. Following the example above, the data would be in this format:

```
Acme Corp Brommel, Inc.

John Smith Jane Jones

212-555-9999 0001000.21 (CR/LF) 303-551-9999 0000201.52 (CR/LF)
```

In this example all fields start at the same offset into the record. The record may or may not have a terminating Carriage Return and Line Feed character. Alliance CrossData/400 supports this format in both low level and high level APIs.

Tab Delimited Files

Many relational database systems are able to read tab delimited files where the fields are of a fixed or variable length and the tab character is used as a field separator. Some examples of these types of systems are database systems by Oracle, Sybase, and Microsoft, and many UNIX applications. Following the example above, the data would be in this format (the tab character is a hex 05 value and is not printable):

```
Acme Corp[tab]John Smith[tab]212-555-9999[tab]0001000.21[tab](CR/LF
```

In this example all fields start at different offsets into the record. The record may or may not have a terminating Carriage Return and Line Feed character. Alliance CrossData/400 supports this format in both low level and high level APIs.

CrossData/400 architecture-

The core of the Alliance CrossData/400 application is a service program named CVTDTA that provides all of the file translation services. This service program is used in high level CrossData/400 applications such as Library Scan and High Level Application Program Interfaces (APIs) that are used by programmers. You can use the CrossData/400 service program in your own ILE applications, or you can use one of the many APIs that provide simpler interfaces. You can use the service program when you have more complex data conversion needs. The high level APIs can generally be used for standard conversion to CSV or fixed length ASCII files. High level APIs can also be used by OPM programs that are not able to incorporate the functions in service programs during compile.

Inbound data conversion

CrossData/400 now supports inbound data conversion. This is the process of converting data in CSV (Comma Separated Values) or Tab-delimited format to an AS/400 relational database file. CrossData/400 will extract the format of the relational database and will convert the CSV or Tab-delimited data to the relational database file. Numeric fields will be appropriately packed for the AS/400 format and the numeric signs will be handled properly. Inbound data conversion support is provided through interactive panels and APIs (Application Program Interfaces).

CrossData/400 high level APIs

The high level APIs in CrossData/400 provide an easy way to convert AS/400 files to standard CSV and fixed length formats. The high-level APIs use the low-level APIs internally but hide some of the complexity of using these APIs. All of the high level APIs can be called from either ILE or OPM programs (RPG, Cobol, CL, etc.). You can convert a database file using one call to a high level API instead of several calls to the low-level APIs. For example, to convert a file named SALES in library ORDERS to an output file in CSV format named SALESCVT in library OUTPUT, you could use this call to API CVFILCSV:

```
call cvfilasc parm('SALES' 'ORDERS' 'SALES' 'SALESCVT' 'OUTPUT' + '
```

The return code will tell you if the conversion was successful. If there was a failure the text field will tell you the cause of the failure. A complete list of the high level APIs is available in a following chapter. There are several high-level APIs that perform standard conversion functions.

CrossData/400 low level APIs

The functions implemented in the CVTDTA service program let you initialize the conversion environment, set the names of the input and output files, set various conversion options such as delimiter and sign characters, change the name of the EBCDIC to ASCII translation table, and actually perform the translation. It is a rich set of APIs that will provide an easy to use and standard interface for data conversion.

All of the low level APIs implemented in the CVTDTA service program accept parameters and return a four byte numeric return code indicating the success or failure of the operation. For example, the CVTInit API is used to initialize the conversion request. It is called like this (using ILE RPG format):

```
callb(d)  'CVTInit'
parm                Return
```

If the return code is zero (0) the request was successful. If it was not successful the request failed and the return code will have a value that indicates the cause of the failure. A complete description of the low-level APIs is included in a separate chapter below, and sample code is provided in the CrossData/400 library.

The APIs in CVTDTA will not be changed from one release to the next. New APIs may be added but existing APIs will not be changed. This will minimize the impact on any applications you develop with Alliance CrossData/400.

IBM Query support

CrossData/400 can now use your IBM Queries as a source of data for outbound data conversion. CrossData/400 can run any query you create and convert the output to CSV, tab-delimited, or ASCII fixed length field format. Interactive queries or queries that produce printed output can be used – CrossData/400 will override the output and create the output file in the format you want. After converting the data CrossData/400 can send the results of the query to an email recipient. The email support requires the IBM SendMail API on your RISC AS/400. Or you can specify that the results of the query be copied to an IFS or Shared Folder directory. This is ideal for automatically updating user files that are accessed through Client Access Shared Folders or AS/400 Network Neighborhood.

ASCII / EBCDIC translation

CrossData/400 provides the ability to translate the output data to ASCII characters. In some cases you will want the data to be in EBCDIC format, and in some cases you will want ASCII data. For example, if you want to send the translated file to an e-mail recipient you may want to leave the data in EBCDIC format and use the SNDDST command to send the file to a recipient as a Final Form Text message. The SMTP support on the AS/400 will convert the Final Form Text file to ASCII when it sends the message. However, if you want to copy the converted document to a Web site you may want to convert to ASCII to make it directly readable on the site. Alliance CrossData/400 gives you the option of doing a conversion or not.

You can use the IBM translation tables or define your own. IBM provides the QASCII table for EBCDIC to ASCII translation. This is the default for CrossData/400. IBM also provides a number of other translation tables for specific code pages. And you can define your own with the OS/400 Create Table (CRTTBL) command. An example and template can be found in the QTBLSRC file in the product library.

ILE and OPM program interfaces

The CrossData/400 low level APIs are implemented in a service program named CVTDTA. You can incorporate the CVTDTA functions into your ILE programs by including it on the appropriate create command as a bound service program. For example:

```
crtpgm pgm(mylib/mypgm) module(mypgm) bndsrvpgm(cvtdta)
```

If you are using an OPM model program such as RPG/400 or Cobol/400 you can use the high level APIs to set the options for the conversion or to convert data directly. APIs such as CVFILCSV provide a simple interface that will accept an input file name and convert the output to a new name you specify. Whether you use OPM or ILE programs on the AS/400 you can use Alliance CrossData/400 for your conversion needs.

System/36 and internally described files

Many AS/400 applications use database files created on the older IBM System/36. These files are internally described. That is, they were created without DDS and the internal formats cannot be extracted by CrossData/400 (or any other application). However, you can still use CrossData/400 for data conversion by specifying the format to use for the database file. In effect you are creating the same format description that CrossData/400 would create if the file were externally described. CrossData/400 provides a function called CVFormat that lets you specify the format string to use for conversion. You should note, however, that this works only if the internally described file contains only one data format. If you have a multiple format file CrossData/400 cannot perform the conversion for you.

CrossData/400 history records

Alliance CrossData/400 keeps track of every database conversion that you request in a history file named AFHIST. You can use the interactive menu option to view history or you can create your own queries on the file to review the results of conversion activity. The success or failure of every conversion attempt is logged in the history file as well as the name of the input file, output file, conversion options, user name, and other information that is important for proper audit trails of application activity. If you use the low-level programmer's toolkit for creating your own specialized conversion applications you can use the available history API to record these activities, too.

CrossData/400 application logs

Alliance CrossData/400 incorporates a detailed application log that can help you analyze problems that may arise from data conversion attempts. The application log can be turned on via an interactive menu option, and then turned off when it is no longer needed. When logging is active CrossData/400 writes detailed status messages to the AFLOGA file. You can use the interactive option to view or print the log, or you can create your own queries on the file. If you encounter problems with CrossData/400 your software supplier may ask you to activate the log, repeat a conversion request, print the log and send the results for problem determination. Your programmers may find the log helpful in analyzing attempts to convert data that are not successful.

Library scanning

To help you with your data distribution needs CrossData/400 includes a library scanning application that will automatically convert files for you and put them in a location you specify. You can set up separate conversion criteria for each file in the library. After converting the file CrossData/400 will delete or move the file to a library you specify.

Additionally, CrossData/400 will optionally execute a command string you specify. You can use this command string to start an FTP session, send e-mail, or do any other task you want to process the converted file. All actions taken by the library scan function are recorded in the history file.

Stream files

Stream files are just binary streams of data that do not have a concept of a logical record format. Most PC, UNIX, and server files are stream files. Alliance CrossData/400 can copy your converted files to your AS/400 server using the Copy To Stream File (CPYTOSTMF) command in the library scan facility. Or you can use this command to create your own copies after converting a file. The CPYTOSTMF command has options for data conversion that you may want to use if you are not using CrossData/400 EBCDIC to ASCII translation. You can use the CPYTOSTMF command to place data on Web sites, into IPCS file systems used by NT, and for other uses.

Electronic mail

CrossData/400 can distribute your converted files via electronic mail if your AS/400 is a RISC platform with the IBM Send Mail API installed. The Send Mail API is QtmmSendMail and is available on all V4R1 systems or later. CrossData/400 supports electronic distribution of converted files on interactive panels, via high level APIs, and via the library scan facility. You can achieve a high level of automation in the distribution of data using one of these mechanisms. Additionally, CrossData/400 gives you an API for sending any file via electronic mail.

If you do not have a RISC system you can still distribute converted file data to e-mail recipients. You can use the Send Distribution (SNDDST) command to do this, or you can use a third party e-mail product like NetMail*400 from Net400, Inc. In any case you can send the output file created by CrossData/400, or you can copy the data to a folder document and send it from there. How you do the distribution depends on how you want your recipient to process the data. If you need more information about the AS/400 and electronic mail you can read about SMTP mail in the TCP/IP Reference manual and the Redbook entitled “AS/400 Electronic Mail Capabilities”.

Web files

The files on a Web site on the AS/400 or any other system include HTML files, image files, CSV files, and many other file types that can be made available to users with Web browsers. You can make data available to these users by copying the output files from CrossData/400 to a Web directory on your system using the CPYTOSTMF command, or you can place them on a remote directory using AS/400 FTP. In either case you can make data available immediately by placing the files in the appropriate directory on your Web server.

If you want to add HTML formatting commands to the converted file you can pre-pend or append the appropriate codes to the output file to make it Web compatible. If you are using the CrossData/400 library scan facility you can add your own procedure to be run after conversion to do this work for you.

PC documents

In some cases you may wish to make CSV files available to Client Access users by copying the data to a folder in the QDLS file system. You can use the Copy To PC Document (CPYTOPCD) command to do this after CrossData/400 has converted the file for you. If you are using the Library Scan function of CrossData/400 there is an option on the file definition that lets you specify a copy command to be run after the file is created. The command can copy the file to a folder automatically. In addition, the new query support can place the results of any query in CSV, tab-delimited, or ASCII format in any user folder or directory.

Using CrossData/400 with Alliance FTP Manager

You can create a powerful data distribution application by combining the data conversion capabilities of CrossData/400 and the file distribution capabilities of Alliance FTP Manager. CrossData/400 can automatically convert files in one or more input scan libraries and place the output in one or more Alliance FTP Manager scan libraries. CrossData/400 will convert the data and pass the output file to Alliance FTP Manager. Alliance FTP Manager will send the converted files to a remote system. You can use specific file names or generic file names for the conversion and transmission.

Web interface for Alliance CrossData/400

Alliance CrossData/400 supports a web interface on the AS/400 using the IBM WebFacing environment. There is no charge for installing the Alliance web interface. In order to use the web interface you must install either the IBM Websphere Application Server Express (or advanced version) or the IBM Apache Tomcat server. The IBM Apache Tomcat server is available at no charge. If you install the Alliance web interface you will not need to have any interactive features installed on your AS/400. See the installation section below for instructions on how to install the Alliance web interface.

Limitations of CrossData/400

CrossData/400 creates files that are compatible with most database systems, but may not be compatible with some systems. By using CrossData/400 you accept responsibility for verifying that data generated by CrossData/400 will be compatible with the receiving system or application.

CrossData/400 handles most data types on the AS/400, but does not handle the following data types:

- Variable length character data
- Double byte character set (DBCS) data

Frequently Asked Questions

How can CrossData/400 help with my AS/400 data distribution requirements?

CrossData/400 is designed to convert your data files into formats that are usable on a variety of platforms including UNIX, Windows PCs, NT Server, Tandem and other systems. The data can be directly imported into many PC applications such as Microsoft Excel, Access, and SQL Server. And CrossData/400 can create data that is compatible with Web site download formats. For almost all eCommerce applications CrossData/400 can create data in the format you need for distribution to network users.

How much programming is required to use CrossData/400?

You can use the interactive panels and the library scan facility to convert database files without any programming at all. Or, if you want to create converted files as a part of your batch routines you can use the CrossData/400 APIs (Application Program Interfaces) to create the files. The programming for this type of conversion is very simple. For more complex data conversion requirements your programmers can use the Programmer's Toolkit to create their own conversion routines.

Can CrossData/400 create Excel files?

Yes. CrossData/400 creates CSV (Comma Separated Values) files from your AS/400 database that can be directly imported to Microsoft Excel. Any file transferred to a PC that has an extension of ".CSV" and that conforms to the rules of CSV files are associated with the Excel application. A user who receives a CSV file can double-click on the file to launch Excel and load the file. Additionally, Excel spreadsheets can execute macros that load and manipulate CSV files created by CrossData/400.

Can CrossData/400 create data for my AS/400 Web site?

Yes, you can create CSV files or fixed record length files in ASCII format that can be placed on your AS/400 web site for access by Web browsers. CrossData/400 can create CSV files in the ASCII character set and copy these files to a directory used by your Web server. It uses the Copy To Stream File (CPYTOSTMF) command to do this task for you. CrossData/400 can even perform this task automatically if you are using the library scan feature. Your interactive or batch programs can create files in a scan library and CrossData/400 will convert the files and distribute to your Web site automatically.

Can CrossData/400 create data for my UNIX applications and UNIX Web site?

Yes. Most UNIX applications can work with ASCII, tab-delimited files as an import format. CrossData/400 can create tab delimited files from AS/400 files that can be used in UNIX applications. After creating the files on the AS/400 you can use FTP or another file transfer mechanism to move the files to a UNIX server. If you use FTP to move the files you can even start a UNIX procedure to process the file.

Can I import Microsoft Excel data to my AS/400 database?

Yes. CrossData/400 supports importing CSV and tab-delimited files into AS/400 database files. CrossData/400 will read the external definition of the database file to create a map of the data. It then loads each row of the CSV file to a record in the AS/400 database file converting character and numeric data to the proper format. Inbound data conversion is available through interactive menus and APIs (Application Program Interfaces).

How is CrossData/400 different than the Client Access Transfer function?

CrossData/400 runs entirely on the AS/400 and requires no other hardware platform or PC to perform the data conversion. With Client Access you must initiate the process on a PC, transfer the data to the PC hard drive, and then move the data to its desired destination on a Web site or server. CrossData/400 removes the requirement for the PC with Client Access and lets you gain full control of the conversion process. You can know immediately if the data is converted without errors and all conversions are logged to a history file. CrossData/400 is an easy-to-use, AS/400-only application for data conversion requirements.

How can I get data from my AS/400 to Oracle database?

Many database vendors include special functions to import AS/400 database files. If you can't use these interfaces you may be able to use CrossData/400 to interface to the database. Oracle, and other database applications from Sybase, Microsoft, and others, include a bulk data copy function that can take an ASCII fixed length file and import it directly to the relational database system. CrossData/400 can create the ASCII fixed length file for you to import into these systems. After creating a converted file you can use FTP or other file transfer application to move the files and initiate the database load process.

How can the library scan function automate my conversion tasks?

The library scan function in CrossData/400 lets you set up automatic data conversion when a file is created in one of the scan libraries. You define which libraries are scanned and which files in the libraries are converted. You can specify the type of conversion and data distribution actions for each file in a library. CrossData/400 library scan gives you a means of data conversion that is automated and requires no programming.

Can I use CrossData/400 with my e-mail application?

Yes. CrossData/400 can create files in CSV, fixed length ASCII, or tab-delimited format that can be distributed by CrossData/400 on the AS/400. CrossData/400 uses the IBM Send Mail API to send MIME compatible SMTP messages to local or Internet recipients. Or you can use NetMail*400 from Net400 or OfficeVision/400 from IBM to distribute your files. If you do not have an RISC AS/400 system you can use the Send Distribution (SNDDST) command to send the converted files.

Can I use CrossData/400 to send files to my PC users?

Yes. CrossData/400 can be used to convert AS/400 data to a format that can be used by most PC applications. As discussed above CrossData/400 can convert an AS/400 database file to CSV format which can be directly used by Microsoft Excel and Word. Many other applications like Symantec's ACT! can also import CSV or ASCII fixed length files. You have several options for getting the converted data to your user's desktops. You can copy the converted file to a folder on the AS/400 that the user accesses with Client Access. Some of the CrossData/400 interfaces will do this copy for you. You can also leave the converted file in a library that the user can access with the Client Access Transfer function. The user can download the data to their PC with no data conversion. If you have a Web site on your AS/400 you can copy the converted data to a directory on the AS/400 that can be accessed by the user with a browser. You can also push the data to your user's PC with FTP. If the user has the Microsoft Personal Web Server FTP application installed on their PC you can push the file to a directory on their hard drive. Lastly, you can send the converted file with your e-mail application.

Can I use CrossData/400 to convert spool file reports?

Yes. CrossData/400 can convert a spool file report to a fixed length ASCII format for you. It expands the text to make the report a true image of the report as it would look on a printer. CrossData/400 also inserts the standard ASCII top-of-form character in the data stream when there is a page break. Many PC applications recognize this special character and will insert a page break when printing. CrossData/400 will not, however, retain any graphical data in the report. For example, you cannot use CrossData/400 to transfer AFP (Advanced Function Print) reports with graphic characters. Only the text content of a spool file will be converted. After converting a spool file to ASCII fixed length format you can distribute the file any way you wish.

How can I use Query with CrossData/400?

IBM Query is a powerful, easy-to-use application for extracting and sorting data in an AS/400 database file. CrossData/400 can use any IBM query as a source of data for conversion. CrossData/400 can run any query you create and convert the results to a CSV, tab-delimited, or ASCII fixed format. You can send the converted file to an email recipient, copy the file to a shared folder or IFS directory, or leave the data in an AS/400 file. If you can create a query you can distribute the data in any format supported by CrossData/400.

What is the Programmer's Toolkit?

The conversion functions of CrossData/400 are implemented in a Service Program called CVTDTA. A programmer can link these functions into their applications and use the functions directly. These functions are documented in the Installation and User Guide and sample code is provided in the product library. Access to the low level APIs in CrossData/400 lets the programmer create applications that meet your special data conversion needs.

Can I use CrossData/400 with my System/36 files?

Yes. CrossData/400 can process a System/36 file (or any internally described file) using the CrossData/400 Format API if there is a single format of data in the file. Normally CrossData/400 reads the DDS layout directly from an externally described file. However, it provides you with a way to define the format of the file if it is internally described. You can use this format feature to describe the types and lengths of fields in the file. CrossData/400 will then use this information to convert the file.

What are the hardware and software requirements?

CrossData/400 runs on any AS/400 hardware model and on V3R2, V3R7, V4R1 or later release of the operating system. No special features of the operating system are required. CrossData/400 does not use any MI interfaces or special operating system calls that will cause security failures on the AS/400.

Alliance CrossData/400 Audit Trails and History

Audit trails

Alliance keeps a history file of all data conversion requests. You can view a list of conversions that have been performed on the AS/400. The audit trail is accumulated in a history file named AFHIST that you can query with IBM's query product or many third party query products.

The history file gives you a way of periodically reviewing use of CrossData/400 on your system. You can identify potentially inappropriate uses of CrossData/400 or attempts to gain unwelcome access to your AS/400 database. You can also use the history file to analyze application usage and plan for growth.

Application logs

To support problem determination and support implementation of Alliance services, you can activate internal logging of the Alliance conversion programs. When logging is active detailed information is written to the log file AFLOGA. You can view this log with the option to Display Application Log on the inquiry menu, or using any file display or query tool. The application log

displays all input parameters to the Alliance control programs and all relevant Alliance options you have configured. If a data conversion fails the reason for the failure will be in the application log.

If you report a problem with Alliance, your software vendor may ask you to activate the application log and print the results. The log is activated with the option to Configure CrossData/400 on the configuration menu.

Before You Begin

OS/400 release level

Before you begin you must verify that your AS/400 is at the proper release level. You must be on release V4R1 or later. You should also be sure that you have applied the latest cumulative PTF tape from IBM.

Installation

Installing for the first time

You may have received the Alliance software product in one of several ways:

- On tape media from your software vendor
- Downloaded from the Internet
- As a binary save file on PC disk

In all cases you will restore the product using the Restore Library (RSTLIB) command.

Installing from tape

If you receive the product on tape media, mount the tape on the appropriate device and restore the library from tape:

```
rstlib savlib(alldta200) dev(tap01)
```

Installing from CD

If you receive the product on CD media, mount the CD on the appropriate CD reader and restore the library from the CD using the Load and Run (LODRUN) command:

```
lodrun dev(opt01)
```

Installing from PC diskette or download

If you received the product on PC disk or downloaded it from the Internet, use these procedures:

1. Create a save file on the AS/400: crtsavf qgpl/alldta200 [Enter].
2. Start an FTP session with the AS/400 from your PC.
3. Change to binary transfer mode: binary [Enter].
4. Transfer the file to the AS/400: put a:\alldta.sav qgpl/alldta200 (replace [Enter].)
5. Close the FTP session and sign on to the AS/400 using the QSECOFR profile.

You can now restore the library from the save file:

```
rstlib savlib(alldta200) dev(*savf) savf(qgpl/alldta200)
```

Note: You must restore the product to the library named ALLDTA200. You cannot change the library to another name.

Installing an upgrade

If you are installing Alliance as an upgrade to a previous release, use these procedures first:

1. Save the current product library to tape or other media.
2. Rename the current product library to a new name.
3. Restore the new release from tape or CD(see above).
4. Run the release update option from the Installation menu.

Note that you may need to re-create your application programs that use the CVTDTA service program if you specified the library for the service program on the create command.

Trial and activation codes

If you are installing an upgrade or new release your trial or activation code will be restored when you run the update procedure. If you are installing Alliance for the first time, you will need a trial code from your software vendor.

Alliance library security

The owner of the Alliance library ALLDTA200 and all programs and objects in the library is QPGMR. The public (*PUBLIC) has all rights to the library and objects in the library. You should secure the library using normal AS/400 security commands.

Planning Your Implementation

Configuring Alliance CrossData/400 is easy, but you will have to make some decisions first about what features you want to use and how to implement them. You should decide the following items before starting:

Separate libraries for converted files

You may want to create separate libraries to hold converted files. This can be helpful in avoiding errors in creating files in production libraries. Alliance CrossData/400 will create any library you specify as an output library if it does not exist, but it is probably better from an administrative point of view if you create these libraries before using CrossData/400.

Using library scan

Alliance CrossData/400 includes a library scan facility that can detect the presence of a file in a library and automatically convert it to a CSV or ASCII format. You should identify the libraries you want to scan and define these to CrossData/400 before you begin. It is recommended that you create separate libraries for the scan functions in order to avoid the accidental conversion of files in an existing application library.

Configuring SMTP Mail

If you are going to use the electronic mail capabilities of CrossData/400 on your RISC AS/400 system you will need to first configure TCP/IP and SMTP. Configuring TCP/IP is beyond the scope of this document, but you can get step-by-step instructions in the AS/400 TCP/IP Reference manual shipped with our AS/400. This manual also describes how to set up SMTP and the POP server. For additional step-by-step instructions and problem solving suggestions, see Appendix A of this manual.

Getting Started

Overview

The following sections provide a step-by-step procedure for setting up Alliance CrossData/400. Some of the steps may not be necessary depending on your objectives outlined above.

Creating libraries for Output files

If you want to create special libraries for converted files you should create them now. You can use the following command as an example:

```
crtlib lib(output) text('Converted files')
```

If you have several different types of conversion requirements or applications and you want to keep the files separate, create other libraries to hold files for each group.

Defining scan libraries

If you are going to activate automatic library scan and data file conversion, you will probably want to create the scan libraries before you start. You can use the following command as an example:

```
crtlib lib(scan) text('Scan and convert library')
```

You must use the CrossData/400 menu option to Work With Scan Libraries to define a library and the files to be processed. After you add the library you must add a list of files to scan for. When you add a file to the scan definition you tell CrossData/400 how to convert the data and where to put the resulting file. There are other options to define for the conversion, too. You should identify which files will be automatically converted and define them now.

Configuring CrossData/400

After defining all of your data conversion needs for the system, you should configure global options for Alliance CrossData/400. Use the Configure CrossData/400 option on the main menu.

Checklist

The following checklist is provided to assist in the configuration and startup process. Some steps may not apply to you, so just skip those steps. See the sections above for more information on the following steps.

STEP	COMPLETE	DESCRIPTION
1		Review OS/400 system requirements before starting.
2		Restore library ALLDTA200 from product media.
3		Enter your temporary or permanent activation code from your soft using the appropriate menu option. To display the menu: `addlible alldta200 [Enter] go dtmain [Enter] Select`
4		Create the output libraries you will use for data conversion. For ex might create a library named OUTPUT for all files you will convert CrossData/400: `Crtlib output text('Data conversion library') [Ente`

STEP	COMPLETE	DESCRIPTION
5		Create the scan libraries you will use for automatic data conversio example, you might create a library named SCAN for all files you v automatically convert with CrossData/400: `Crtlib scan text('CrossData/400 scan library') [Ente`
6		Configure Alliance CrossData/400 global options. Use the option t CrossData/400 on the Alliance configuration menu.
7		Test data conversion by using the menu options on the CrossData Conversion menu. Verify that data is being converted correctly.
8		If you are going to use the library scan facility prepare a test file in library you defined above. Start the ALLDTA200 subsystem and m the library scan facility converted the file properly.
9		Review history logs on the AS/400. Use the View History option fr menu. If there are no errors the installation is complete.

On Line Help

Every panel in Alliance has on-line help available. Press the Help or F1 key on your keyboard to access help. You will be able to review a description of the panel's function, each field's function and use, and a list of active function keys and what they do.

Main Menu

The main menu for Alliance CrossData/400 is DTMAIN. You can access this menu by placing the library ALLDTA200 in your library list and using the GO command:

```
addlible alldta200 [Enter]
go dtmain [Enter]
```

The following menu is displayed. All of the Alliance programs are available from this menu and sub-menus.

```
*****************************************************************
01 * DTMAIN                  Alliance CrossData/400 Main Menu
```

Select an option to begin working with Alliance CrossData/400.

Installation

After you restore the CrossData/400 product from tape media you must perform certain configuration tasks. Display the Installation menu for a list of all installation options. From this menu you can enter a trial or activation code, and upgrade from a previous release.

```
*******************************************************************************
*******************************************************************************
```

Select an option to begin work.

Enter Trial Code

To use Alliance CrossData/400 you must have a trial code or permanent code. A trial code was provided with your shipment and will be active for approximately 30 days. Use this panel to enter the trial code.

```
*******************************************************************************
01 * DT5013A                          Enter Trial Code                ROBB
```

Trial code

Enter the trial code provided by your software vendor.

Enter Permanent Activation

After you have licensed Alliance for your AS/400 you will receive a permanent activation code. Enter the permanent code in this panel.

```
*******************************************************************************
01 * DT5012A                        Enter Permanent Code              ROBB
```

Activation code

Enter the permanent activation code provided by your software vendor. If you upgrade to a new system or model you will need to obtain a new activation code.

Apply Release Upgrade

If you receive a release upgrade from your software vendor you will use this option to re-establish your application options and other configurations you developed on the earlier release. Please be sure to read the installation cover letter that accompanies the release upgrade before using this option. When you select this option the following panel is displayed.

```
*******************************************************************
01 * AL4090A                              Update Version
```

Previous version library

Enter the name of the library that contains the previous release.

Configuration

When you select option 2 from the main Alliance menu the following menu is displayed. You can configure all of the Alliance CrossData/400 options and tables from this menu. Select an option to begin work.

```
*******************************************************************
01 * DTMNU2              Alliance CrossData/400 Configuration Menu
```

Use option 1 from the menu to configure system-wide options for CrossData/400. Use option 2 to configure scan libraries for automatic data conversion. With option 3 you can store conversion information into a profile name and use it in API's and Cross Data panels. Enter option 10 to start the CrossData/400 subsystem for library scan functions. Enter option 11 to terminate the library scan subsystem. Enter option 12 to work with all jobs in the library scan subsystem.

Configure CrossData/400

Use this panel to configure global options for Alliance CrossData/400 You can activate or de-activate application logging, and define the message queues used for notification.

```
*******************************************************************
01 * DT6040A                    Configure CrossData/400          ROBB
```

Enable logging

Enter 1 for Yes if you want to enable logging. Enter 2 for No if you do not want to use application logging. This should normally be 2 to not produce application logs as the logs can become quite large.

Notification message queue 1 and library

Enter the name of the message queue you want to use for error notification. You must enter a value in this field. The other two message queues are optional.

Notification message queue 2 and library

Enter the name of the message queue you want to use for error notification.

Notification message queue 3 and library

Enter the name of the message queue you want to use for error notification.

ASCII to EBCDIC table

Enter the name of the ASCII to EBCDIC translation table you want to use. The default is the IBM-supplied QEBCDIC table in library QSYS. You can use a different table or create your own translation table.

EBCDIC to ASCII table

Enter the name of the EBCDIC to ASCII translation table you want to use. The default is the IBM-supplied QASCII table in library QSYS. You can use a different table or create your own translation table.

Default conversion profile

Enter the name of the default conversion profile. The default is DEFAULT. You can create your own profiles using the configuration option to Work With Conversion Profiles.

Use profile as default

Enter 1 for Yes to use the default profile as the default for interactive conversion displays.
Enter 2 for No to not use the default profile on Interactive conversion panels.

After pressing Enter the second panel is displayed for configuration

```
*****************************************************************
*****************************************************************
```

Email interface

Enter 1 to use the Alliance SMTP client interface. Enter 2 to use the older IBM QtmmSendMail interface. The default is to use the Alliance SMTP client interface.

Work with library scan definitions

Use this option to scan a library and convert the files in the library. You must first add a library to the scan list and then use option 12 to add file actions. It is strongly recommended that you do NOT define a production library as a scan library. You should create a new library to serve as a scan library for purposes of data conversion. Using this approach will minimize the chances of accidental deletion of production files.

You can add as many scan libraries as you wish. You must add file actions for each library you add.

Note that you may wish to update your IPL start up job to automatically start the library scan subsystem. You can determine the name of the IPL program by displaying the system value QSTRUPPGM. It is normally QSTRUP in library QSYS. You can retrieve the source using the RTVCLSRC command and modify it to start the ALLDTA200 subsystem. Insert the following lines in the appropriate section of the start up program:

```
strsbs alldta200/alldta200
monmsg cpf0000
```

```
*****************************************************************
01 * DT5080-A                    Work With Scan Libraries        ROBB
```

Position to

To position the list to a specific record enter the key value in this field and press Enter or F16.

Options

2=Change

Use this option to change information about the item in the list.

4=Delete

Use this option to delete an item from the list and all associated detail records.

6=Print

Use this option to print a record. The printed report will be placed in the output queue that is the default for your job.

12=Work with file actions

Use this option to work with detail file actions for this library.

Add/Change Scan Library

Use this panel to change information about a scan library.

```
**************************************************************************
01 * DT5081-A                      Change Scan Library              ROBB
```

Description

Enter a description for this scan library.

Status

Enter 1 for Active if you want to activate scanning for this library. Enter 2 for Inactive if you do not want to scan this library. The status takes effect immediately. Libraries will not, however, be scanned until you activate the scan library subsystem.

Work With File Actions

After you add a library to the list of scanned libraries you must add file actions. File actions are based on the file name that is found in the library and specifies a conversion action to take for the file. You can also define a user command to execute before the file is processed, a command to execute after a file is processed, a copy action for the converted file, and the post conversion action for the file. A file will not be processed if the file action is not found for it.

```
**************************************************************************
01 * DT5090-A                    Work With File Actions             ROBB
```

Use option 2 to change a file action. Use option 4 to delete a file action. Use option 6 to print the details of a file action. Use F6 to add a new file action.

Add/Change File Action

Use this panel to change the actions you want to take for converting a file. These actions apply only to the library scan facility. You must enter an action record for each file you want to convert.

```
*****************************************************************
19 * Copy to stream file . . . . . . . 2             1=Yes, 2=No
```

Description

Enter a description for this file action.

Status

Enter 1 for Active if you want to process actions for this file. Enter 2 for Inactive if you do not want to process this file during the library scan. The status takes effect immediately.

Conversion type

Enter 1 to convert the file to fixed length fields in the EBCDIC character set. Enter 2 to convert the file to CSV (Comma Separated Values) format in the EBCDIC character set. Enter 3 to convert the file to ASC (Fixed length format) in the ASCII character set. Enter 4 to convert the file to CSV format in the ASCII character set. CSV format in the ASCII character set. Enter 5 to convert the file to Tab-delimited format in the EBCDIC character set. Enter 6 to convert the file to Tab-delimited format in the ASCII character set. Enter 7 to convert the file using a conversion profile. If you use this option you must create a conversion profile, or use an existing conversion profile. Enter 8 to not perform any conversion from AS/400 database to output format.

Conversion profile

If you enter 7 for the conversion type you must enter the name of a conversion profile.

Output file, library, member

Enter the name of the file, library, and member to receive the converted data. It is recommended that this file not exist at the time the data conversion takes place. CrossData/400 will create the file for you with the correct record length. You can also use the special value of *GEN for the file name. When you specify *GEN CrossData/400 will create a file and member name to be used for the output file. This can be helpful if you want to create a *ANY file action record and process all files with one definition. The output file name format will be “Fnnnnnnnnn” where “nnnnnnnnn” is a sequence number.

Copy to stream file

Enter 1 for Yes to copy the output file to a stream file in the QDLS folder system or a Web site or other directory on your AS/400. Enter 2 to not copy the output file to a stream file.

Stream file

Enter the full path and stream file name. For example, you could copy the output file to a stream file named “data.csv” in the folder SMITH with the following path:

```
/QDLS/SMITH/data.csv
```

To copy to a directory on your AS/400 web site enter the full path. For example, it might be a path like this:

```
/www/webdocs/data.csv.
```

Your web path may be different than this example. Contact your system administrator for the actual path to your web directory.

After pressing Enter the second panel is displayed for add/change file action.

```
*****************************************************************
01 * DT5091-B                          Change File Action
```

Before command

If you want to execute a command or call a program before the file conversion is performed you can enter it here. For example, if you want to run a program named ORDERS before the file conversion takes place you can enter the following command:

```
call orders
```

The before command is executed immediately before the file is converted.

After command

If you want to execute a command or call a program after the file conversion is performed you can enter it here. For example, if you want to run a program named PRINT after the file conversion takes place you can enter the following command:

```
call print
```

The after command is executed immediately after the file is converted.

Post conversion action

This option refers to what you want to do with the input file after it has been processed. Enter 1 if you want to delete the file from the scan library. Enter 2 if you want to move the file to a backup library. If you enter 2 you must specify a library in the following field. Enter 3 if you want to rename the file to a unique name and then move it to a backup library. You must choose one of these options in order to prevent the file from being continually converted.

Library

If you select post conversion option 2 or 3 you must enter the name of a backup library. The library will be created if it does not exist.

Send via e-mail

Enter 1 for Yes if you want to send the converted data to an e-mail recipient. Enter 2 for No if you do not want to send the file via e-mail. If you specify Yes you must have a RISC AS/400, the IBM Send Mail API QtmmSendMail installed, a valid SMTP configuration. You can send the file with up to five lines of message text.

To:

Enter the e-mail address of the recipient. This must be a valid SMTP address with a user and domain. For example:

```
Bill_Jones@acme.com
```

After pressing Enter the third panel is displayed for add/change file action

```
*******************************************************************
01 * DT5091-C                        Change File Action
```

From

Enter your e-mail address in this field. This must be a valid SMTP address with a user and domain portion of the address. For example:

```
Jane_Smith@as400.acme.com
```

Reply to

Enter your reply-to address if you want replies to return to a different address than you From address. This field is optional.

Subject

Enter a subject for this message.

Attachment name

Enter the name of the attachment. This is normally a file name with the extension you want to use.

Message

Enter up to five lines of text for a message to accompany this file.

After pressing Enter the fourth panel is displayed for add/change file action

```
*******************************************************************
01 * DT5091-D                        Change File Action
```

Inbound translation

Enter 1 to convert a delimited file to an AS/400 database file. Enter 2 for no inbound conversion.

Convert to file, library and member

If you selected Yes to inbound conversion, enter the name of the database file, library and member to receive the data.

Add or replace

Enter 1 to add data to the inbound file. Enter 2 to replace the data in the file. If you specify Replace the data in the file will be cleared before the conversion takes place. This data cannot be retrieved.

Conversion profile

Enter the name of a conversion profile for the inbound translation.

Work with conversion profiles

```
**********************************************************************
**********************************************************************
```

Use this option to create a conversion profile. A conversion profile stores the parameters used to convert data. You can then reference the conversion profile on APIs and commands in CrossData/400. Note that you can also define the name of the default conversion profile for the product.

Use this panel to add a new conversion profile

```
**********************************************************************
**********************************************************************
```

Description

Enter a description for this conversion profile.

New line option

Enter 1 if you do not want to use new line characters. Enter 2 if you want to add Carriage Return and New Line characters at the end of each line. Enter 3 if you want to add just a Carriage Return character at the end of the line. Enter 4 if you want to add just a New Line character at the end of each line. Enter a character value to use for the new line character.

Field delimiter

Enter the character you want to use to separate fields. This is normally a comma for Excel CSV (Comma Separated Values) files. If you leave the field blank fields will not be separated by a delimiter.

Numeric decimal

Enter the character you want to use for numeric decimals.

Numeric negative sign

Enter the character you want to use to indicate a negative value in a numeric field. If you leave this field blank no character will be used to indicate a negative field and all negative fields will appear positive.

Numeric zero display

Enter 1 to use blanks for zero value numeric fields. Enter 2 to use zeros for zero value numeric fields.

Numeric leading blanks

Enter 1 to preserve leading blanks or zeros in numeric fields. Enter 2 to suppress leading blanks or zeros. Option 2 will eliminate the leading blanks or zeros and is best used with delimited conversions. Enter 3 to suppress leading blanks and not provide a zero value place holder. Enter option 4 to replace leading zeros with blanks and preserve a zero value as a place holder in zero-value fields.

Character field quote

Enter the character you want to use to surround character fields. If you leave this field blank character fields will not be surrounded.

Blank truncation

Enter 1 for Yes if you want to truncate the trailing blanks in character fields. Enter 2 for No if you do not want to truncate trailing blanks.

Column titles

Enter 1 if you do not want to use column titles in your output file. Enter 2 if you want to use the field name as the column title. Enter 3 if you want to use the column header data defined for the fields.

Translate to ASCII

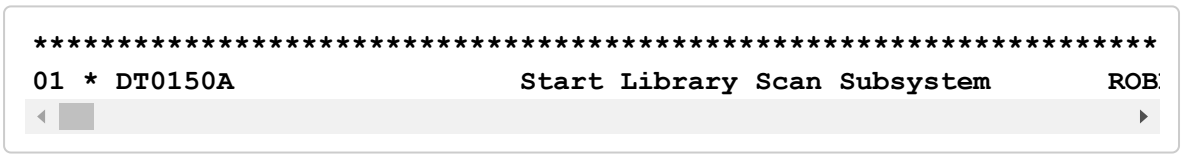
Enter 1 for Yes if you want to translate the output data to ASCII. Enter 2 for No if you do not want to translate the data to ASCII. If you specify Yes for this option you must specify a table name and library in the following fields.

Table and library

If you specified Yes to translate to ASCII you must specify a translation table and library. The default is the IBM supplied table QASCII in library QSYS, but you can specify any valid translation table. You can use the DTEB2AS table in library ALLDTA200, or you can create your own translation table. There is a template in file QTBLSRC in the Alliance product library.

Start Scan Library Subsystem

Use this option to start the CrossData/400 subsystem named ALLDTA200. The subsystem runs the library scan functions. If you are not using the library scan functions you do not need to start this subsystem.



There are no input fields on this panel. Press Enter to start the subsystem.

End Library Scan Subsystem

Use this option to end the CrossData/400 subsystem named ALLDTA200. The subsystem runs the library scan functions.

```
********************************************************************************
01 * DT0160A                    End Library Scan Subsystem        ROBB
```

There are no input fields for this panel. Press Enter to end the subsystem.

Work With Active Library Scan Jobs

Use this option to view the active jobs in the subsystem ALLDTA200.

```
********************************************************************************
01 *                                Work with Active Jobs
```

Maintenance

Use this menu to perform routine application maintenance. History and log files will be purged and internal database files will be re-organized. You should perform maintenance weekly or monthly depending on the level of use of the CrossData/400 application. Select an option to begin work.

```
********************************************************************************
01 * DTMNU3             Alliance CrossData/400 Maintenance Menu
```

Application Maintenance

This module is used to remove records from Alliance history and logging files and to reorganize database files to recover database space. It is recommended that you run maintenance weekly or monthly depending on the level of use of the security application.

```
********************************************************************************
01 * DT4095A                        Application Maintenance
```

Retain history after date

Enter the date from which you wish to retain history records. The date is entered in YYYYMMDD format. Records may be removed from AFHIST (history file) and AFLOGA (application log file.)

Data Conversion

Alliance CrossData/400 includes both interactive and program interfaces to data conversion services. You can access the interactive services on the Data Conversion menu:

```
********************************************************************************
01 * DTMNU4              Alliance CrossData/400 Data Conversion Menu
```

Select an option to begin work.

Convert file to CSV/EBCDIC

Use this option to convert an existing database file to a CSV (Comma Separated Values) file. The output will be in the EBCDIC character set. If you specify an output file that already exists the output will be appended to the file. You should be aware, however, that the data may be invalid if the existing record length is not correct for the converted data. It is recommended that you allow CrossData/400 to create the file for you.

```
********************************************************************************
01 * DT1100A                     Convert File to CSV/EBCDIC          ROBB
```

File name

Enter the name of the file you want to convert. This must be an existing externally described database file on the AS/400.

Library name

Enter the name of the library containing the file you want to convert. This can be the special value *LIBL or *CURLIB.

Member name

Enter the name of the member in the file that you want to convert. This can be the special value *FIRST.

Convert to file

Enter the name of the output file for the converted data. CrossData/400 will create this file for you during the conversion process, and will calculate the correct length of the records in the file.

Library name

Enter the name of the library that will contain the converted data. This must be the name of an AS/400 library.

Member name

Enter the name of the member that will contain the converted data. This must be an actual member name.

Copy to stream file

Enter 1 for Yes if you want to copy the converted data to a stream file. The stream file can exist in a Web directory on your AS/400 system, or in the QDLS folder system. Enter 2 for No if you do not want to copy the converted data to a stream file.

Stream file

Enter the full path to the stream file. If your Web files are stored in a directory called “webdocs” in the directory “www” your full path name might look like this:

```
/www/webdocs/data.csv
```

To copy a converted file to the folder file system your path might look like this:

```
/QDLS/docs/data.csv
```

Send via e-mail

Enter 1 for Yes if you want to send the converted file to an email recipient. Enter 2 for No to not send the converted file. If you enter Yes you will be prompted for the email address of the recipient, your e-mail address, the subject, and up to five lines of message text. Note that your AS/400 must be configured for SMTP mail and must have the IBM Send Mail API installed.

To

Enter the e-mail address of the recipient. This must be a valid SMTP address of a local or Internet recipient.

If you entered Yes to send the converted file via e-mail, the following panel will be displayed.

```
*****************************************************************
01 * DT1100B                  Convert File to CSV/EBCDIC       ROBB
```

From

Enter the From address for this message. This is normally your e-mail address at this AS/400. The first time you use this option the From address will be blank. After the first time CrossData/400 will remember your From address and use it as the default. This must be a valid SMTP address. If you are sending to an Internet recipient you should be aware of the fact that most Internet Service Providers will not deliver your message if the domain name in your From address is not registered on the Internet. This is done to reduce unsolicited e-mail (SPAM).

Reply to

Enter the Reply To address if it is different that the From address. This must also be a valid SMTP mail address.

Subject

Enter a subject for this message. It is recommended that you keep the subject to no more than 44 characters to remain compatible with most email applications.

Attachment name

Enter the file name you want to use for the converted file. The default is the AS/400 file name with the .CSV extension. You can use any name that is a valid file name on the receiving system.

Message

Enter up to five lines of message text. These lines of text are optional.

If you pressed F18 to change defaults on the first panel the following panel is displayed. The defaults for the current translation option are shown and you can change any of the default values or options. Note that if you change the data conversion defaults you will not longer have a standard CSV, tab-delimited, or fixed ASCII field file. After you change the defaults you will be returned to the first panel and the new values will be used for the conversion.

```
**************************************************************************
01 * DT1100C                          Change Defaults                ROBB
```

New line option

Enter 1 if you do not want to use new line characters. Enter 2 if you want to add Carriage Return and New Line characters at the end of each line. Enter 3 if you want to add just a Carriage Return character at the end of the line. Enter 4 if you want to add just a New Line character at the end of each line.

Field delimiter

Enter the character you want to use to separate fields. This is normally a comma for Excel CSV (Comma Separated Values) files. If you leave the field blank fields will not be separated by a delimiter.

Numeric decimal

Enter the character you want to use for numeric decimals. This can be a period (“.”), comma (“,”) or any other character. If you leave this field blank no decimal value will be used.

Numeric negative sign

Enter the character you want to use to indicate a negative value in a numeric field. This can be a dash (“-“) or any other character. If you leave this field blank no character will be used to indicate a negative field and all negative fields will appear positive.

Numeric zero display

Enter 1 to use blanks for zero value numeric fields. Enter 2 to use zeros for zero value numeric fields.

Numeric leading blanks

Enter 1 for Yes to leave leading blanks or zeros at the beginning of numeric fields (the default). Enter 2 for No to remove leading blanks or zeros. Whether a numeric field will have leading blanks or zeros depends on the value of the Numeric Zero Display option in the previous field.

Character field quote

Enter the character you want to use to surround character fields. You can use the double quote (“) character or any other character. If you leave this field blank character fields will not be surrounded.

Blank truncation

Enter 1 for Yes if you want to truncate the trailing blanks in character fields. Enter 2 for No if you do not want to truncate trailing blanks.

Column titles

Enter 1 if you do not want to use column titles in your output file. Enter 2 if you want to use the field name as the column title. Enter 3 if you want to use the column header data defined for the fields.

Translate to ASCII

Enter 1 for Yes if you want to translate the output data to ASCII. Enter 2 for No if you do not want to translate the data to ASCII. If you specify Yes for this option you must specify a table name and library in the following fields.

Translation table and library

If you specified Yes to translate to ASCII you must specify a translation table and library. The default is the IBM supplied table QASCII in library QSYS, but you can specify any valid translation table. You can use the DTEB2AS table in library ALLDTA200, or you can create your own translation table. There is a template in file QTBLSRC in the Alliance product library.

Convert file to ASC/EBCDIC

Use this option to convert an existing database file to a ASC (fixed length values) file. The output will be in the EBCDIC character set. If you specify an output file that already exists the output will be appended to the file. You should be aware, however, that the data may be invalid if the existing record length is not correct for the converted data. It is recommended that you allow CrossData/400 to create the file for you.

```
********************************************************************
01 * DT1110A                  Convert File to ASC/EBCDIC       ROBB
```

File name

Enter the name of the file you want to convert. This must be an existing externally described database file on the AS/400.

Library name

Enter the name of the library containing the file you want to convert. This can be the special value *LIBL or *CURLIB.

Member name

Enter the name of the member in the file that you want to convert. This can be the special value *FIRST.

Convert to file

Enter the name of the output file for the converted data. CrossData/400 will create this file for you during the conversion process, and will calculate the correct length of the records in the file.

Library name

Enter the name of the library that will contain the converted data. This must be the name of an AS/400 library.

Member name

Enter the name of the member that will contain the converted data. This must be an actual member name.

Copy to stream file

Enter 1 for Yes if you want to copy the converted data to a stream file. The stream file can exist in a Web directory on your AS/400 system, or in the QDLS folder system. Enter 2 for No if you do not want to copy the converted data to a stream file.

Stream file

Enter the full path to the stream file. If your Web files are stored in a directory called “webdocs” in the directory “www” your full path name might look like this:

```
/www/webdocs/data.csv
```

To copy a converted file to the folder file system your path might look like this:

```
/QDLS/docs/data.csv
```

Send via e-mail

Enter 1 for Yes if you want to send the converted file to an email recipient. Enter 2 for No to not send the converted file. If you enter Yes you will be prompted for the email address of the recipient, your e-mail address, the subject, and up to five lines of message text. Note that your AS/400 must be configured for SMTP mail and must have the IBM Send Mail API installed.

*To**

Enter the e-mail address of the recipient. This must be a valid SMTP address of a local or Internet recipient.

If you entered Yes to send the converted file via e-mail, the following panel will be displayed.

```
**************************************************************************
01 * DT1110B                    Convert File to ASC/EBCDIC        ROBB
```

From

Enter the From address for this message. This is normally your e-mail address at this AS/400. The first time you use this option the From address will be blank. After the first time CrossData/400 will remember your From address and use it as the default. This must be a valid SMTP address. If you are sending to an Internet recipient you should be aware of the fact that most Internet Service Providers will not deliver your message if the domain name in your From address is not registered on the Internet. This is done to reduce unsolicited e-mail (SPAM).

Reply to

Enter the Reply To address if it is different that the From address. This must also be a valid SMTP mail address.

Subject

Enter a subject for this message. It is recommended that you keep the subject to no more than 44 characters to remain compatible with most email applications.

Attachment name

Enter the file name you want to use for the converted file. The default is the AS/400 file name with the .CSV extension. You can use any name that is a valid file name on the receiving system.

Message

Enter up to five lines of message text. These lines of text are optional.

If you pressed F18 to change defaults on the first panel the following panel is displayed. The defaults for the current translation option are shown and you can change any of the default values or options. Note that if you change the data conversion defaults you will not longer have a standard CSV, tab-delimited, or fixed ASCII field file. After you change the defaults you will be returned to the first panel and the new values will be used for the conversion.

```
**************************************************************************
01 * DT1110C                       Change Defaults                ROBB
```

New line option

Enter 1 if you do not want to use new line characters. Enter 2 if you want to add Carriage Return and New Line characters at the end of each line. Enter 3 if you want to add just a Carriage Return character at the end of the line. Enter 4 if you want to add just a New Line character at the end of each line.

Field delimiter

Enter the character you want to use to separate fields. This is normally a comma for Excel CSV (Comma Separated Values) files. If you leave the field blank fields will not be separated by a delimiter.

Numeric decimal

Enter the character you want to use for numeric decimals. This can be a period (“.”), comma (“,”) or any other character. If you leave this field blank no decimal value will be used.

**Numeric negative sign*

Enter the character you want to use to indicate a negative value in a numeric field. This can be a dash (“-“) or any other character. If you leave this field blank no character will be used to indicate a negative field and all negative fields will appear positive.

Numeric zero display

Enter 1 to use blanks for zero value numeric fields. Enter 2 to use zeros for zero value numeric fields.

Numeric leading blanks

Enter 1 for Yes to leave leading blanks or zeros at the beginning of numeric fields (the default). Enter 2 for No to remove leading blanks or zeros. Whether a numeric field will have leading blanks or zeros depends on the value of the Numeric Zero Display option in the previous field.

Character field quote

Enter the character you want to use to surround character fields. You can use the double quote (“) character or any other character. If you leave this field blank character fields will not be surrounded.

Blank truncation

Enter 1 for Yes if you want to truncate the trailing blanks in character fields. Enter 2 for No if you do not want to truncate trailing blanks.

Column titles

Enter 1 if you do not want to use column titles in your output file. Enter 2 if you want to use the field name as the column title. Enter 3 if you want to use the column header data defined for the fields.

Translate to ASCII

Enter 1 for Yes if you want to translate the output data to ASCII. Enter 2 for No if you do not want to translate the data to ASCII. If you specify Yes for this option you must specify a table name and library in the following fields.

Translation table and library

If you specified Yes to translate to ASCII you must specify a translation table and library. The default is the IBM supplied table QASCII in library QSYS, but you can specify any valid translation table. You can use the DTEB2AS table in library ALLDTA200, or you can create your own translation table. There is a template in file QTBLSRC in the Alliance product library.

Convert file to TAB/EBCDIC

Use this option to convert an existing database file to a TAB (Tab-delimited fields) file. The output will be in the EBCDIC character set. If you specify an output file that already exists the output will be appended to the file. You should be aware, however, that the data may be invalid if the existing record length is not correct for the converted data. It is recommended that you allow CrossData/400 to create the file for you.

```
*****************************************************************
01 * DT1150A                  Convert File to TAB/EBCDIC       ROBB
```

File name

Enter the name of the file you want to convert. This must be an existing externally described database file on the AS/400.

Library name

Enter the name of the library containing the file you want to convert. This can be the special value *LIBL or *CURLIB.

Member name

Enter the name of the member in the file that you want to convert. This can be the special value *FIRST.

Convert to file

Enter the name of the output file for the converted data. CrossData/400 will create this file for you during the conversion process, and will calculate the correct length of the records in the file.

Library name

Enter the name of the library that will contain the converted data. This must be the name of an AS/400 library.

Member name

Enter the name of the member that will contain the converted data. This must be an actual member name.

Copy to stream file

Enter 1 for Yes if you want to copy the converted data to a stream file. The stream file can exist in a Web directory on your AS/400 system, or in the QDLS folder system. Enter 2 for No if you do not want to copy the converted data to a stream file.

Stream file

Enter the full path to the stream file. If your Web files are stored in a directory called “webdocs” in the directory “www” your full path name might look like this:

```
/www/webdocs/data.tab
```

To copy a converted file to the folder file system your path might look like this:

```
/QDLS/docs/data.tab
```

Send via e-mail

Enter 1 for Yes if you want to send the converted file to an email recipient. Enter 2 for No to not send the converted file. If you enter Yes you will be prompted for the email address of the recipient, your e-mail address, the subject, and up to five lines of message text. Note that your AS/400 must be configured for SMTP mail and must have the IBM Send Mail API installed.

To

Enter the e-mail address of the recipient. This must be a valid SMTP address of a local or Internet recipient.

If you entered Yes to send the converted file via e-mail, the following panel will be displayed.

```
****************************************************************
01 * DT1150B                  Convert File to TAB/EBCDIC       ROBB
```

From

Enter the From address for this message. This is normally your e-mail address at this AS/400. The first time you use this option the From address will be blank. After the first time CrossData/400 will remember your From address and use it as the default. This must be a valid SMTP address. If you are sending to an Internet recipient you should be aware of the fact that most Internet Service Providers will not deliver your message if the domain name in your From address is not registered on the Internet. This is done to reduce unsolicited e-mail (SPAM).

Reply to

Enter the Reply To address if it is different that the From address. This must also be a valid SMTP mail address.

Subject

Enter a subject for this message. It is recommended that you keep the subject to no more than 44 characters to remain compatible with most email applications.

Attachment name

Enter the file name you want to use for the converted file. The default is the AS/400 file name with the .TAB extension. You can use any name that is a valid file name on the receiving system.

Message

Enter up to five lines of message text. These lines of text are optional.

If you pressed F18 to change defaults on the first panel the following panel is displayed. The defaults for the current translation option are shown and you can change any of the default values or options. Note that if you change the data conversion defaults you will not longer have a standard CSV, tab-delimited, or fixed ASCII field file. After you change the defaults you will be returned to the first panel and the new values will be used for the conversion.

```
*****************************************************************
01 * DT1150C                        Change Defaults              ROBB
```

New line option

Enter 1 if you do not want to use new line characters. Enter 2 if you want to add Carriage Return and New Line characters at the end of each line. Enter 3 if you want to add just a Carriage Return character at the end of the line. Enter 4 if you want to add just a New Line character at the end of each line.

Field delimiter

Enter the character you want to use to separate fields. This is normally a comma for Excel CSV (Comma Separated Values) files. If you leave the field blank fields will not be separated by a delimiter.

Numeric decimal

Enter the character you want to use for numeric decimals. This can be a period (“.”), comma (“,”) or any other character. If you leave this field blank no decimal value will be used.

Numeric negative sign

Enter the character you want to use to indicate a negative value in a numeric field. This can be a dash (“-“) or any other character. If you leave this field blank no character will be used to indicate a negative field and all negative fields will appear positive.

Numeric zero display

Enter 1 to use blanks for zero value numeric fields. Enter 2 to use zeros for zero value numeric fields.

Numeric leading blanks

Enter 1 for Yes to leave leading blanks or zeros at the beginning of numeric fields (the default). Enter 2 for No to remove leading blanks or zeros. Whether a numeric field will have leading blanks or zeros depends on the value of the Numeric Zero Display option in the previous field.

Character field quote

Enter the character you want to use to surround character fields. You can use the double quote (“) character or any other character. If you leave this field blank character fields will not be surrounded.

Blank truncation

Enter 1 for Yes if you want to truncate the trailing blanks in character fields. Enter 2 for No if you do not want to truncate trailing blanks.

Column titles

Enter 1 if you do not want to use column titles in your output file. Enter 2 if you want to use the field name as the column title. Enter 3 if you want to use the column header data defined for the fields.

Translate to ASCII

Enter 1 for Yes if you want to translate the output data to ASCII. Enter 2 for No if you do not want to translate the data to ASCII. If you specify Yes for this option you must specify a table name and library in the following fields.

Translation table and library

If you specified Yes to translate to ASCII you must specify a translation table and library. The default is the IBM supplied table QASCII in library QSYS, but you can specify any valid translation table. You can use the DTEB2AS table in library ALLDTA200, or you can create your own translation table. There is a template in file QTBLSRC in the Alliance product library.

Convert file to CSV/ASCII

Use this option to convert an existing database file to a CSV (Comma Separated Values) file. The output will be in the ASCII character set. If you specify an output file that already exists the output will be appended to the file. You should be aware, however, that the data may be invalid if the existing record length is not correct for the converted data. It is recommended that you allow CrossData/400 to create the file for you.



File name

Enter the name of the file you want to convert. This must be an existing externally described database file on the AS/400.

Library name

Enter the name of the library containing the file you want to convert. This can be the special value *LIBL or *CURLIB.

Member name

Enter the name of the member in the file that you want to convert. This can be the special value *FIRST.

Convert to file

Enter the name of the output file for the converted data. CrossData/400 will create this file for you during the conversion process, and will calculate the correct length of the records in the file.

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Library name

Enter the name of the library that will contain the converted data. This must be the name of an AS/400 library.

Member name

Enter the name of the member that will contain the converted data. This must be an actual member name.

Copy to stream file

Enter 1 for Yes if you want to copy the converted data to a stream file. The stream file can exist in a Web directory on your AS/400 system, or in the QDLS folder system. Enter 2 for No if you do not want to copy the converted data to a stream file.

Stream file

Enter the full path to the stream file. If your Web files are stored in a directory called “webdocs” in the directory “www” your full path name might look like this:

/www/webdocs/data.csv

To copy a converted file to the folder file system your path might look like this:

/QDLS/docs/data.csv

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If you pressed F18 to change defaults on the first panel the following panel is displayed. The defaults for the current translation option are shown and you can change any of the default values or options. Note that if you change the data conversion defaults you will not longer have a standard CSV, tab-delimited, or fixed ASCII field file. After you change the defaults you will be returned to the first panel and the new values will be used for the conversion.

```
*****************************************************************
01 * DT1130C                      Change Defaults             ROBB
```

New line option

Enter 1 if you do not want to use new line characters. Enter 2 if you want to add Carriage Return and New Line characters at the end of each line. Enter 3 if you want to add just a Carriage Return character at the end of the line. Enter 4 if you want to add just a New Line character at the end of each line.

Field delimiter

Enter the character you want to use to separate fields. This is normally a comma for Excel CSV (Comma Separated Values) files. If you leave the field blank fields will not be separated by a delimiter.

Numeric decimal

Enter the character you want to use for numeric decimals. This can be a period (“.”), comma (“,”) or any other character. If you leave this field blank no decimal value will be used.

Numeric negative sign

Enter the character you want to use to indicate a negative value in a numeric field. This can be a dash (“-“) or any other character. If you leave this field blank no character will be used to indicate a negative field and all negative fields will appear positive.

Numeric zero display

Enter 1 to use blanks for zero value numeric fields. Enter 2 to use zeros for zero value numeric fields.

Numeric leading blanks

Enter 1 for Yes to leave leading blanks or zeros at the beginning of numeric fields (the default). Enter 2 for No to remove leading blanks or zeros. Whether a numeric field will have leading blanks or zeros depends on the value of the Numeric Zero Display option in the previous field.

Character field quote

Enter the character you want to use to surround character fields. You can use the double quote (“) character or any other character. If you leave this field blank character fields will not be surrounded.

Blank truncation

Enter 1 for Yes if you want to truncate the trailing blanks in character fields. Enter 2 for No if you do not want to truncate trailing blanks.

Column titles

Enter 1 if you do not want to use column titles in your output file. Enter 2 if you want to use the field name as the column title. Enter 3 if you want to use the column header data defined for the fields.

Translate to ASCII

Enter 1 for Yes if you want to translate the output data to ASCII. Enter 2 for No if you do not want to translate the data to ASCII. If you specify Yes for this option you must specify a table name and library in the following fields.

Translation table and library

If you specified Yes to translate to ASCII you must specify a translation table and library. The default is the IBM supplied table QASCII in library QSYS, but you can specify any valid translation table. You can use the DTEB2AS table in library ALLDTA200, or you can create your own translation table. There is a template in file QTBLSRC in the Alliance product library.

Convert file to ASC/ASCII

Use this option to convert an existing database file to an ASC (Fixed Length Format) file. The output will be in the ASCII character set. If you specify an output file that already exists the output will be appended to the file. You should be aware, however, that the data may be invalid if the existing record length is not correct for the converted data. It is recommended that you allow CrossData/400 to create the file for you.

```
*****************************************************************
01 * DT1140A                    Convert File to ASC/ASCII       ROBB
```

File name Enter the name of the file you want to convert. This must be an existing externally described database file on the AS/400.

Library name

Enter the name of the library containing the file you want to convert. This can be the special value *LIBL or *CURLIB.

Member name

Enter the name of the member in the file that you want to convert. This can be the special value *FIRST.

Convert to file

Enter the name of the output file for the converted data. CrossData/400 will create this file for you during the conversion process, and will calculate the correct length of the records in the file.

Library name

Enter the name of the library that will contain the converted data. This must be the name of an AS/400 library.

Member name

Enter the name of the member that will contain the converted data. This must be an actual member name.

Copy to stream file

Enter 1 for Yes if you want to copy the converted data to a stream file. The stream file can exist in a Web directory on your AS/400 system, or in the QDLS folder system. Enter 2 for No if you do not want to copy the converted data to a stream file.

Stream file

Enter the full path to the stream file. If your Web files are stored in a directory called “webdocs” in the directory “www” your full path name might look like this:

```
/www/webdocs/data.csv
```

To copy a converted file to the folder file system your path might look like this:

```
/QDLS/docs/data.csv
```

If you pressed F18 to change defaults on the first panel the following panel is displayed. The defaults for the current translation option are shown and you can change any of the default values or options. Note that if you change the data conversion defaults you will not longer have

a standard CSV, tab-delimited, or fixed ASCII field file. After you change the defaults you will be returned to the first panel and the new values will be used for the conversion.



New line option

Enter 1 if you do not want to use new line characters. Enter 2 if you want to add Carriage Return and New Line characters at the end of each line. Enter 3 if you want to add just a Carriage Return character at the end of the line. Enter 4 if you want to add just a New Line character at the end of each line.

Field delimiter

Enter the character you want to use to separate fields. This is normally a comma for Excel CSV (Comma Separated Values) files. If you leave the field blank fields will not be separated by a delimiter.

Numeric decimal

Enter the character you want to use for numeric decimals. This can be a period (“.”), comma (“,”) or any other character. If you leave this field blank no decimal value will be used.

Numeric negative sign

Enter the character you want to use to indicate a negative value in a numeric field. This can be a dash (“-“) or any other character. If you leave this field blank no character will be used to indicate a negative field and all negative fields will appear positive.

Numeric zero display

Enter 1 to use blanks for zero value numeric fields. Enter 2 to use zeros for zero value numeric fields.

Numeric leading blanks

Enter 1 for Yes to leave leading blanks or zeros at the beginning of numeric fields (the default). Enter 2 for No to remove leading blanks or zeros. Whether a numeric field will have leading blanks or zeros depends on the value of the Numeric Zero Display option in the previous field.

Character field quote

Enter the character you want to use to surround character fields. You can use the double quote (“) character or any other character. If you leave this field blank character fields will not be surrounded.

Blank truncation

Enter 1 for Yes if you want to truncate the trailing blanks in character fields. Enter 2 for No if you do not want to truncate trailing blanks.

Column titles

Enter 1 if you do not want to use column titles in your output file. Enter 2 if you want to use the field name as the column title. Enter 3 if you want to use the column header data defined for the fields.

Translate to ASCII

Enter 1 for Yes if you want to translate the output data to ASCII. Enter 2 for No if you do not want to translate the data to ASCII. If you specify Yes for this option you must specify a table name and library in the following fields.

Translation table and library

If you specified Yes to translate to ASCII you must specify a translation table and library. The default is the IBM supplied table QASCII in library QSYS, but you can specify any valid translation table. You can use the DTEB2AS table in library ALLDTA200, or you can create your own translation table. There is a template in file QTBLSRC in the Alliance product library.

Convert file to TAB/ASCII

Use this option to convert an existing database file to a TAB (Tab-delimited fields) file. The output will be in the ASCII character set. If you specify an output file that already exists the output will be appended to the file. You should be aware, however, that the data may be invalid if the existing record length is not correct for the converted data. It is recommended that you allow CrossData/400 to create the file for you.

```
*****************************************************************
01 * DT1160A                  Convert File to TAB/ASCII        ROBB
```

File name

Enter the name of the file you want to convert. This must be an existing externally described database file on the AS/400.

Library name

Enter the name of the library containing the file you want to convert. This can be the special value *LIBL or *CURLIB.

Member name

Enter the name of the member in the file that you want to convert. This can be the special value *FIRST.

Convert to file

Enter the name of the output file for the converted data. CrossData/400 will create this file for you during the conversion process, and will calculate the correct length of the records in the file.

Library name

Enter the name of the library that will contain the converted data. This must be the name of an AS/400 library.

Member name

Enter the name of the member that will contain the converted data. This must be an actual member name.

Copy to stream file

Enter 1 for Yes if you want to copy the converted data to a stream file. The stream file can exist in a Web directory on your AS/400 system, or in the QDLS folder system. Enter 2 for No if you do not want to copy the converted data to a stream file.

Stream file

Enter the full path to the stream file. If your Web files are stored in a directory called “webdocs” in the directory “www” your full path name might look like this:

```
/www/webdocs/data.tab
```

To copy a converted file to the folder file system your path might look like this:

```
/QDLS/docs/data.tab
```

```
If you entered Yes to send the converted file via e-mail, the follo

*****************************************************************
01 * DT1160B                  Convert File to TAB/ASCII        ROBB
```

From

Enter the From address for this message. This is normally your e-mail address at this AS/400. The first time you use this option the From address will be blank. After the first time CrossData/400 will remember your From address and use it as the default. This must be a valid SMTP address. If you are sending to an Internet recipient you should be aware of the fact that most Internet Service Providers will not deliver your message if the domain name in your From address is not registered on the Internet. This is done to reduce unsolicited e-mail (SPAM).

Reply to

Enter the Reply To address if it is different that the From address. This must also be a valid SMTP mail address.

Subject

Enter a subject for this message. It is recommended that you keep the subject to no more than 44 characters to remain compatible with most email applications.

Attachment name

Enter the file name you want to use for the converted file. The default is the AS/400 file name with the .TAB extension. You can use any name that is a valid file name on the receiving system.

Message

Enter up to five lines of message text. These lines of text are optional.

If you pressed F18 to change defaults on the first panel the following panel is displayed. The defaults for the current translation option are shown and you can change any of the default values or options. Note that if you change the data conversion defaults you will not longer have a standard CSV, tab-delimited, or fixed ASCII field file. After you change the defaults you will be returned to the first panel and the new values will be used for the conversion.



New line option

Enter 1 if you do not want to use new line characters. Enter 2 if you want to add Carriage Return and New Line characters at the end of each line. Enter 3 if you want to add just a Carriage Return character at the end of the line. Enter 4 if you want to add just a New Line character at the end of each line.

Field delimiter

Enter the character you want to use to separate fields. This is normally a comma for Excel CSV (Comma Separated Values) files. If you leave the field blank fields will not be separated by a delimiter.

Numeric decimal

Enter the character you want to use for numeric decimals. This can be a period (“.”), comma (“,”) or any other character. If you leave this field blank no decimal value will be used.

Numeric negative sign

Enter the character you want to use to indicate a negative value in a numeric field. This can be a dash (“-“) or any other character. If you leave this field blank no character will be used to indicate a negative field and all negative fields will appear positive.

Numeric zero display

Enter 1 to use blanks for zero value numeric fields. Enter 2 to use zeros for zero value numeric fields.

Numeric leading blanks

Enter 1 for Yes to leave leading blanks or zeros at the beginning of numeric fields (the default). Enter 2 for No to remove leading blanks or zeros. Whether a numeric field will have leading blanks or zeros depends on the value of the Numeric Zero Display option in the previous field.

Character field quote

Enter the character you want to use to surround character fields. You can use the double quote (“) character or any other character. If you leave this field blank character fields will not be surrounded.

Blank truncation

Enter 1 for Yes if you want to truncate the trailing blanks in character fields. Enter 2 for No if you do not want to truncate trailing blanks.

Column titles

Enter 1 if you do not want to use column titles in your output file. Enter 2 if you want to use the field name as the column title. Enter 3 if you want to use the column header data defined for the fields.

Translate to ASCII

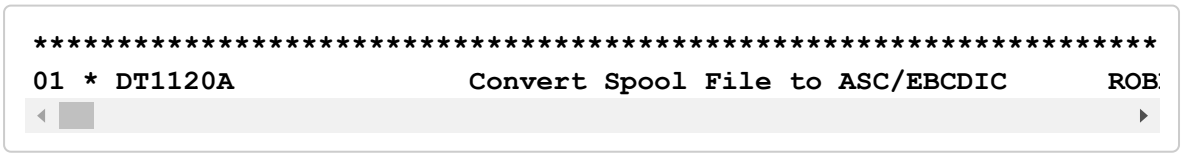
Enter 1 for Yes if you want to translate the output data to ASCII. Enter 2 for No if you do not want to translate the data to ASCII. If you specify Yes for this option you must specify a table name and library in the following fields.

Translation table and library

If you specified Yes to translate to ASCII you must specify a translation table and library. The default is the IBM supplied table QASCII in library QSYS, but you can specify any valid translation table. You can use the DTEB2AS table in library ALLDTA200, or you can create your own translation table. There is a template in file QTBLSRC in the Alliance product library.

Convert spool file to ASC/EBCDIC**

Use this option to convert a spool file to an ASC (Fixed length values) file. The output will be in the EBCDIC character set. If you specify an output file that already exists the output will be appended to the file. You should be aware, however, that the data may be invalid if the existing record length is not correct for the converted data. It is recommended that you allow CrossData/400 to create the file for you.



Spool file name

Enter the name of the spool file you want to convert. This must be an existing spool file in an output queue on the AS/400.

Job name

Enter the name of the job that created the spool file. You can use the Work With Spool Files (WRKSPLF) command or the Work With Output Queues (WRKOUTQ) command to view the name of the job that creates a spool file.

User name

Enter the name of the user who created the spool file.

Job number

Enter the job number of the job that created the spool file.

Spool file number

Enter the spool file number of the spool file. Spool file numbers are assigned sequentially in a job. The first spool file created is 0001, the second 0002, etc.

Convert to file

Enter the name of the output file for the converted data. CrossData/400 will create this file for you during the conversion process, and will calculate the correct length of the records in the file.

Library name

Enter the name of the library that will contain the converted data. This must be the name of an AS/400 library.

Member name

Enter the name of the member that will contain the converted data. This must be an actual member name.

Copy to stream file

Enter 1 for Yes if you want to copy the converted data to a stream file. The stream file can exist in a Web directory on your AS/400 system, or in the QDLS folder system. Enter 2 for No if you do not want to copy the converted data to a stream file.

Stream file

Enter the full path to the stream file. If your Web files are stored in a directory called “webdocs” in the directory “www” your full path name might look like this:

```
/www/webdocs/data.csv
```

To copy a converted file to the folder file system your path might look like this:

```
/QDLS/docs/data.csv
```

Query to CSV/EBCDIC

Use this option to run a query and convert the output to the CSV (Comma Separated Values) format. If the query is defined to print the output or display it interactively, the output will be overridden to a file. You can specify any existing query to select and convert data. In addition to converting the data this option will let you copy the converted data to a shared folder or IFS directory. You can also send the converted data to an email recipient.

```
************************************************************************
01 * DT1170A                    Query File to CSV/EBCDIC         ROBB
```

Query name and library

Enter the name of the query you want use as a source for the data, and the library that contains the query definition. This can be any existing IBM query.

File name, library, and member

Enter the name of the file to use for the query. You can enter *SAME to use the same file that was used to create the query, or enter a file name. If you enter *SAME you must leave the library and member names blank. If you specify a file name you must also specify a library and member name. The definition of the file you select must be the same as the file that was used to create the query.

Convert to file

Enter the name of the output file for the converted data. CrossData/400 will create this file for you during the conversion process, and will calculate the correct length of the records in the file.

Library name

Enter the name of the library that will contain the converted data. This must be the name of an AS/400 library.

Member name

Enter the name of the member that will contain the converted data. This must be an actual member name.

Copy to stream file

Enter 1 for Yes if you want to copy the converted data to a stream file. The stream file can exist in a Web directory on your AS/400 system, or in the QDLS folder system. Enter 2 for No if you do not want to copy the converted data to a stream file.

Stream file

Enter the full path to the stream file. If your Web files are stored in a directory called “webdocs” in the directory “www” your full path name might look like this:

```
/www/webdocs/data.csv
```

To copy a converted file to the folder file system your path might look like this:

```
/QDLS/docs/data.csv
```

Send via e-mail

Enter 1 for Yes if you want to send the converted file to an email recipient. Enter 2 for No to not send the converted file. If you enter Yes you will be prompted for the email address of the recipient, your e-mail address, the subject, and up to five lines of message text. Note that your AS/400 must be configured for SMTP mail and must have the IBM Send Mail API installed.

To

Enter the e-mail address of the recipient. This must be a valid SMTP address of a local or Internet recipient.

If you entered Yes to send the converted file via e-mail, the following panel will be displayed.

```
************************************************************************
01 * DT1170B                    Query File to CSV/EBCDIC          ROBB
```

From

Enter the From address for this message. This is normally your e-mail address at this AS/400. The first time you use this option the From address will be blank. After the first time CrossData/400 will remember your From address and use it as the default. This must be a valid SMTP address. If you are sending to an Internet recipient you should be aware of the fact that most Internet Service Providers will not deliver your message if the domain name in your From address is not registered on the Internet. This is done to reduce unsolicited e-mail (SPAM).

Reply to

Enter the Reply To address if it is different that the From address. This must also be a valid SMTP mail address.

Subject

Enter a subject for this message. It is recommended that you keep the subject to no more than 44 characters to remain compatible with most email applications.

Attachment name

Enter the file name you want to use for the converted file. The default is the AS/400 file name with the .CSV extension. You can use any name that is a valid file name on the receiving system.

Message

Enter up to five lines of message text. These lines of text are optional.

Query to ASC/EBCDIC

Use this option to run a query and convert the output to the ASC (ASCII fixed length) format. If the query is defined to print the output or display it interactively, the output will be overridden to a file. You can specify any existing query to select and convert data. The resulting data will be in the EBCDIC character set. In addition to converting the data this option will let you copy the converted data to a shared folder or IFS directory. You can also send the converted data to an email recipient.

```
************************************************************************
01 * DT1180A                    Query File to ASC/EBCDIC          ROBB
```

Query name and library

Enter the name of the query you want use as a source for the data, and the library that contains the query definition. This can be any existing IBM query.

File name, library, and member

Enter the name of the file to use for the query. You can enter *SAME to use the same file that was used to create the query, or enter a file name. If you enter *SAME you must leave the library and member names blank. If you specify a file name you must also specify a library and member name. The definition of the file you select must be the same as the file that was used to create the query.

Convert to file

Enter the name of the output file for the converted data. CrossData/400 will create this file for you during the conversion process, and will calculate the correct length of the records in the file.

Library name

Enter the name of the library that will contain the converted data. This must be the name of an AS/400 library.

Member name

Enter the name of the member that will contain the converted data. This must be an actual member name.

Copy to stream file

Enter 1 for Yes if you want to copy the converted data to a stream file. The stream file can exist in a Web directory on your AS/400 system, or in the QDLS folder system. Enter 2 for No if you do not want to copy the converted data to a stream file.

Stream file

Enter the full path to the stream file. If your Web files are stored in a directory called “webdocs” in the directory “www” your full path name might look like this:

```
/www/webdocs/data.prn
```

To copy a converted file to the folder file system your path might look like this:

```
/QDLS/docs/data.prn
```

Send via e-mail

Enter 1 for Yes if you want to send the converted file to an email recipient. Enter 2 for No to not send the converted file. If you enter Yes you will be prompted for the email address of the recipient, your e-mail address, the subject, and up to five lines of message text. Note that your AS/400 must be configured for SMTP mail and must have the IBM Send Mail API installed.

To

Enter the e-mail address of the recipient. This must be a valid SMTP address of a local or Internet recipient.

If you entered Yes to send the converted file via e-mail, the following panel will be displayed.

```
*****************************************************************
01 * DT1180B                    Query File to ASC/EBCDIC        ROBB
```

From

Enter the From address for this message. This is normally your e-mail address at this AS/400. The first time you use this option the From address will be blank. After the first time CrossData/400 will remember your From address and use it as the default. This must be a valid SMTP address. If you are sending to an Internet recipient you should be aware of the fact that most Internet Service Providers will not deliver your message if the domain name in your From address is not registered on the Internet. This is done to reduce unsolicited e-mail (SPAM).

Reply to

Enter the Reply To address if it is different that the From address. This must also be a valid SMTP mail address.

Subject

Enter a subject for this message. It is recommended that you keep the subject to no more than 44 characters to remain compatible with most email applications.

Attachment name

Enter the file name you want to use for the converted file. The default is the AS/400 file name with the .CSV extension. You can use any name that is a valid file name on the receiving system.

Message

Enter up to five lines of message text. These lines of text are optional.

Query to TAB/EBCDIC

Use this option to run a query and convert the output to the TAB (tab-delimited values) format. If the query is defined to print the output or display it interactively, the output will be overridden to a file. You can specify any existing query to select and convert data. The resulting data will be in the EBCDIC character set. In addition to converting the data this option will let you copy the converted data to a shared folder or IFS directory. You can also send the converted data to an email recipient.

```
*****************************************************************
01 * DT1190A                    Query File to TAB/EBCDIC        ROBB
```

Query name and library

Enter the name of the query you want use as a source for the data, and the library that contains the query definition. This can be any existing IBM query.

File name, library, and member

Enter the name of the file to use for the query. You can enter *SAME to use the same file that was used to create the query, or enter a file name. If you enter *SAME you must leave the library and member names blank. If you specify a file name you must also specify a library and member name. The definition of the file you select must be the same as the file that was used to create the query.

Convert to file

Enter the name of the output file for the converted data. CrossData/400 will create this file for you during the conversion process, and will calculate the correct length of the records in the file.

Library name

Enter the name of the library that will contain the converted data. This must be the name of an AS/400 library.

Member name

Enter the name of the member that will contain the converted data. This must be an actual member name.

Copy to stream file

Enter 1 for Yes if you want to copy the converted data to a stream file. The stream file can exist in a Web directory on your AS/400 system, or in the QDLS folder system. Enter 2 for No if you do not want to copy the converted data to a stream file.

Stream file

Enter the full path to the stream file. If your Web files are stored in a directory called “webdocs” in the directory “www” your full path name might look like this:

```
/www/webdocs/data.csv
```

To copy a converted file to the folder file system your path might look like this:

```
/QDLS/docs/data.csv
```

Send via e-mail

Enter 1 for Yes if you want to send the converted file to an email recipient. Enter 2 for No to not send the converted file. If you enter Yes you will be prompted for the email address of the recipient, your e-mail address, the subject, and up to five lines of message text. Note that your AS/400 must be configured for SMTP mail and must have the IBM Send Mail API installed.

To

Enter the e-mail address of the recipient. This must be a valid SMTP address of a local or Internet recipient.

If you entered Yes to send the converted file via e-mail, the following panel will be displayed.

```
*****************************************************************
01 * DT1190B                    Query File to TAB/EBCDIC        ROBB
```

From

Enter the From address for this message. This is normally your e-mail address at this AS/400. The first time you use this option the From address will be blank. After the first time CrossData/400 will remember your From address and use it as the default. This must be a valid SMTP address. If you are sending to an Internet recipient you should be aware of the fact that most Internet Service Providers will not deliver your message if the domain name in your From address is not registered on the Internet. This is done to reduce unsolicited e-mail (SPAM).

Reply to

Enter the Reply To address if it is different that the From address. This must also be a valid SMTP mail address.

Subject

Enter a subject for this message. It is recommended that you keep the subject to no more than 44 characters to remain compatible with most email applications.

Attachment name

Enter the file name you want to use for the converted file. The default is the AS/400 file name with the .CSV extension. You can use any name that is a valid file name on the receiving system.

Message

Enter up to five lines of message text. These lines of text are optional.

Query to CSV/ASCII

Use this option to run a query and convert the output to the CSV (Comma Separated Values) format. If the query is defined to print the output or display it interactively, the output will be overridden to a file. You can specify any existing query to select and convert data. The output file will be in the ASCII character set. In addition to converting the data this option will let you copy the converted data to a shared folder or IFS directory.

```
*****************************************************************
01 * DT1200A                    Query File to CSV/ASCII         ROBB
```

Query name and library

Enter the name of the query you want use as a source for the data, and the library that contains the query definition. This can be any existing IBM query.

File name, library, and member

Enter the name of the file to use for the query. You can enter *SAME to use the same file that was used to create the query, or enter a file name. If you enter *SAME you must leave the library and member names blank. If you specify a file name you must also specify a library and member name. The definition of the file you select must be the same as the file that was used to create the query.

Convert to file

Enter the name of the output file for the converted data. CrossData/400 will create this file for you during the conversion process, and will calculate the correct length of the records in the file.

Library name

Enter the name of the library that will contain the converted data. This must be the name of an AS/400 library.

Member name

Enter the name of the member that will contain the converted data. This must be an actual member name.

Copy to stream file

Enter 1 for Yes if you want to copy the converted data to a stream file. The stream file can exist in a Web directory on your AS/400 system, or in the QDLS folder system. Enter 2 for No if you do not want to copy the converted data to a stream file.

Stream file

Enter the full path to the stream file. If your Web files are stored in a directory called “webdocs” in the directory “www” your full path name might look like this:

```
/www/webdocs/data.csv
```

To copy a converted file to the folder file system your path might look like this:

```
/QDLS/docs/data.csv
```

Query to ASC/ASCII

Use this option to run a query and convert the output to the ASC (ASCII fixed length field) format. If the query is defined to print the output or display it interactively, the output will be overridden to a file. You can specify any existing query to select and convert data. The output file will be in the ASCII character set. In addition to converting the data this option will let you copy the converted data to a shared folder or IFS directory.

```
*******************************************************************
01 * DT1210A                    Query File to ASC/ASCII          ROBB
```

Query name and library

Enter the name of the query you want use as a source for the data, and the library that contains the query definition. This can be any existing IBM query.

File name, library, and member

Enter the name of the file to use for the query. You can enter *SAME to use the same file that was used to create the query, or enter a file name. If you enter *SAME you must leave the library and member names blank. If you specify a file name you must also specify a library and member name. The definition of the file you select must be the same as the file that was used to create the query.

Convert to file

Enter the name of the output file for the converted data. CrossData/400 will create this file for you during the conversion process, and will calculate the correct length of the records in the file.

Library name

Enter the name of the library that will contain the converted data. This must be the name of an AS/400 library.

Member name

Enter the name of the member that will contain the converted data. This must be an actual member name.

Copy to stream file

Enter 1 for Yes if you want to copy the converted data to a stream file. The stream file can exist in a Web directory on your AS/400 system, or in the QDLS folder system. Enter 2 for No if you do not want to copy the converted data to a stream file.

Stream file

Enter the full path to the stream file. If your Web files are stored in a directory called “webdocs” in the directory “www” your full path name might look like this:

```
/www/webdocs/data.prn
```

To copy a converted file to the folder file system your path might look like this:

```
/QDLS/docs/data.prn
```

Query to TAB/ASCII

Use this option to run a query and convert the output to the TAB (tab-delimited values) format. If the query is defined to print the output or display it interactively, the output will be overridden to a file. You can specify any existing query to select and convert data. The output file will be in

the ASCII character set. In addition to converting the data this option will let you copy the converted data to a shared folder or IFS directory.

```
*****************************************************************
01 * DT1220A                    Query File to TAB/ASCII          ROBB
```

Query name and library

Enter the name of the query you want use as a source for the data, and the library that contains the query definition. This can be any existing IBM query.

File name, library, and member

Enter the name of the file to use for the query. You can enter *SAME to use the same file that was used to create the query, or enter a file name. If you enter *SAME you must leave the library and member names blank. If you specify a file name you must also specify a library and member name. The definition of the file you select must be the same as the file that was used to create the query.

Convert to file

Enter the name of the output file for the converted data. CrossData/400 will create this file for you during the conversion process, and will calculate the correct length of the records in the file.

Library name

Enter the name of the library that will contain the converted data. This must be the name of an AS/400 library.

Member name

Enter the name of the member that will contain the converted data. This must be an actual member name.

Copy to stream file

Enter 1 for Yes if you want to copy the converted data to a stream file. The stream file can exist in a Web directory on your AS/400 system, or in the QDLS folder system. Enter 2 for No if you do not want to copy the converted data to a stream file.

Stream file

Enter the full path to the stream file. If your Web files are stored in a directory called “webdocs” in the directory “www” your full path name might look like this:

```
/www/webdocs/data.tab
```

To copy a converted file to the folder file system your path might look like this:

```
/QDLS/docs/data.tab
```

Convert CSV/EBCDIC to Database

Use this option to convert a file with data in CSV (Comma Separated Values) format to an AS/400 relational database file. The target AS/400 database file must be an externally described file and must have a data definition that corresponds with the CSV fields. The CSV input file must be in the EBCDIC character set and must have Carriage Return and Line Feed characters at the end of each record.

```
*****************************************************************
01 * DT1230A                 Convert CSV/EBCDIC to Database      ROB
```

CSV/EBCDIC file, library, member

Enter the name of the file containing the data in CSV format. This must be an existing file on the AS/400.

AS/400 database file, library, member

Enter the name of the AS/400 database file to receive the converted data. This must be an externally described file and each field in the file must correspond to the field in the CSV file.

Option

Enter 1 to add the converted data to the file. Enter 2 to replace the data in the file with the converted data.

Convert TAB/EBCDIC to Database

Use this option to convert a file with data in TAB (tab-delimited values) format to an AS/400 relational database file. The target AS/400 database file must be an externally described file and must have a data definition that corresponds with the tab-delimited fields. The TAB input file must be in the EBCDIC character set and must have Carriage Return and Line Feed characters at the end of each record.

```
*****************************************************************
01 * DT1240A                 Convert TAB/EBCDIC to Database      ROB
```

TAB/EBCDIC file, library, member

Enter the name of the file containing the data in TAB format. This must be an existing file on the AS/400.

AS/400 database file, library, member

Enter the name of the AS/400 database file to receive the converted data. This must be an externally described file and each field in the file must correspond to the field in the TAB file.

Option

Enter 1 to add the converted data to the file. Enter 2 to replace the data in the file with the converted data.

Convert CSV/ASCII to Database

Use this option to convert a file with data in CSV (Comma Separated Values) format to an AS/400 relational database file. The target AS/400 database file must be an externally described file and must have a data definition that corresponds with the CSV fields. The CSV input file must be in the ASCII character set and must have Carriage Return and Line Feed characters at the end of each record.



CSV/ASCII file, library, member

Enter the name of the file containing the data in CSV format. This must be an existing file on the AS/400.

AS/400 database file, library, member

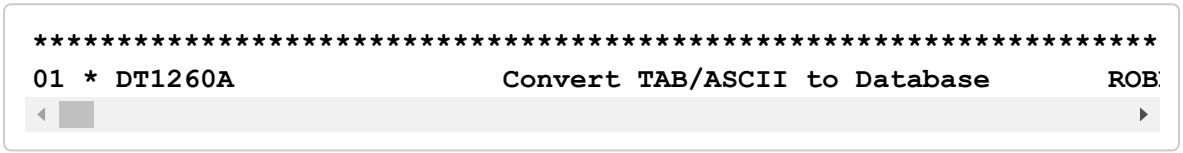
Enter the name of the AS/400 database file to receive the converted data. This must be an externally described file and each field in the file must correspond to the field in the CSV file.

Option

Enter 1 to add the converted data to the file. Enter 2 to replace the data in the file with the converted data.

Convert TAB/ASCII to Database

Use this option to convert a file with data in TAB (tab-delimited values) format to an AS/400 relational database file. The target AS/400 database file must be an externally described file and must have a data definition that corresponds with the tab-delimited fields. The TAB input file must be in the ASCII character set and must have Carriage Return and Line Feed characters at the end of each record.



TAB/ASCII file, library, member

Enter the name of the file containing the data in TAB format. This must be an existing file on the AS/400.

AS/400 database file, library, member

Enter the name of the AS/400 database file to receive the converted data. This must be an externally described file and each field in the file must correspond to the field in the TAB file.

Option

Enter 1 to add the converted data to the file. Enter 2 to replace the data in the file with the converted data.

Work with Queries

********************************* 01 * Work with Queries * 01 02 * * 02 03 * Type choices, press Enter. * 03 04 * * 04 05 * Option . . . . . 1=Create, 2=Change, 3=Copy, 4=Delete * 05 06 * 5=Display, 6=Print definition * 06 07 * 8=Run in batch, 9=Run * 07 08 * Query . . . . . Name, F4 for list * 08 09 * Library . . . . . ALLDTA200 Name, *LIBL, F4 for list * 09 10 * * 10 11 * * 11 12 * * 12 13 * * 13 14 * * 14 15 * * 15 16 * * 16 17 * * 17 18 * * 18 19 * * 19 20 * * 20 21 * * 21 22 * * 22 23 * F3=Exit F4=Prompt F5=Refresh F12=Cancel * 23 24 * (C) COPYRIGHT IBM CORP. 1988 * 24 *********************************

Type the number of the task you want to do.

Create

Create (define) a new query. When this option is processed, the Define the Query display is shown to begin query definition.

Change

Change an existing query definition.

Copy

Copy an existing query definition.

Delete

Delete an existing query definition.

Display

Show the definition of a query without being able to change it.

Print definition

Print a query definition

Column list merge

If you came to Query/400 from OfficeVision/400, you can use this option to copy information defined in an existing query definition into the document you are working with. The needed query definition information is copied onto a split display and can then be copied into the document as Office data field instructions.

Inquiry

Use this menu to access the inquiry functions of Alliance CrossData/400. You can display the application log, conversion history log, and work with library scan jobs.

```
************************************************************************
01 * DTMNU5                  Alliance CrossData/400 Inquiry Menu
```

Select an option to begin work.

Work with application logs

You can use this option to view a list of conversion actions and the results as recorded by Alliance. From this panel you can also view the details of an entry and print the log.

The application log is intended to provide the system administrator with diagnostic information about the conversion module, and should only be created when there is a need for such information. Application log entries will only be created if you enable logging with the Configure CrossData/400 option from the main menu. Entries are removed from the log when you use the Maintenance option on the main menu and specify a date for removing old records. You can also clear the application log with the CLRPFM command:

```
clrpfm afloga
```

```
************************************************************************
01 * DT0360-A                  Work With Application Logs        ROB
```

Enter the option corresponding to the function you want to perform. A list of valid options follows:

5=Display

Use this option to display detailed information about a record.

6=Print

Use this option to print a record. The printed report will go to the output queue that is the default for your job.

View application log detail

Use this panel to view the detail information for an application log.

```
*****************************************************************
01 * DT0450-A                  View Application Log Entries      ROB
```

Date

For each line, this is the date the line was created in YYYYMMDD format.

Time

For each line this is the time the line was created in HHMMSS format.

Phase

The phase describes the high level activity of the security module at the time the log was written.

Step

The step describes the activity within a phase.

Routine

The routine records the program routine in which the log entry was created.

Description

The description provides diagnostic information about the activity and/or errors encountered by the Alliance program. The reasons for rejecting a request will appear in the descriptive text.

View data conversion history

This module provides a summary view of the status of all conversion requests made with CrossData/400. It shows the date, time, status, input file, output file, user and job information, and other details of the conversion action. The system administrator can use this panel to view history of conversion activity and to assist in problem determination.

The history file is AFHIST and is an externally described file. The file can be queried using any file utility or query product such as IBM Query, etc. Records are removed from this file only when the Maintenance option is run from the main menu. Records are removed based on the date entered from that panel. You can also use the standard audit trail and exception reports to print information in the history file.

```
*****************************************************************
01 * DT6060-A                View Data Conversion History       ROBB
```

Position to

To position the list to a specific record enter the key value in this field and press Enter or F16.

Enter the option corresponding to the function you want to perform. A list of valid options follows:

5=Display

Use this option to display detailed information about the history record.

View history detail

Use this panel to view the details for a history log entry.



Date / time

The date and time field records the date and time of the log record. Date is in YYYYMMDD format and time is in HHMMSS format.

Job

The job number, user name, and job name are recorded for the history record.

Ending status

This field will contain a “1” if the conversion request was successful. It will contain a “2” if the conversion failed.

Reason code and text

The reason code will contain “0000” if the conversion was successful. It will contain a different value of the conversion failed. The text will contain a description of the reason code.

Input file, library and member

The name of the input file will be displayed in these fields.

Output file, library and member

The name of the output file will be displayed in these fields.

Scan library

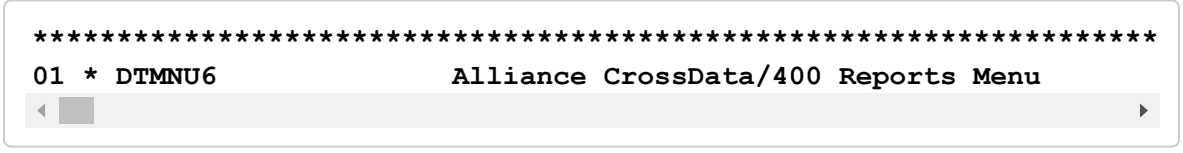
If the library scan function was used to convert the file this field will contain the name of the library.

Copy to stream file**

If the option was taken to copy the file to a stream file on the AS/400, this value will be “1” and the stream file will be displayed.

Application Reports

Use this menu to display the report options available in CrossData/400.



Select an option to begin work.

Submit CrossData/400 Audit Report

The CrossData/400 audit report prints a detailed listing of conversion activity for the period you specify. In addition to selecting the date and time windows for the report, you can select by job name and user name.

```
*****************************************************************
01 * DT8010A                 Submit CrossData/400 Audit Report ROBBN
```

From date

Enter the starting date for records to be included in the audit report. The default starting date is one month prior to today's date. Enter the date in YYYYMMDD format.

From time

Enter the from time for the report. The report will start at the time you enter. Enter the time in 24-hour HHMMSS format.

To date

Enter the ending date for the records to be included in the report. The default is today's date. Enter the date in YYYYMMDD format.

To time

Enter the ending time on the ending date for this report. Enter the time in 24-hour HHMMSS format.

Ending status

Enter 1 to include only records of CrossData/400 history where the request was successful. Enter 2 to include only records of CrossData/400 history where the request failed. Enter 3 to include all records.

Job name

Enter a job name if you want to include only records for this job. Leave this field blank to include all jobs in the report.

User name

Enter a user name if you want to include only records for this user. Leave this field blank to include all users in the report.

Support

Use this menu to access all Alliance Support functions. You will normally only use these functions when working with your software vendor to resolve questions or problems on the use of Alliance CrossData/400.

```
********************************************************************
01 * DTMNU5                 Alliance CrossData/400 Support Menu
```

Select an option to begin work.

Start Remote Support

This option is provided to allow your software vendor to perform problem analysis or software updates. When you use this option a line description, controller description, and device description will be created for a dial-in connection to your AS/400. A user profile named PATINC will also be created. The connection will use your normal ECS modem. Your software vendor will request the phone number of your ECS modem to make the connection.

When software support is complete you should run the End Remote Support to disable the user profile PATINC and vary the line, controller, and device off.

```
********************************************************************
01 * DT4080A                       Start Remote Support
```

Resource name

Enter the resource name of your ECS line. This is usually LIN011 or CMN01. You can determine the resource name by displaying the line description QESLINE:

```
dsplind qesline [Enter]
```

Line speed

Enter the speed of your ECS modem. This is usually 9600 baud. For an IBM 5853 modem, the speed is 2400. For an IBM 7855 the speed is 9600. For an IBM 7856 modem the speed is 14400.

End Remote Support

Use this option when your software vendor has finished with remote support. This option will disable the support user profile named PATINC and vary off the line, controller, and device description.

```
********************************************************************
01 * DT4081A                       End Remote Support
```

There are no input fields on this panel. Press Enter to end remote support.

Application Program Interfaces – High level

Overview

The CrossData/400 high level application program interfaces provide an easy way to convert AS/400 data to a standard ASCII data format. Several APIs are provided that let you easily create CSV or ASCII fixed length files. You can convert an AS/400 database file by simply calling one of these APIs if you need data in one of the standard formats. The APIs are programs that you can call from any RPG, Cobol, CL, or other high-level language program.

API call format

The format for the call to the API is always in the form:

```
call pgm(api-program) parm(&parm1 &parm2 &return &text)
```

The API program will accept one or more input parameters depending on the function it performs. It will return a 4 byte return code and text field that will indicate the success or failure of the API. If the return code is “0000” the request was successful. Any other value indicates an error. If there is a failure you can use the information returned in the text field to determine the cause.

High Level API Index

The following tables provide a list of all of the available APIs and a brief description of their function.

API NAME	DESCRIPTION
CVCRTFMT	Create a data conversion format string from an AS/400 externally described database file. The format string can be used to convert a CSV, Tab-delimited, or other type of file to an AS/400 database.
CVCSADBF	Convert CSV file to AS/400 database file. Input file is in ASCII.
CVCSVDBF	Convert CSV file to AS/400 database file. Input file is in EBCDIC.

API NAME	DESCRIPTION
CVEXTDBF	Convert delimited file to AS/400 database file. Input file is in EBCDIC.
CVFILASA	Convert AS/400 database file to ASC fixed length field format. The output is in ASCII.
CVFILASC	Convert AS/400 database file to ASC fixed length field format. The output is in EBCDIC.
CVFILCSV	Convert AS/400 database file to CSV. The output is in EBCDIC.
CVFILCSA	Convert AS/400 database file to CSV. The output is in ASCII.
CVFILTAA	Convert AS/400 database file to Tab-delimited format. The output is in ASCII.
CVFILTAB	Convert AS/400 database file to Tab-delimited format. The output is in EBCDIC
CVMAILMSG	Send an Internet/SMTP email message with an attached file.
CVQRYASC	Run query and convert output to fixed length field format. Output is in EBCDIC.
CVQRYASA	Run query and convert output to fixed length field format. Output is in ASCII.
CVQRYCSA	Run query and convert output to CSV format. Output is in ASCII.
CVQRYCSV	Run query and convert output to CSV format. Output is in EBCDIC.
CVQRYEML	Run query, convert output to CSV format and send it as an attachment to an email message.
CVQRYTAA	Run query and convert output to Tab-delimited format. Output is in ASCII.
CVQRYTAB	Run query and convert output to Tab-delimited format. Output is in EBCDIC.
CVSPLASA	Convert spool file to fixed length field format. Output is in ASCII

API NAME	DESCRIPTION
CVSPLASC	Convert spool file to fixed length field format. Output is in EBCDIC.
CVTAADBF	Convert tab-delimited file to AS/400 database file. Input file is in ASCII.
CVTABDBF	Convert tab-delimited file to AS/400 database file. Input file is in EBCDIC.

CVCRTFMT – Create Conversion Format from Database File

Description

The CVCRTFMT API accepts a database file name and generates a conversion format string. You should normally not need to use this API unless you are using the low level APIs for data conversion. This API reads the external definition of a file to generate the format string. You can get more information on the format string in the section on Low Level APIs.

Calling format

You can call the API like this:

```
Call pgm(cvcrtfmt) parm(&file &lib &format &return)
```

Parameters

&file (Character, 10)

The name of the file to use as a basis for the format string. This must be an existing externally described physical file.

&library (Character, 10)

The name of the library containing the file for the format string.

&format (Character, 32767)

The format string created by the API. This field is output only. Any data in the field passed to the API will be cleared.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

Sample programs

The program SAMPCRTFMT in source file QRPGLESRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVCSADBF – Convert ASCII CSV File to AS/400 Database

Description

The CVCSADBF API converts a CSV file to an AS/400 database file. The input CSV file must be an ASCII file. The target AS/400 database file must be an externally described physical file. The API uses the definition of the database file for mapping the data from the CSV file to the database file.

Calling format

You can call the API like this:

```
Call pgm(cvcsadbf) parm(&fromfile &fromlib &frommbr &tofile &tolib
```

Parameters

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing file on the AS/400 and contain data in CSV format.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member.

&tofile (Character, 10)

The name of the file to use for the converted data. This file must exist before the API is called. Data will be appended to the file.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

Sample programs

The program SAMPCSADBF in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVCSVDBF – Convert EBCDIC CSV File to AS/400 Database

Description

The CVCSVDBF API converts a CSV file to an AS/400 database file. The input CSV file must be an EBCDIC file. The target AS/400 database file must be an externally described physical file. The API uses the definition of the database file for mapping the data from the CSV file to the database file.

Calling format

You can call the API like this:

```
Call pgm(cvcsvdbf) parm(&fromfile &fromlib &frommbr &tofile &tolib
```

Parameters

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing file on the AS/400 and contain data in CSV format.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member.

&tofile (Character, 10)

The name of the file to use for the converted data. This file must exist before the API is called. Data will be appended to the file.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

Sample programs

The program SAMPCSVDBF in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVEXTDBF – Convert EBCDIC delimited File to AS/400 Database

Description

The CVEXTDBF API converts a delimited file to an AS/400 database file. The input file must be an EBCDIC file. The target AS/400 database file must be an externally described physical file. The API uses the definition of the database file for mapping the data from the delimited file to the database file.

Calling format

You can call the API like this:

```
Call pgm(cvextdbf) parm(&fromfile &fromlib &frommbr &tofile &tolib
```

Parameters

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing file on the AS/400 and contain data in delimited format.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member.

&tofile (Character, 10)

The name of the file to use for the converted data. This file must exist before the API is called. Data will be appended to the file.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

Sample programs

The program SAMPEXTDBF in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVFILCSV – Create CSV file in EBCDIC

Description

The CVFILCSV API converts an input file to a CSV (Comma Separated Values) output file. The results are in EBCDIC. This is ideal for creating files that you are going to distribute via e-mail. You may also want to use this API during the development phase to easily determine how CrossData/400 will convert the file. After running the API you can display the file using the Display Physical File Member (DSPPFM) command to view the results. Since the output is in EBCDIC it is easy to determine what the result will look like.

Calling format

You can call the API like this:

```
Call pgm(cvfilcsv) parm(&fromfile &fromlib &frommbr &tofile &tolib
```

Parameters

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPCSVCL in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVFILCSA – Create CSV file in ASCII

Description

The CVFILCSA API converts an input file to a CSV (Comma Separated Values) output file. The results are in the ASCII character set. The translation table used for the EBCDIC to ASCII conversion is the IBM default QASCII table. This API is ideal for creating files that you are going to distribute via your web site or FTP to a non-IBM system. After creating the file you can place it in a directory on your AS/400 web site using the Copy to Stream File (CPYTOSTMF) command. You can specify no translation to copy the file without change to the directory. If you want to share this file with PCs that use the Client Access application you can copy the ASCII file to a shared folder using the Copy To PC Document (CPYTOPCD) command. The file can be used by any PC user with read access to the folder.

Calling format

You can call the API like this:

```
Call pgm(cvfilcsa) parm(&fromfile &fromlib &frommbr &tofile &tolib
```

Parameters

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data, the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPCSACL in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library. This library will be replaced when upgrades are made to the application.

CVFILASC – Create fixed length file in EBCDIC

Description

The CSFILASC API converts an input file to an ASC (fixed length field) output file. The results are in EBCDIC. This is ideal for creating files that you are going to distribute to external database systems such as Oracle, Sybase, Microsoft SQL Server, or Btrieve. You may also want to use this API during the development phase to easily determine how CrossData/400 will convert the file. After running the API you can display the file using the Display Physical File Member (DSPPFM) command to view the results. Since the output is in EBCDIC it is easy to determine what the result will look like.

Calling format

You can call the API like this:

```
Call pgm(cvfilasc) parm(&fromfile &fromlib &frommbr &tofile &tolib
```

Parameters

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data, the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPASCCL in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library. This library will be replaced when upgrades are made to the application.

CVFILASA – Create fixed length file in ASCII

Description

The CVFILASA API converts an input file to an ASC (fixed length field) output file. The results are in the ASCII character set. The translation table used for the EBCDIC to ASCII conversion is the IBM default QASCII table. This API is ideal for creating files that you are going to distribute via your web site or FTP to a non-IBM system where the file needs to be fixed length. After creating the file you can place it in a directory on your AS/400 web site using the Copy to Stream File (CPYTOSTMF) command. You can specify no translation to copy the file without change to the directory. If you want to share this file with PCs that use the Client Access application you can copy the ASCII file to a shared folder using the Copy To PC Document (CPYTOPCD) command. The file can be used by any PC user with read access to the folder.

Calling format

You can call the API like this:

```
Call pgm(cvfilasa) parm(&fromfile &fromlib &frommbr &tofile &tolib
```

Parameters

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPASACL in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library. This library will be replaced when upgrades are made to the application.

CVFILEXT – Convert file with extended options

Description

The CVFILEXT API converts an input file to an AS/400 output file in the format you specify. This API provides access to almost all of the conversion parameters available in CrossData/400. You can specify field delimiters, numeric decimal values, quoted field characteristics, data translation, end-of-record terminators and much more. This API is provided for developers who want an easy-to-use but highly functional API without using the Programmer's Toolkit.

Calling format

You can call the API like this:

```
call pgm(cvfilext) parm(&fromfil &fromlib &frommbr &tofil &tolib &t
&table &tablelib &zerodisp &title &title1 &title2 &title3 &truncate
```

Parameters

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&delimiter (Character, 1)

The character to use to separate fields in the output file. For CSV (comma separated values) files you would use the comma character (“,”). If you specify “T” for this parameter CrossData/400 will use the tab character (hex value 05). If you leave this field blank delimiters will not be used.

&decimal (Character, 1)

The character to use for the numeric decimal value delimiter. The English language group uses the period character (“.”). Other language groups may use the comma character (“,”). If you leave this field blank the decimal values will not be delimited in numeric fields. This will cause numbers like 123.05 to appear as the larger number 12305. Use caution if you leave this field blank.

&newline (Character, 3)

The end of line terminator characters to use in the output file. This can be a maximum of two characters followed by a null character (hex 00). To specify a that lines should be terminated with Carriage Return and Line Feed you would specify hex value 0D2500. In a CL program you can define it like this:

```
CHGVAR VAR(&NEWLINE) VALUE(X'0D2500')
```

In an ILE RPG program you can define it like this:

```
eval    newline = x'0D2500'
```

If you do not want to use line terminators you should pass all null values (hex 000000).

&numsign (Character, 1)

This one character field is the numeric negative sign character to use. You would normally want to specify the dash character (“-“) for this field. If you leave this field blank numeric fields with a negative value will appear to be positive. Use caution when leaving this field blank.

&quote (Character, 1)

This one character field contains the character to use to surround character fields in delimited field files. This is normally the double quote character (“””) for CSV files. For tab-delimited or fixed field length files you will want to leave this field blank. If the field is blank character fields will not be enclosed in any surrounding characters.

&table (Character, 10)

The name of the EBCDIC to ASCII translation table to use for data conversion. If you do not want to do data conversion you should leave this field blank.

&tablelib (Character, 10)

The name of the library containing the translation table you specified above. If you do not want to use EBCDIC to ASCII conversion you should leave this field blank.

&zerodisp (Character, 1)

Specify “0” for this field if you want zero value numeric fields to be blank filled. Specify “1” for this field if you want zero value numeric fields to be zero-filled.

&title (Character, 1)

Enter “1” in this field if you do not want to use column titles in your output file. Enter a “2” value in this field if you want to use the field description (TEXT keyword in the DDS for a file) as column title in your output file. Enter “3” in this field if you want to use the file’s column headers (COLHDG keyword in the DDS for a file) for the column titles in your output file. Enter “4” in this field if you want to specify the column headers to use. If you use option “4” you must specify at least one of the following fields.

&title1 (Character, 2048)

If you specified option 4 for column titles in the previous field you must specify the column titles to use in this field. If the output file requires column titles amounting to more than the 2,048 bytes allowed on this parameter you cannot use this API to specify them. You must use the Programmer’s Toolkit APIs. See the description of the CVTTITLES API in the programmer’s Toolkit for a description of how to specify column titles in this field.

&title2 (Character, 2048)

If you specified option 4 for column titles in the &titles field you can specify the second row of titles in this field.

&title3 (Character, 2048)

If you specified option 4 for column titles in the &titles field you can specify the third row of titles in this field.

&truncate (Character, 1)

Specify “0” in this field if you do not want to use character field blank truncation. Specify a “1” for this field if you want to use character field blank truncation. Blank truncation may significantly reduce the size of the output file after conversion if you have many character fields that are blank or are mostly blanks.

&truncnum (Character, 1)

Specify “0” in this field if you do not want to use numeric field leading blank or zero truncation. Specify a “1” for this field if you want to use numeric field leading blank truncation. Leading blank or zero truncation may significantly reduce the size of the output file after conversion if you have many numeric fields that have a zero value. Use the value “2” to suppress leading blanks or zeros. Use value “3” to suppress leading zeros or blanks, but to supply a “0” character for zero value fields. Use value “4” to float the negative sign to the first position before a number when the value of the field is numeric.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPEXT in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVFILEX2 – Convert file with extended options including characters escape

Description

The CVFILEX2 API converts an input file to an AS/400 output file in the format you specify. This API provides access to almost all of the conversion parameters available in CrossData/400. You can specify field delimiters, numeric decimal values, quoted field characteristics, data translation, end-of-record terminators and much more. You can also specify that you want to escape quote characters in the AS/400 data. This provides a way to use AS/400 character fields that already have quote marks in them. This API is provided for developers who want an easy-to-use but highly functional API without using the Programmer's Toolkit.

Calling format

You can call the API like this:

```
call pgm(cvfilex2) parm(&fromfil &fromlib &frommbr &tofil &tolib &t
&table &tablelib &zerodisp &title &title1 &title2 &title3 &truncate
```

Parameters

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&delimiter (Character, 1)

The character to use to separate fields in the output file. For CSV (comma separated values) files you would use the comma character (“,”). If you specify “T” for this parameter CrossData/400 will use the tab character (hex value 05). If you leave this field blank delimiters will not be used.

&decimal (Character, 1)

The character to use for the numeric decimal value delimiter. The English language group uses the period character (“.”). Other language groups may use the comma character (“,”). If you leave this field blank the decimal values will not be delimited in numeric fields. This will cause numbers like 123.05 to appear as the larger number 12305. Use caution if you leave this field blank.

&newline (Character, 3)

The end of line terminator characters to use in the output file. This can be a maximum of two characters followed by a null character (hex 00). To specify a that lines should be terminated with Carriage Return and Line Feed you would specify hex value 0D2500. In a CL program you can define it like this:

```
CHGVAR VAR(&NEWLINE) VALUE(X'0D2500')
```

In an ILE RPG program you can define it like this:

```
eval    newline = x'0D2500'
```

If you do not want to use line terminators you should pass all null values (hex 000000).

&numsign (Character, 1)

This one character field is the numeric negative sign character to use. You would normally want to specify the dash character (“-“) for this field. If you leave this field blank numeric fields with a negative value will appear to be positive. Use caution when leaving this field blank.

&quote (Character, 1)

This one character field contains the character to use to surround character fields in delimited field files. This is normally the double quote character (“”) for CSV files. For tab-delimited or fixed field length files you will want to leave this field blank. If the field is blank character fields will not be enclosed in any surrounding characters.

&table (Character, 10)

The name of the EBCDIC to ASCII translation table to use for data conversion. If you do not want to do data conversion you should leave this field blank.

&tablelib (Character, 10)

The name of the library containing the translation table you specified above. If you do not want to use EBCDIC to ASCII conversion you should leave this field blank.

&zerodisp (Character, 1)

Specify “0” for this field if you want zero value numeric fields to be blank filled. Specify “1” for this field if you want zero value numeric fields to be zero-filled.

&title (Character, 1)

Enter “1” in this field if you do not want to use column titles in your output file. Enter a “2” value in this field if you want to use the field description (TEXT keyword in the DDS for a file) as column title in your output file. Enter “3” in this field if you want to use the file’s column headers (COLHDG keyword in the DDS for a file) for the column titles in your output file. Enter “4” in this field if you want to specify the column headers to use. If you use option “4” you must specify at least one of the following fields.

&title1 (Character, 2048)

If you specified option 4 for column titles in the previous field you must specify the column titles to use in this field. If the output file requires column titles amounting to more than the 2,048 bytes allowed on this parameter you cannot use this API to specify them. You must use the Programmer’s Toolkit APIs. See the description of the CVTTITLES API in the programmer’s Toolkit for a description of how to specify column titles in this field.

&title2 (Character, 2048)

If you specified option 4 for column titles in the &titles field you can specify the second row of titles in this field.

&title3 (Character, 2048)

If you specified option 4 for column titles in the &titles field you can specify the third row of titles in this field.

&truncate (Character, 1)

Specify “0” in this field if you do not want to use character field blank truncation. Specify a “1” for this field if you want to use character field blank truncation. Blank truncation may significantly reduce the size of the output file after conversion if you have many character fields that are blank or are mostly blanks.

&truncnum (Character, 1)

Specify “0” in this field if you do not want to use numeric field leading blank or zero truncation. Specify a “1” for this field if you want to use numeric field leading blank truncation. Leading blank or zero truncation may significantly reduce the size of the output file after conversion if

you have many numeric fields that have a zero value. Use the value “2” to suppress leading blanks or zeros. Use value “3” to suppress leading zeros or blanks, but to supply a “0” character for zero value fields. Use value “4” to float the negative sign to the first position before a number when the value of the field is numeric.

&escape (Character, 1)

Specify “0” in this field it you do not want to escape character fields. Specify “1” if you do want to escape character fields. If you specify “1” to escape character fields any occurrence of the quote mark (“) will be converted to two quote marks (“”). You may also wish to specify additional length to add to the output record length.

&escapenum (Character, 4)

If you specified “1” in the previous field to escape quote characters, you can specify the number of characters to increase the output record length in this field. This can be helpful if you are not truncating blanks from character fields and want to avoid data wrapping in the output file. Specify the value as 4 character digits from “0000” to “9999”. The value of “0000” indicates that the record will not be expanded.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPEX2 in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVFILEX3 – Convert file with extended options including characters escape and packed numeric expansion

Description

The CVFILEX3 API converts an input file to an AS/400 output file in the format you specify. This API provides access to almost all of the conversion parameters available in CrossData/400. You can specify field delimiters, numeric decimal values, quoted field characteristics, data translation, end-of-record terminators, expanded packed numeric fields, and much more. You can also specify that you want to escape quote characters in the AS/400 data. This provides a way to use AS/400 character fields that already have quote marks in them. This API is provided for developers who want an easy-to-use but highly functional API without using the Programmer’s Toolkit.

Calling format

You can call the API like this:

```
call pgm(cvfilex3) parm(&fromfil &fromlib &frommbr &tofil &tolib &t
&table &tablelib &zerodisp &title &title1 &title2 &title3 &truncate
```

Parameters

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&delimiter (Character, 1)

The character to use to separate fields in the output file. For CSV (comma separated values) files you would use the comma character (“,”). If you specify “T” for this parameter CrossData/400 will use the tab character (hex value 05). If you leave this field blank delimiters will not be used.

&decimal (Character, 1)

The character to use for the numeric decimal value delimiter. The English language group uses the period character (“.”). Other language groups may use the comma character (“,”). If you leave this field blank the decimal values will not be delimited in numeric fields. This will cause numbers like 123.05 to appear as the larger number 12305. Use caution if you leave this field blank.

&newline (Character, 3)

The end of line terminator characters to use in the output file. This can be a maximum of two characters followed by a null character (hex 00). To specify a that lines should be terminated with Carriage Return and Line Feed you would specify hex value 0D2500. In a CL program you can define it like this:

```
CHGVAR VAR(&NEWLINE) VALUE(X'0D2500')
```

In an ILE RPG program you can define it like this:

```
eval    newline = x'0D2500'
```

If you do not want to use line terminators you should pass all null values (hex 000000).

&numsign (Character, 1)

This one character field is the numeric negative sign character to use. You would normally want to specify the dash character (“-“) for this field. If you leave this field blank numeric fields with a negative value will appear to be positive. Use caution when leaving this field blank.

&quote (Character, 1)

This one character field contains the character to use to surround character fields in delimited field files. This is normally the double quote character (“””) for CSV files. For tab-delimited or fixed field length files you will want to leave this field blank. If the field is blank character fields will not be enclosed in any surrounding characters.

&table (Character, 10)

The name of the EBCDIC to ASCII translation table to use for data conversion. If you do not want to do data conversion you should leave this field blank.

&tablelib (Character, 10)

The name of the library containing the translation table you specified above. If you do not want to use EBCDIC to ASCII conversion you should leave this field blank.

&zerodisp (Character, 1)

Specify “0” for this field if you want zero value numeric fields to be blank filled. Specify “1” for this field if you want zero value numeric fields to be zero-filled.

&title (Character, 1)

Enter “1” in this field if you do not want to use column titles in your output file. Enter a “2” value in this field if you want to use the field description (TEXT keyword in the DDS for a file) as column title in your output file. Enter “3” in this field if you want to use the file’s column headers

(COLHDG keyword in the DDS for a file) for the column titles in your output file. Enter “4” in this field if you want to specify the column headers to use. If you use option “4” you must specify at least one of the following fields.

&title1 (Character, 2048)

If you specified option 4 for column titles in the previous field you must specify the column titles to use in this field. If the output file requires column titles amounting to more than the 2,048 bytes allowed on this parameter you cannot use this API to specify them. You must use the Programmer’s Toolkit APIs. See the description of the CVTTITLES API in the programmer’s Toolkit for a description of how to specify column titles in this field.

&title2 (Character, 2048)

If you specified option 4 for column titles in the &titles field you can specify the second row of titles in this field.

&title3 (Character, 2048)

If you specified option 4 for column titles in the &titles field you can specify the third row of titles in this field.

&truncate (Character, 1)

Specify “0” in this field if you do not want to use character field blank truncation. Specify a “1” for this field if you want to use character field blank truncation. Blank truncation may significantly reduce the size of the output file after conversion if you have many character fields that are blank or are mostly blanks.

&truncnum (Character, 1)

Specify “0” in this field if you do not want to use numeric field leading blank or zero truncation. Specify a “1” for this field if you want to use numeric field leading blank truncation. Leading blank or zero truncation may significantly reduce the size of the output file after conversion if you have many numeric fields that have a zero value. Use the value “2” to suppress leading blanks or zeros. Use value “3” to suppress leading zeros or blanks, but to supply a “0” character for zero value fields. Use value “4” to float the negative sign to the first position before a number when the value of the field is numeric.

&escape (Character, 1)

Specify “0” in this field it you do not want to escape character fields. Specify “1” if you do want to escape character fields. If you specify “1” to escape character fields any occurrence of the quote mark (“) will be converted to two quote marks (“”). You may also wish to specify additional length to add to the output record length.

&escapenum (Character, 4)

If you specified “1” in the previous field to escape quote characters, you can specify the number of characters to increase the output record length in this field. This can be helpful if you are not truncating blanks from character fields and want to avoid data wrapping in the output file. Specify the value as 4 character digits from “0000” to “9999”. The value of “0000” indicates that the record will not be expanded.

&exppack (Character, 1)

Specify “0” in this field it you do not want to expand packed numeric fields. Specify “1” if you do want to expand packed numeric fields by one byte.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPEX3 in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVIFSEXT – Convert to IFS file with extended options

Description

The CVIFSEXT API converts an input file to an AS/400 IFS output file in the format you specify. This API provides access to almost all of the conversion parameters available in CrossData/400. You can specify field delimiters, numeric decimal values, quoted field characteristics, data translation, end-of-record terminators and much more. This API is provided for developers who want an easy-to-use but highly functional API without using the Programmer’s Toolkit.

Calling format

You can call the API like this:

```
call pgm(cvifsext) parm(&fromfil &fromlib &frommbr &toifs &delimite
&table &tablelib &zerodisp &title &title1 &title2 &title3 &truncate
```

Parameters

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member.

&toifs (Character, 128)

The fully qualified name of the IFS file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created. If the file does exist it will be replaced. The IFS file will be in normal stream file format that is compatible with most PC and UNIX systems.

&delimiter (Character, 1)

The character to use to separate fields in the output file. For CSV (comma separated values) files you would use the comma character (“,”). If you specify “T” for this parameter CrossData/400 will use the tab character (hex value 05). If you leave this field blank delimiters will not be used.

&decimal (Character, 1)

The character to use for the numeric decimal value delimiter. The English language group uses the period character (“.”). Other language groups may use the comma character (“,”). If you leave this field blank the decimal values will not be delimited in numeric fields. This will cause numbers like 123.05 to appear as the larger number 12305. Use caution if you leave this field blank.

&newline (Character, 3)

The end of line terminator characters to use in the output file. This can be a maximum of two characters followed by a null character (hex 00). To specify a that lines should be terminated with Carriage Return and Line Feed you would specify hex value 0D2500. In a CL program you can define it like this:

```
CHGVAR VAR(&NEWLINE) VALUE(X'0D2500')
```

In an ILE RPG program you can define it like this:

```
eval    newline = x'0D2500'
```

If you do not want to use line terminators you should pass all null values (hex 000000).

&numsign (Character, 1)

This one character field is the numeric negative sign character to use. You would normally want to specify the dash character (“-“) for this field. If you leave this field blank numeric fields with a negative value will appear to be positive. Use caution when leaving this field blank.

&quote (Character, 1)

This one character field contains the character to use to surround character fields in delimited field files. This is normally the double quote character (“””) for CSV files. For tab-delimited or fixed field length files you will want to leave this field blank. If the field is blank character fields will not be enclosed in any surrounding characters.

&table (Character, 10)

The name of the EBCDIC to ASCII translation table to use for data conversion. You can use the table QASCII in library QSYS provide by IBM. Or you can use the CrossData/400 translation table DTEB2AS in the product library. The source for this table is in source file QTBLSRC in the product library. This translation table provides better translation of certain EBCDIC characters. If you want to create your own table you can use the template in the QTBLSRC file in the product library and use the Create Table (CRTTBL) command to create it. Since this API copies the converted data to the IFS file system you will normally want to specify a translation table.

&tablelib (Character, 10)

The name of the library containing the translation table you specified above. If you do not want to use EBCDIC to ASCII conversion you should leave this field blank.

&zerodisp (Character, 1)

Specify “0” for this field if you want zero value numeric fields to be blank filled. Specify “1” for this field if you want zero value numeric fields to be zero-filled.

&title (Character, 1)

Enter “1” in this field if you do not want to use column titles in your output file. Enter a “2” value in this field if you want to use the field description (TEXT keyword in the DDS for a file) as column title in your output file. Enter “3” in this field if you want to use the file’s column headers (COLHDG keyword in the DDS for a file) for the column titles in your output file. Enter “4” in this field if you want to specify the column headers to use. If you use option “4” you must specify at least one of the following fields.

&title1 (Character, 2048)

If you specified option 4 for column titles in the previous field you must specify the column titles to use in this field. If the output file requires column titles amounting to more than the 2,048 bytes allowed on this parameter you cannot use this API to specify them. You must use the Programmer’s Toolkit APIs. See the description of the CVTTITLES API in the programmer’s Toolkit for a description of how to specify column titles in this field.

&title2 (Character, 2048)

If you specified option 4 for column titles in the &titles field you can specify the second row of titles in this field.

&title3 (Character, 2048)

If you specified option 4 for column titles in the &titles field you can specify the third row of titles in this field.

&truncate (Character, 1)

Specify “1” in this field if you do not want to use character field blank truncation. Specify a “2” for this field if you want to use character field blank truncation. Blank truncation may significantly reduce the size of the output file after conversion if you have many character fields that are blank or are mostly blanks.

&truncnum (Character, 1)

Specify “0” in this field if you do not want to use numeric field leading blank or zero truncation. Specify a “1” for this field if you want to use numeric field leading blank truncation. Leading blank or zero truncation may significantly reduce the size of the output file after conversion if

you have many numeric fields that have a zero value.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPIFSEXT in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVIFSCSA – Create CSV file in ASCII in IFS directory

Description

The CVIFSCSA API converts an input file to a CSV (Comma Separated Values) output file in the IFS directory. The results are in ASCII. This is ideal for creating files that you are going to copy to a web site, distribute via e-mail, or copy to a relational database for import. After running the API you can access the file using the Work With Link (WRKLNK) command. The output is in ASCII so you can copy the file to a PC or other ASCII system to view the results.

Calling format

You can call the API like this:

```
Call pgm(cvifscsa) parm(&fromfile &fromlib &frommbr &ifsfile &retur
```

Parameters

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member.

&ifsfile (Character, 128)

The name of the IFS file to use for the converted data. This file should exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPIFSACL in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVIFSCSV – Create CSV file in EBCDIC in IFS directory

Description

The CVIFSCSV API converts an input file to a CSV (Comma Separated Values) output file in the IFS directory. The results are in EBCDIC. This is ideal for creating files that you are going to distribute via e-mail. You may also want to use this API during the development phase to easily determine how CrossData/400 will convert the file. After running the API you can access the file using the Work With Link (WRKLNK) command. The output is in EBCDIC so you may need to copy it to a physical work file on the AS/400, or copy to a PC and view in a hex editor that supports EBCDIC data.

Calling format

You can call the API like this:

```
Call pgm(cvifscsv) parm(&fromfile &fromlib &frommbr &ifsfile &retur
```

Parameters

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member.

&ifsfile (Character, 128)

The name of the IFS file to use for the converted data. This file should exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPIFSCCL in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVSPLASC – Create fixed length file in EBCDIC from Spool File

Description

The CVSPLASC API converts an input spool file to an ASC (fixed length field) output file. The results are in the EBCDIC character set. This API is ideal for creating files that you are going to distribute via your web site, e-mail, or FTP to a non-IBM system where the file needs to be fixed length. After creating the file you can place it in a directory on your AS/400 web site using the Copy to Stream File (CPYTOSTMF) command. You can specify no translation to copy the file without change to the directory. If you want to share this file with PCs that use the Client Access application you can copy the EBCDIC file to a shared folder using the Copy To PC Document (CPYTOPCD) command. The file can be used by any PC user with read access to the folder.

Calling format

You can call the API like this:

```
call pgm(cvsplasc) parm(&splfname &jobname &username &jobnumber &sp
```

Parameters

&splfname (Character, 10)

The name of the spool file you want to convert. This must be an existing spool file in an AS/400 output queue and must not be in the process of writing when the API is called.

&jobname (Character, 10)

The name of the job that created the spool file. You can use the Work With Output Queues (WRKOUTQ) or Work With Spool Files (WRKSPLF) command to determine the name of the job.

&username (Character, 10)

The name of the user who created the spool file.

&jobnumber (Character, 6)

The number of the job that created the spool file.

&splfno (Character, 4)

The spool file number of the report. This is a sequential number from 0001 to 9999.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPSPLCL in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVSPLASA – Create fixed length file in ASCII from Spool File

Description

The CVSPLASA API converts an input spool file to an ASC (fixed length field) output file. The results are in the ASCII character set. This API is ideal for creating files that you are going to distribute via your web site, or FTP to a non-IBM system where the file needs to be a fixed length. After creating the file you can place it in a directory on your AS/400 web site using the Copy to Stream File (CPYTOSTMF) command. You can specify no translation to copy the file without change to the directory. If you want to share this file with PCs that use the Client Access application you can copy the ASCII file to a shared folder using the Copy To PC Document (CPYTOPCD) command. The file can be used by any PC user with read access to the folder.

Calling format

You can call the API like this:

```
call pgm(cvsplasa) parm(&splfname &jobname &username &jobnumber &sp
```

Parameters

&splfname (Character, 10)

The name of the spool file you want to convert. This must be an existing spool file in an AS/400 output queue and must not be in the process of writing when the API is called.

&jobname (Character, 10)

The name of the job that created the spool file. You can use the Work With Output Queues (WRKOUTQ) or Work With Spool Files (WRKSPLF) command to determine the name of the job.

&username (Character, 10)

The name of the user who created the spool file.

&jobnumber (Character, 6)

The number of the job that created the spool file.

&splfno (Character, 4)

The spool file number of the report. This is a sequential number from 0001 to 9999.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPSPACL in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVFILTAB – Create Tab delimited file in EBCDIC

Description

The CVFILTAB API converts an input file to a tab-delimited output file. The results are in EBCDIC. This is ideal for creating files that you are going to distribute via E-mail. You may also want to use this API during the development phase to easily determine how CrossData/400 will convert the file. After running the API you can display the file using the Display Physical File Member (DSPPFM) command to view the results. Since the output is in EBCDIC it is easy to determine what the result will look like. The tab character in EBCDIC has a value of hex 05. When converted to ASCII the tab character has a hex 09 value.

Calling format

You can call the API like this:

```
Call pgm(cvfiltab) parm(&fromfile &fromlib &frommbr &tofile &tolib
```

Parameters

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPTABCL in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library. This library will be replaced when upgrades are made to the application.

CVFILTAA – Create Tab delimited file in ASCII

Description

The CVFILTAA API converts an input file to a tab delimited output file. The results are in the ASCII character set. The translation table used for the EBCDIC to ASCII conversion is the IBM default QASCII table. This API is ideal for creating files that you are going to distribute via your web site or FTP to a non-IBM system. After creating the file you can place it in a directory on your AS/400 web site using the Copy to Stream File (CPYTOSTMF) command. You can specify no translation to copy the file without change to the directory. If you want to share this file with PCs that use the Client Access application you can copy the ASCII file to a shared folder using the Copy To PC Document (CPYTOPCD) command. The file can be used by any PC user with read access to the folder.

Calling format

You can call the API like this:

```
call pgm(cvfiltaa) parm(&fromfile &fromlib &frommbr &tofile &tolib
```

Parameters

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPTAACL in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library. This library will be replaced when upgrades are made to the application.

CVMAILMSG – Send simple e-mail message with MIME attachment

Description

The CVMAILMSG API uses the IBM Send Mail API to send a simple message with a MIME attachment. The API accepts up to five lines of message text and the name of a file to send as an attachment. You can specify the MIME attachment headers for maximum control of the format of the output. Binary encoding is supported via the Base64 encoding scheme. This API can only be used on RISC systems that have the Send Mail API installed.

Calling format

You can call the API like this:

```
call pgm(cvmailmsg) parm(&from &to &replyto &subject &msg01 &msg02
```

Parameters

&from (Character, 128)

The from address of the sender. This must be a valid SMTP address with a user and domain name.

&to (Character, 128)

The address of the recipient of this message. This must be a valid SMTP address.

&replyto (Character, 128)

The reply to address of the sender. This can be different than the from address. It must be a valid SMTP address with a user and domain name.

&subject (Character, 128)

The subject of this message. It is recommended that you restrict the length of the subject to 44 characters to be compatible with most e-mail applications.

&msg01, &msg02, &msg03, &msg04, &msg05 (Character, 75)

Up to five lines of message text. The message is optional. If there is no message the API will assign a simple line of text for the message indicating the attachment name.

&file (Character, 10)

The name of the file you want to attach to the message. This must be an existing file and must have at least one record. The file can contain binary information you have transferred from a PC or UNIX platform. The file should be transferred with FTP or other application that will retain the binary characteristics of the data.

&library (Character, 10)

The library of the attachment file. This must be a valid library name and cannot be special values like *LIBL or *CURLIB.

&member (Character, 10)

The member name of the file. This must be a valid member and cannot be the special value of *FIRST.

&extnam (Character, 128)

The name of the attachment. This can be any valid file name for mail recipient's computer system.

&contype (Character, 128)

The MIME content type of the attachment. This can be any valid MIME content type. Some examples are:

```
application/msexcel; name="spreadsheet.csv" application/acrobat; na
```

&conencode (Character, 128)

The MIME content encoding type of the attachment. This can be any valid MIME content type. For binary encoding use the "base64" content encoding type. Binhex and other binary encoding types are not supported. Some examples are:

```
quoted-printable 7bit
base64
```

&condisp (Character, 128)

The MIME content disposition of the attachment. This field is used to indicate to the recipient's mail system how to process the attachment. Some examples are:

```
attachment; filename="myfile.pdf" inline; filename="myfile.txt"
```

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is "0000" the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPMAILMS in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library. This library will be replaced when upgrades are made to the application.

CVQRYASA – Create ASC file in ASCII from Query

Description

The CVQRYASA API converts an the output of a query to an ASC (fixed field length) output file. The results are in ASCII. This is ideal for creating files that you are going to distribute to a relational database system like Oracle or Sybase. You must first create the query that is used by this API. The query can have interactive or printed output. CrossData/400 will use the IBM Run Query (RUNQRY) command and override the output to a database file.

Calling format

You can call the API like this:

```
call pgm(cvqryasa) parm(&query &querylib &fromfile &fromlib &frommb
```

Parameters

&query (Character, 10)

The name of the query to be used. This must be an existing query on the AS/400.

&querylib (Character, 10)

The name of the library containing the query definition. This must be an existing library on the AS/400.

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400. You can use the special value “*SAME” to indicate that the file defined in the query is to be used for the input data.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400. You can leave this field blank if the file name is the special value *SAME.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member. You can leave this field blank if the file name is the special value *SAME.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPQRYASA in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVQRYASC – Create ASC file in EBCDIC from Query

Description

The CVQRYASC API converts an the output of a query to an ASC (fixed field length) output file. The results are in EBCDIC. This is ideal for creating files that you are going to import to database systems such as Oracle or Sybase. You may want to use this API during the development phase to easily determine how CrossData/400 will convert the file. After running the API you can display the file using the Display Physical File Member (DSPPFM) command to view the results. Since the output is in EBCDIC it is easy to determine what the resulting file will look like. You must first create the query that is used by this API. The query can have interactive or printed output. CrossData/400 will use the IBM Run Query (RUNQRY) command and override the output to a database file.

Calling format

You can call the API like this:

```
Call pgm(cvqryasc) parm(&query &querylib &fromfile &fromlib &frommb
```

Parameters

&query (Character, 10)

The name of the query to be used. This must be an existing query on the AS/400.

&querylib (Character, 10)

The name of the library containing the query definition. This must be an existing library on the AS/400.

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400. You can use the special value “*SAME” to indicate that the file defined in the query is to be used for the input data.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400. You can leave this field blank if the file name is the special value *SAME.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member. You can leave this field blank if the file name is the special value *SAME.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will

be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPQRYASC in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVQRYCSA – Create CSV file in ASCII from Query

Description

The CVQRYCSA API converts an the output of a query to a CSV (Comma Separated Values) output file. The results are in ASCII. This is ideal for creating files that you are going to distribute to a web site or send via FTP to an ASCII host. You must first create the query that is used by this API. The query can have interactive or printed output. CrossData/400 will use the IBM Run Query (RUNQRY) command and override the output to a database file.

Calling format

You can call the API like this:

```
call pgm(cvqrycsa) parm(&query &querylib &fromfile &fromlib &frommb
```

Parameters

&query (Character, 10)

The name of the query to be used. This must be an existing query on the AS/400.

&querylib (Character, 10)

The name of the library containing the query definition. This must be an existing library on the AS/400.

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400. You can use the special value “*SAME” to indicate that the file defined in the query is to be used for the input data.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400. You can leave this field blank if the file name is the special value *SAME.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member. You can leave this field blank if the file name is the special value *SAME.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPQRYCSA in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVQRYCSV – Create CSV file in EBCDIC from Query

Description

The CVQRYCSV API converts an the output of a query to a CSV (Comma Separated Values) output file. The results are in EBCDIC. This is ideal for creating files that you are going to distribute via e-mail. You may also want to use this API during the development phase to easily determine how CrossData/400 will convert the file. After running the API you can display the file using the Display Physical File Member (DSPPFM) command to view the results. Since the output is in EBCDIC it is easy to determine what the result will look like. You must first create the query that is used by this API. The query can have interactive or printed output. CrossData/400 will use the IBM Run Query (RUNQRY) command and override the output to a database file.

Calling format

You can call the API like this:

```
Call pgm(cvqrycsv) parm(&query &querylib &fromfile &fromlib &frommb
```

Parameters

&query (Character, 10)

The name of the query to be used. This must be an existing query on the AS/400.

&querylib (Character, 10)

The name of the library containing the query definition. This must be an existing library on the AS/400.

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400. You can use the special value “*SAME” to indicate that the file defined in the query is to be used for the input data.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400. You can leave this field blank if the file name is the special value *SAME.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member. You can leave this field blank if the file name is the special value *SAME.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPQRYCSV in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVQRYTAA – Create TAB file in ASCII from Query

Description

The CVQRYTAA API converts an the output of a query to a TAB (tab delimited values) output file. The results are in ASCII. This is ideal for creating files that you are going to distribute to a UNIX host or application that can read tab-delimited files. You must first create the query that is used by this API. The query can have interactive or printed output. CrossData/400 will use the IBM Run Query (RUNQRY) command and override the output to a database file.

Calling format

You can call the API like this:

```
call pgm(cvqrytaa) parm(&query &querylib &fromfile &fromlib &frommb
```

Parameters

&query (Character, 10)

The name of the query to be used. This must be an existing query on the AS/400.

&querylib (Character, 10)

The name of the library containing the query definition. This must be an existing library on the AS/400.

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400. You can use the special value “*SAME” to indicate that the file defined in the query is to be used for the input data.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400. You can leave this field blank if the file name is the special value *SAME.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member. You can leave this field blank if the file name is the special value *SAME.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPQRYTAA in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

CVQRYTAB – Create TAB file in EBCDIC from Query

Description

The CVQRYTAB API converts an the output of a query to a TAB (tab-delimited values) output file. The results are in EBCDIC. This is ideal for creating files that you are going to distribute to UNIX or other ASCII host systems and applications. You may also want to use this API during the development phase to easily determine how CrossData/400 will convert the file. After running the API you can display the file using the Display Physical File Member (DSPPFM) command to view the results. Since the output is in EBCDIC it is easy to determine what the result will look like. You must first create the query that is used by this API. The query can have interactive or printed output. CrossData/400 will use the IBM Run Query (RUNQRY) command and override the output to a database file.

Calling format

You can call the API like this:

```
call pgm(cvqrytab) parm(&query &querylib &fromfile &fromlib &frommb
```

Parameters

&query (Character, 10)

The name of the query to be used. This must be an existing query on the AS/400.

&querylib (Character, 10)

The name of the library containing the query definition. This must be an existing library on the AS/400.

&fromfile (Character, 10)

The name of the input file to be converted. This must be an existing externally described file on the AS/400. You can use the special value “*SAME” to indicate that the file defined in the query is to be used for the input data.

&fromlib (Character, 10)

The name of the library containing the input file. This must be an existing library on the AS/400. You can leave this field blank if the file name is the special value *SAME.

&frommbr (Character, 10)

The name of the member in the file to be processed. This must be an existing member. You can leave this field blank if the file name is the special value *SAME.

&tofile (Character, 10)

The name of the file to use for the converted data. This file does not have to exist before the API is called. If the file does not exist it will be created with the record length long enough to hold the converted data and any column headers. If the record exists the converted data will be appended to the file. Caution should be used, however, if adding converted data to an existing file. If the record length is not long enough to hold the converted data the data may be truncated and some data lost. It is best to specify an output file that does not exist and let CrossData/400 create the file for you.

&tolib (Character, 10)

The name of the library for the converted data file.

&tombr (Character, 10)

The name of the member in the output file to use for the converted data.

&return (Character, 4)

This value is returned by the API to indicate the success or failure of the operation. If the returned value is “0000” the request was successful. If the value is anything else the request failed. If the request failed you can use the following text field to determine the cause of the failure.

&text (Character, 80)

A text description of the success or failure of the conversion text.

Sample programs

The program SAMPQRYTAB in source file QCLSRC in library ALLDTA200 shows an example of using this API. You can copy this source to your own program name and modify it for your own needs. Be sure to put your programs in a library other than the Alliance CrossData/400 library, as this library will be replaced when upgrades are made to the application.

QRYTOEML – Query to E-mail Command

Description

The Query To Email (QRYTOEML) command can be used to run a query and send the output to an email recipient. Up to five lines of text can be sent with the message. The converted file will be in CSV format and attached to the message as a MIME attachment. This command uses the IBM SendMail API and is available only on RISC AS/400s. Context sensitive help is available for this command. You can type the name of the command and press F4 for prompting, then press F1 on any field for command help.

Calling format

You can use the command like this:

```
qrytoeml qry(query-name) qryfile(*same | file-name) attach(attachme
msgline3(message-line-3) msgline4(message-line-4) msgline5(message-
```

Parameters

qry (Character, 20)

The name of the query and library to use to create the output file. This must be an existing query.

qryfile (Character, 30)

The file to use for the query. The special value *SAME can be used to indicate that the file used to create the query should be used as the source of the data. You can use any file name that you like that is valid for the query.

attach (Character, 40)

The name of the attachment file. This is the name used with the email message for the attachment file name. It can be any name that is valid on the receiving user's computer system.

to (Character, 80)

The email address of the recipient. This should be a valid SMTP address in the format user@domain.

from (Character, 80)

The address of the sender. This should be a valid SMTP address in the format of user@domain. The special value of *PRV can be used to indicate that the from address used on the last message should be used. CrossData/400 remembers the from address for each user profile that executes an email interface and will use it as a default.

replyto (Character, 80)

The reply-to address of the sender. This should be a valid SMTP address in the format of user@domain. The special value of *PRV can be used to indicate that the from address used on the last message should be used. The special value *FROMADDR can be used to indicate that the reply-to address should be the same as the From address. CrossData/400 remembers the reply-to address for each user profile that executes an email interface and will use it as a default.

subject (Character, 40)

The subject for the email message.

Msgline1 to msgline5 (Character, 75)

Up to five lines of message text. The message text will be used in the body of the email message. The output file from the query will be attached to the message.

QRYTOIFS – Query to IFS FILE Command

Description

The Query To IFS File (QRYTOIFS) command can be used to run a query and copy the output in CSV format to an IFS directory or shared folder in the QDLS file system. The converted file will be in CSV format. Context sensitive help is available for this command. You can type the name of the command and press F4 for prompting, then press F1 on any field for command help.

Calling format

You can use the command like this:

```
qrytoifs qry(query-name) qryfile(*same | file-name) cvttype(convers
```

Parameters

qry (Character, 20)

The name of the query and library to use to create the output file. This must be an existing query.

qryfile (Character, 30)

The file to use for the query. The special value *SAME can be used to indicate that the file used to create the query should be used as the source of the data. You can use any file name that you like that is valid for the query.

cvttype (Character, 10)

The conversion type for the data. You can specify *CSV to create the file in comma separated values format, or *TAB to create the file in tab-delimited format, or *ASC to convert the file to fixed field length format.

ifsname (Character, 80)

The full path and file name for the output file. The directory path must exist before the command is executed. To copy to the IFS file system specify a file name like this:

```
/users/myname/myfile.csv
```

You can also copy a file to a shared folder that is accessible with Client Access shared folder or network neighborhood support. To copy a file to a shared folder specify a file name like this:

```
/QDLS/myname/myfile.csv
```

API200 – Email message

Description

This is a generalized API that can be used to send a short message or a long message stored in a database file to an SMTP/Internet recipient. For a short message you can send up to 15 lines of message text. For a long message you specify a database file with the message text. If you use the long message option you must first create the message in a database file before calling this API. This API also lets you specify an attachment for the message.

Note that the call to this API may be successful, but the message may not be delivered to the recipient. There can be many reasons for this including an invalid recipient address, spam controls implemented to Internet Service Providers, or lack of availability of the recipient's mail system. A failure to deliver the message may not be reported back to the sender depending on the reason for the non-delivery.

Call Format

```
CALL PGM(API200) PARM(&TYPE &TOADDR &FROMADDR &REPLYTO &ORGANIZ &SM
&LOCALHOST &SUBJECT &MSG01 &MSG02 &MSG03 + &MSG04 &MSG05 &MSG06 &MS
```

Parameters

The following parameters are passed to the API. They must be specified in the following order and each field passed to the API must have the type and length specified.

Type (Character, 1)

Specify the value of "1" for a short message or the value "2" for a long message stored in a database file. If you specify "1" for a short message you can provide the message in the 15 message lines in the following parameters. If you specify "2" for a long message in a database file you must specify the file, library, and member name that contains the message.

To Address (Character, 128)

The email address of the recipient. This must be a valid SMTP address in the form user@domain.

From address (Character, 128)

The address of the sender. This must be a valid SMTP address, but does not have to the actual address of the user that sends the message. The address must be in the format of user@domain. Please note that most Internet Service Providers will not deliver mail with a sender address with an unregistered domain name. You should also be aware that it is illegal in some jurisdictions to provide a false address on unsolicited email.

Reply-to address (Character, 128)

The address for the reply message. This address can be the same as the From address or can be different. You can specify a return address that is on a different email system and a different domain. See the notes about Internet Service Providers above.

Organization (Character, 128)

Enter the name of your organization. You can provide any text in this field that you like. This would normally be your company name.

SMTP host name (Character, 128)

The name of the SMTP mail server that will be used to deliver this mail. This is normally the name of your mail relay host or ISP mail server and is the domain format. For example: “mail.my-isp.com”. If you do not specify an SMTP host name you must specify an SMTP IP address.

SMTP IP address (Character, 15)

The IP address of your mail relay host or ISP mail server. If you do not specify an SMTP host name you must specify an IP address. The address is in dotted decimal format, for example “206.14.53.98”.

SMTP port (Character, 5)

The port number of the mail server. This is normally “00025” but may be different if you are using an internal mail server or firewall.

Local host (Character, 128)

The name of your local host. This name is sent to the mail server to identify the sending entity. It must be a valid domain name, for example: “as400.acme.com”.

Subject (Character, 50)

The subject for this message. You can supply an text you wish for the subject. The text should not contain any non-printable characters that may cause problems for the recipient’s email application.

Message lines 1 to 15 (Character, 132)

Enter up to 15 lines of message text for a short message. You can leave blank lines in the message, or can leave all lines blank. It is recommended that you use no more than 74 characters in each line in order to stay compatible with most email applications.

File name (Character, 10)

The name of the file containing the message you want to send. The file may be internally or externally described, but must contain only character data. Only the first 132 bytes of the data in each record will be used for the message. Since many email applications expect message line lengths of 74 characters or less it is recommended that you restrict the line length to 74 characters.

Library name (Character, 10)

The name of the library in which the message file resides.

Member (Character, 10)

The name of the member of the message file

Attachment type (Character, 1)

Enter one of the following values for the attachment type. The parameters that follow relating to attachments will be edited based on this parameter. The valid values are:

- 0 = No attachment
- 1 = QDLS document
- 2 = IFS file
- 3 = Database file
- 4 = Save file
- 5 = Source file

Attachment external name (Character, 64)

Specify the name of the attachment. This can be any valid name for the recipient's file system. If the recipient is on a DOS computer you should specify this name in the 8.3 format common to DOS. If the recipient is on a Windows 95/98/NT or UNIX system you can specify a long name.

Attachment path (Character, 64)

Specify the attachment path. This is required for attachment types 1 and 2 (QDLS and IFS).

Attachment name (Character, 64)

Specify the name of the attachment in the QDLS or IFS directory. This parameter is required for attachment types 1 and 2.

Attachment file (Character, 10)

If you specified attachment types 3, 4 or 5 you must specify the name of the file that contains the attachment.

Attachment library (Character, 10)

If you specified attachment type 3, 4 or 5 you must specify the name of the library that contains the attachment file.

Attachment member (Character, 10)

If you specified attachment type 3 or 5 you must specify the member name of the attachment. Note that you do not need to specify a member name for Save files and any value you specify for save files will be ignored.

Return Code (Character, 4)

This field is returned as an indication of the success or failure of the call to the API program. If the call was successful and a name is returned, the value of this parameter will be "0000". Any other value indicates an error in processing the request. If an error occurs there may be information in the following field describing the error.

Text (Character, 80)

This field contains text that describes the success or failure of the API. The field may be blank if the API call was successful.

Problem Analysis

If an error occurs in the API an error indicator is returned in the four byte return code. Any return code other than “0000” indicates an error condition. Generally if there is an error condition there will be text in the Text field that describes the error. You can also view a description of the error by viewing the associated error in the MSG001 message file in the application library. The message ID will be “TMS” plus the return code. For example, if the return code is “1646” you can view the error message associates with this return code with the following command:

```
dspmsgd range(tms1646) msgf(msg001)
```

You can also enable the NetMail application trace flag in data area TMICFT. When this flag is enabled the API will generate additional diagnostic information to the job log of the calling program. To enable application tracing change the data area TMICFT to the value “Y”:

```
chgdtaara tmicft value(“Y”)
```

After calling the API display the job log and review the messages for diagnostic information. You can use the DSPJOBLOG command to view the job log. If the job was submitted to batch you can use the WRKSBMJOB or WRKUSRJOB commands to view the printed job log. Note that you may have to change the job description used to submit the job to produce the printed job log. Remember to change the data area TMICFT to “N” when you are finished to avoid unnecessary printed output and the performance impact of additional logging.

If you think the error may be occurring in the SMTP interface or with the connection to your SMTP mail server, you can turn on the application logging function and review the resulting log in file TMLOGA. The application logging function is turned on from the administration menu. After turning on the logging function you can view the log using the DSPPFM command. There may be diagnostic information in this file that describe the problem. All of the responses from the SMTP mail server are logging in this file. Note that you will want to turn off application logging as soon as you are finished to avoid accumulating logging records and to improve performance. You can clear the log with the CLRPFM command:

```
clrpfm tmloga
```

API201 – Email message with Spool file

Description

This is a generalized API that can be used to send a short message or a long message stored in a database file to an SMTP/Internet recipient with a spool file attachment. For a short message you can send up to 15 lines of message text. For a long message you specify a database file with the message text. If you use the long message option you must first create the message in a database file before calling this API. This API also lets you specify an attachment for the message.

Note that the call to this API may be successful, but the message may not be delivered to the recipient. There can be many reasons for this including an invalid recipient address, spam controls implemented to Internet Service Providers, or lack of availability of the recipient's mail system. A failure to deliver the message may not be reported back to the sender depending on the reason for the non-delivery.

Call Format

```
CALL PGM(API201) PARM(&TYPE &TOADDR &FROMADDR &REPLYTO + &ORGANIZ &
```

Parameters

The following parameters are passed to the API. They must be specified in the following order and each field passed to the API must have the type and length specified.

Type (Character, 1)

Specify the value of "1" for a short message or the value "2" for a long message stored in a database file. If you specify "1" for a short message you can provide the message in the 15 message lines in the following parameters. If you specify "2" for a long message in a database file you must specify the file, library, and member name that contains the message.

To Address (Character, 128)

The email address of the recipient. This must be a valid SMTP address in the form user@domain.

From address (Character, 128)

The address of the sender. This must be a valid SMTP address, but does not have to the actual address of the user that sends the message. The address must be in the format of user@domain. Please note that most Internet Service Providers will not deliver mail with a sender address with an unregistered domain name. You should also be aware that it is illegal in some jurisdictions to provide a false address on unsolicited email.

Reply-to address (Character, 128)

The address for the reply message. This address can be the same as the From address or can be different. You can specify a return address that is on a different email system and a different domain. See the notes about Internet Service Providers above.

Organization (Character, 128)

Enter the name of your organization. You can provide any text in this field that you like. This would normally be your company name.

SMTP host name (Character, 128)

The name of the SMTP mail server that will be used to deliver this mail. This is normally the name of your mail relay host or ISP mail server and is the domain format. For example: "mail.my-isp.com". If you do not specify an SMTP host name you must specify an SMTP IP address.

SMTP IP address (Character, 15)

The IP address of your mail relay host or ISP mail server. If you do not specify an SMTP host name you must specify an IP address. The address is in dotted decimal format, for example “206.14.53.98”.

SMTP port (Character, 5)

The port number of the mail server. This is normally “00025” but may be different if you are using an internal mail server or firewall.

Local host (Character, 128)

The name of your local host. This name is sent to the mail server to identify the sending entity. It must be a valid domain name, for example: “as400.acme.com”.

Subject (Character, 50)

The subject for this message. You can supply an text you wish for the subject. The text should not contain any non-printable characters that may cause problems for the recipient’s email application.

Message lines 1 to 15 (Character, 132)

Enter up to 15 lines of message text for a short message. You can leave blank lines in the message, or can leave all lines blank. It is recommended that you use no more than 74 characters in each line in order to stay compatible with most email applications.

File name (Character, 10)

The name of the file containing the message you want to send. The file may be internally or externally described, but must contain only character data. Only the first 132 bytes of the data in each record will be used for the message. Since many email applications expect message line lengths of 74 characters or less it is recommended that you restrict the line length to 74 characters.

Library name (Character, 10)

The name of the library in which the message file resides.

Member (Character, 10)

The name of the member of the message file

Attachment external name (Character, 64)

Specify the name of the attachment. This can be any valid name for the recipient’s file system. If the recipient is on a DOS computer you should specify this name in the 8.3 format common to DOS. If the recipient is on a Windows 95/98/NT or UNIX system you can specify a long name.

Spool file name (Character, 10)

Specify the name of the spool file you want to attach to this message. The spool file must exist at the time the API is called.

Job name (Character, 10)

Specify the name of the job that created the spool file.

User name (Character, 10)

Specify the name of the user (user profile) that created the spool file.

Job number (Character, 6)

Specify the job number for the job that created the spool file

Spool file number (Character, 6)

Specify the number of the spool file. This must be 4 digits in the range of 0001 to 9999, or the special values of *LAST or *ONLY.

Return Code (Character, 4)

This field is returned as an indication of the success or failure of the call to the API program. If the call was successful and a name is returned, the value of this parameter will be “0000”. Any other value indicates an error in processing the request. If an error occurs there may be information in the following field describing the error.

Text (Character, 80)

This field contains text that describes the success or failure of the API. The field may be blank if the API call was successful.

Problem Analysis

If an error occurs in the API an error indicator is returned in the four byte return code. Any return code other than “0000” indicates an error condition. Generally if there is an error condition there will be text in the Text field that describes the error. You can also view a description of the error by viewing the associated error in the MSG001 message file in the application library. The message ID will be “TMS” plus the return code. For example, if the return code is “1646” you can view the error message associates with this return code with the following command:

```
dspmsgd range(tms1646) msgf(msg001)
```

You can also enable the NetMail application trace flag in data area TMICFT. When this flag is enabled the API will generate additional diagnostic information to the job log of the calling program. To enable application tracing change the data area TMICFT to the value “Y”:

```
chgdtaara tmicft value(“Y”)
```

After calling the API display the job log and review the messages for diagnostic information. You can use the DSPJOBLOG command to view the job log. If the job was submitted to batch you can use the WRKSBMJOB or WRKUSRJOB commands to view the printed job log. Note that you may have to change the job description used to submit the job to produce the printed job log. Remember to change the data area TMICFT to “N” when you are finished to avoid unnecessary printed output and the performance impact of additional logging.

If you think the error may be occurring in the SMTP interface or with the connection to your SMTP mail server, you can turn on the application logging function and review the resulting log in file TMLOGA. The application logging function is turned on from the administration menu. After turning on the logging function you can view the log using the DSPPFM command. There

may be diagnostic information in this file that describe the problem. All of the responses from the SMTP mail server are logging in this file. Note that you will want to turn off application logging as soon as you are finished to avoid accumulating logging records and to improve performance. You can clear the log with the CLRPFM command:

```
clrpfm tmloga
```

API216 – Send File Message or Short Message To Multiple SMTP/Internet Recipients with Optional Attachment

Description

This is a generalized API that can be used to send a short message or a long message stored in a database file to one or more SMTP/Internet recipients. For a short message you can send up to 15 lines of message text. For a long message you specify a database file with the message text. If you use the long message option you must first create the message in a database file before calling this API. This API also lets you specify an attachment for the message. This API is similar to API200 with the exception that is allows you to specify multiple To, CC and BCC recipients.

Note that the call to this API may be successful, but the message may not be delivered to the recipient. There can be many reasons for this including an invalid recipient address, spam controls implemented to Internet Service Providers, or lack of availability of the recipient's mail system. A failure to deliver the message may not be reported back to the sender depending on the reason for the non-delivery.

Call Format

```
CALL PGM(API216) PARM(&TYPE &TOADDR &CCADDR &BCADDR &FROMADDR &REPL
&ATCEXTNAM &ATCPATH &ATCNAME &ATCFILE + &ATCLIBR &ATCMEMB &RETURN &
```

Parameters

The following parameters are passed to the API. They must be specified in the following order and each field passed to the API must have the type and length specified.

Type (Character, 1)

Specify the value of "1" for a short message or the value "2" for a long message stored in a database file. If you specify "1" for a short message you can provide the message in the 15 message lines in the following parameters. If you specify "2" for a long message in a database file you must specify the file, library, and member name that contains the message.

To Address (Character, 9999)

The email addresses of the recipient. This must be one or more valid SMTP addresses in the form user@domain. If there are multiple addresses you should separate them with a comma. For example:

```
User1@acme.com, User2@acme.com, User3@acme.com
```

CC Addresses (Character, 9999)

The email addresses of the CC recipients. This must be one or more valid SMTP addresses in the form user@domain. If there are multiple addresses you should separate them with a comma. For example:

```
User1@acme.com, User2@acme.com, User3@acme.com
```

BCC Addresses (Character, 9999)

The email addresses of the BCC (Blind Carbon Copy) recipients. This must be one or more valid SMTP addresses in the form user@domain. If there are multiple addresses you should separate them with a comma. For example:

```
User1@acme.com, User2@acme.com, User3@acme.com
```

From address (Character, 128)

The address of the sender. This must be a valid SMTP address, but does not have to the actual address of the user that sends the message. The address must be in the format of user@domain. Please note that most Internet Service Providers will not deliver mail with a sender address with an unregistered domain name. You should also be aware that it is illegal in some jurisdictions to provide a false address on unsolicited email.

Reply-to address (Character, 128)

The address for the reply message. This address can be the same as the From address or can be different. You can specify a return address that is on a different email system and a different domain. See the notes about Internet Service Providers above.

Organization (Character, 128)

Enter the name of your organization. You can provide any text in this field that you like. This would normally be your company name.

SMTP host name (Character, 128)

The name of the SMTP mail server that will be used to deliver this mail. This is normally the name of your mail relay host or ISP mail server and is the domain format. For example: “mail.my-isp.com”. If you do not specify an SMTP host name you must specify an SMTP IP address.

SMTP IP address (Character, 15)

The IP address of your mail relay host or ISP mail server. If you do not specify an SMTP host name you must specify an IP address. The address is in dotted decimal format, for example “206.14.53.98”.

SMTP port (Character, 5)

The port number of the mail server. This is normally “00025” but may be different if you are using an internal mail server or firewall.

Local host (Character, 128)

The name of your local host. This name is sent to the mail server to identify the sending entity. It must be a valid domain name, for example: “as400.acme.com”.

Subject (Character, 50)

The subject for this message. You can supply an text you wish for the subject. The text should not contain any non-printable characters that may cause problems for the recipient’s email application.

Message lines 1 to 15 (Character, 132)

Enter up to 15 lines of message text for a short message. You can leave blank lines in the message, or can leave all lines blank. It is recommended that you use no more than 74 characters in each line in order to stay compatible with most email applications.

File name (Character, 10)

The name of the file containing the message you want to send. The file may be internally or externally described, but must contain only character data. Only the first 132 bytes of the data in each record will be used for the message. Since many email applications expect message line lengths of 74 characters or less it is recommended that you restrict the line length to 74 characters.

Library name (Character, 10)

The name of the library in which the message file resides.

Member (Character, 10)

The name of the member of the message file

Attachment type (Character, 1)

Enter one of the following values for the attachment type. The parameters that follow relating to attachments will be edited based on this parameter. The valid values are:

- 0 = No attachment
- 1 = QDLS document
- 2 = IFS file
- 3 = Database file
- 4 = Save file
- 5 = Source file

Attachment external name (Character, 64)

Specify the name of the attachment. This can be any valid name for the recipient's file system. If the recipient is on a DOS computer you should specify this name in the 8.3 format common to DOS. If the recipient is on a Windows 95/98/NT or UNIX system you can specify a long name.

Attachment path (Character, 64)

Specify the attachment path. This is required for attachment types 1 and 2 (QDLS and IFS).

Attachment name (Character, 64)

Specify the name of the attachment in the QDLS or IFS directory. This parameter is required for attachment types 1 and 2.

Attachment file (Character, 10)

If you specified attachment types 3, 4 or 5 you must specify the name of the file that contains the attachment.

Attachment library (Character, 10)

If you specified attachment type 3, 4 or 5 you must specify the name of the library that contains the attachment file.

Attachment member (Character, 10)

If you specified attachment type 3 or 5 you must specify the member name of the attachment. Note that you do not need to specify a member name for Save files and any value you specify for save files will be ignored.

Return Code (Character, 4)

This field is returned as an indication of the success or failure of the call to the API program. If the call was successful and a name is returned, the value of this parameter will be "0000". Any other value indicates an error in processing the request. If an error occurs there may be information in the following field describing the error.

Text (Character, 80)

This field contains text that describes the success or failure of the API. The field may be blank if the API call was successful.

Problem Analysis

If an error occurs in the API an error indicator is returned in the four byte return code. Any return code other than "0000" indicates an error condition. Generally if there is an error condition there will be text in the Text field that describes the error. You can also view a description of the error by viewing the associated error in the MSG001 message file in the application library. The message ID will be "TMS" plus the return code. For example, if the return code is "1646" you can view the error message associates with this return code with the following command:

```
dspmsgd range(tms1646) msgf(msg001)
```

You can also enable the NetMail application trace flag in data area TMICFT. When this flag is enabled the API will generate additional diagnostic information to the job log of the calling program. To enable application tracing change the data area TMICFT to the value “Y”:

```
chgdtaara tmicft value(“Y”)
```

After calling the API display the job log and review the messages for diagnostic information. You can use the DSPJOBLOG command to view the job log. If the job was submitted to batch you can use the WRKSBMJOB or WRKUSRJOB commands to view the printed job log. Note that you may have to change the job description used to submit the job to produce the printed job log. Remember to change the data area TMICFT to “N” when you are finished to avoid unnecessary printed output and the performance impact of additional logging.

If you think the error may be occurring in the SMTP interface or with the connection to your SMTP mail server, you can turn on the application logging function and review the resulting log in file TMLOGA. The application logging function is turned on from the administration menu. After turning on the logging function you can view the log using the DSPPFM command. There may be diagnostic information in this file that describe the problem. All of the responses from the SMTP mail server are logging in this file. Note that you will want to turn off application logging as soon as you are finished to avoid accumulating logging records and to improve performance. You can clear the log with the CLRPFM command:

```
clrpfm tmloga
```

Programmer’s Toolkit – low level APIs

Overview

The low level APIs are implemented in a service program named CVTDTA in library ALLDTA200. You can call the functions in the service program using the CALLB operation code in an ILE RPG program, or by using the similar function in ILE Cobol. Like high level API programs the functions accept parameters and return a result. The low- level APIs return just a 4 byte binary numeric error code. They do not return a text field. Your program must inspect the return code to determine if the request was successful. If the value is 0 the request was successful. Any other value indicates a failure in the API.

Compiling ILE RPG Programs

To compile an ILE RPG program and bind the service program CVTDTA requires two steps. You must first create an RPG module of your ILE RPG program using the Create RPG Module (CRTRPGMOD) command, and then use the Create Program (CRTPGM) command to create

the executable program with the CVTDTA service program. For example, if you have an ILE RPG program called MYPGM1 you could use the following steps to create it and use the CVTDTA functions in it:

```
crtrpgmod module(mylib/mypgm1) srcfile(mylib/qrpglesrc)

crtpgm pgm(mylib/mypgm1) module(mylib/mypgm1) bndsrvpgm(cvtdta)
```

Service program CVTDTA

The CrossData/400 outbound conversion routines are implemented in the CVTDTA service program. They must be called in a specific order. You must first call the initialization function CVTInit before calling any other function. Then you must call the CVTInFile and CVTOutFile functions to set the names of the input and output files. You can then call other functions to set the delimiter characters, column heading options, and so forth. Lastly you will call the CVTConvert function to perform the actual conversion of the data. If the CVTInit function is not called first, or the CVTConvert function is called before the input and output files are defined the conversion will fail with an appropriate return code. The API function calls will be in roughly this sequence:

```
CVTInit (Initialize the environment)
CVTInFile (Specify the input file name)
CVTOutFile (Specify the output file name)
CVTConvert  (Perform the data conversion)
```

There are many other APIs you can call that affect the conversion results. These are documented in the sections below.

Compiling ILE programs with CVTDTA

When you compile an ILE RPG program you can first create the RPG module and then create the program binding the CVTDTA service program. If you wanted to create a program called CONVERT you could use these commands:

```
crtrpgmod module(convert) srcfile(mylib/qrpglesrc)

crtpgm pgm(convert) module(convert) bndsrvpgm(cvtdta)
```

For each language (RPG, Cobol, C, etc.) there is a separate command to create the module. After creating the module you can use the Create Program (CRTPGM) command to build the program and bind the CVTDTA service program.

Functions in CVTDTA

The following functions are implemented in the service program CVTDTA. Other functions may be added in the future.

FUNCTION	DESCRIPTION
CVTInit	Initialize the conversion environment. It is always called first.
CVTDelim	Specifies the delimiter character to use between output fields, if any.
CVTFormat	Specifies a user-defined format string for the input file. This is only used if the input file is internally described or a System/36 file.
CVTInFile	Specifies the input file that will be converted. This is a required function call.
CVTOutFile	Specifies the output file for the converted data. This is a required parameter.
CVTDecimal	Specifies the decimal character to be used in numeric fields. If not called the default of “.” is used.
CVTNewLine	Specifies the Carriage Return and Line Feed characters to be used at the end of an output record, if any.
CVTNumSign	Specifies the negative sign character for negative numeric fields. The sign character precedes the number in the output field.
CVTQuote	Specifies the quote character to use for character fields, if any. The default is the double quote character (“).
CVTTable	Specifies the EBCDIC to ASCII conversion table and library. The default is QASCII in library QSYS.
CVTZeroDsp	Specifies the format for numeric fields with zero values. The default is to display all zeros in output fields. Can be used to display blanks in the output fields.
CVTTruncat	Specifies whether or not character fields will be blank truncated. The default is not to truncate trailing blanks in character fields.
CVTPk2Stor	Specifies whether or not to expand packed numeric values by one digit. The default is to not expand packed numerics. You may enable this option in order to be compatible with certain PC based transfer applications.

FUNCTION	DESCRIPTION
CVTTruncNr	Specifies whether or not numeric fields will have leading blanks or zeros suppressed. The default is not to suppress leading blanks or zeros.
CVTTitles	Specifies the column headers used for the CSV file, if any. The default is not to use column titles in the output file.
CVTConvert	Instructs CrossData/400 to perform the file conversion. Minimally the CVTInit, CVTInFile, and CVTOutFile must have been called before CVTConvert is called.
CVTTo400	Convert data to AS/400 format. Requires data to be converted and format string. Output is AS/400 data.
CVTFldSep	The field separator character for converting inbound data.
CVTHist	Adds a history record to the CrossData/400 history file. Provided for user written programs that use the Programmer's Toolkit.

Required calling sequence of CVTDTA functions

The first low-level function you must call is CVTInit. This routine allocates memory for the conversion process and initializes the environment. If you do not call this function first all other functions will fail. CVTInit calls many of the other APIs to establish the defaults for the application.

After calling the CVTInit function you can call any of the other functions except CVTConvert. CVTConvert must be the last function called. You must call the CVTInFile and CVTOutFile functions to define the input file for the conversion and the output file for the converted data. You can call any of the other functions to set the translation table and various delimiter and format options. See the description of each function below to determine which ones you need to call for your conversion requirements.

When you have called the above functions to define the conversion files, you must call CVTConvert to perform the data conversion. The output file will be created if it does not exist.

After running the CVTConvert function you can call CVTInFile and CVTOutFile and do another conversion. It is not necessary to call the CVTInit function again.

CVTInit – Initialize CrossData/400 conversion

Description

This function initializes the conversion environment. It allocates memory, establishes defaults, adds the product to the library list, and checks the license state of the product. You must call this function before any other function. There are no input parameters to this function. The one

parameter is a return code indicating the success or failure of the request.

Call format

You can call the function in an ILE RPG program like this:

```
d TS    DS
d RETRB 1 4B 0
c*
c       Eval RETRB = 0
c       callb(d) 'CVTInit'
c       parm RETRB
```

Parameters

Return (Binary, 4)

The return code indicating the success or failure of the initialization. If the return code is 0 the initialization was successful.

CVTInFile – Specify input file

Description

This function defines the input file to be converted. You must define the file name, library name, and member name for the input file.

Call format

You can call the function in an ILE RPG program like this:

```
d TS DS
d FRMFIL11 1 11A
d FRMLIB11 12 22A
d FRMMBR11 23 33A
d RETRB 34 37B 0
d*
c Eval FRMLIB11 = 'ALLDTA200 ' + x'00'
c Eval FRMFIL11 = 'TEST01 ' + x'00'
c Eval FRMMBR11 = 'TEST01 ' + x'00'
c*
c callb(d) 'CVTInFile'
c parm FRMLIB11
c parm FRMFIL11
c parm FRMMBR11
c parm RETRB
c*
```

Parameters

Library (Character, 11)

The name of the library that contains the file to be converted. This must be a 10 character field terminated by a null character (hex 00).

File name (Character, 11)

The name of the input file. This is a required field and must be null terminated.

Member name (Character, 11)

The name of the member in the file. This is a required field and must be null terminated.

Return (Binary, 4)

The return code indicating the success or failure of the initialization. If the return code is 0 the request was successful.

CVTOutFile – Specify output file

Description

This function defines the output file to receive the converted data. You must define the file name, library name, and member name for the output file.

Call format

You can call the function in an ILE RPG program like this:

```
d TS DS
d FRMFIL11 1 11A
d FRMLIB11 12 22A
d FRMMBR11 23 33A
d TOFIL11 34 44A
d TOLIB11 45 55A
d TOMBR11 56 66A
d RETRB 67 70B 0
c*
c Eval TOLIB11 = 'ALLDTA200W' + x'00'
c Eval TOFIL11 = 'TESTOUT ' + x'00'
c Eval TOMBR11 = 'TESTOUT ' + x'00'
c*
c callb(d) 'CVTOutFile'
c parm TOLIB11
c parm TOFIL11
c parm TOMBR11
c parm RETRB
c*
```

Parameters

To Library (Character, 11)

The name of the library that contains the file to be converted. This must be a 10 character field terminated by a null character (hex 00).

To File name (Character, 11)

The name of the input file. This is a required field and must be null terminated.

To Member name (Character, 11)

The name of the member in the file. This is a required field and must be null terminated.

Return (Binary, 4)

The return code indicating the success or failure of the initialization. If the return code is 0 the request was successful.

CVTDelim – Set delimiter options

Description

This function establishes the delimiters to use between fields, if desired. The default when you call CVTInit is for the comma character (“,”) to be used as the field delimiter.

Call format

You can call the function in an ILE RPG program like this:

```
d TS    DS
d DELIM 1 2A
d SET   3 6B 0
d RETRB 7 10B 0
c*
c       Eval DELIM = ',' + x'00'
c       Eval SET = 1
c*
c       callb(d) 'CVTDelim'
c       parm SET
c       parm DELIM
c       parm RETRB
c*
```

Parameters

Set value (Binary, 4)

This value is equal to 1 to set the delimiter character to the next field, or 0 to indicate that no delimiters are used.

Delimiter character (Character, 2)

This field contains the delimiter character followed by a null (Hex 00). It can be any value. By default the value is a comma character.

Return (Binary, 4)

The return code indicating the success or failure of the request. If the return code is 0 the request was successful.

CVTFormat – Set data format

Description

This function establishes the format of the data in the input file. CVTFormat is normally only used when your input file is an internally described file as is the case for System/36 database files. CrossData/400 can determine the format of an externally described file automatically and you would not use this API function to define the file.

Call format

You can call the function in an ILE RPG program like this:

```
d TS DS
d FORMAT 1 128A
d SET 129 132B 0
d RETRB 133 136B 0
c*
c Eval FORMAT = 'C0002000C0004000' + x'00'
c Eval SET = 1
c*
c callb(d) 'CVTFormat'
c parm SET
c parm FORMAT
c parm RETRB
c*
```

Parameters

Set value (Binary, 4)

This value is equal to 1 to set the format string to the next field, or 0 to indicate that no format is defined.

Format string (Character, Variable, minimum 9 bytes)

This field contains the format string followed by a null (Hex 00). The format string is a series of characters that defines the type of data, length of the data, and, for numeric fields, the number of decimal places. The structure of the field definition is as follows:

Position 1 to 1: Data type. This is the same value as you would find if you used the DSPFFD command. So, a character field would have the value “A”, a numeric packed field would be “P,” etc.

Position 2 to 6: The length of the field, zero filled. If the field is 10 bytes long you would specify “00010”.

Position 7 to 8: The number of decimal positions. If the field is a character field use “00”. If the field is numeric enter the number of decimal positions.

In the above code example there are two fields being defined by the format. The first field is a character field 20 bytes long and the second field is a character field 40 bytes long.

Please note that the L (date) format type is special. The first character is “L” to indicate a target database date field. The next 5 digits indicate the length of the field. This must be “00010” for an L field type. Data must be in ISO date format “CCYY-MM-DD”. For example: “2008-01-22”. Alliance CrossData/400 will change all delimiters to the dash (“-“) character. For example, the input data “2008.01.22” will be changed to “2008-01-22”.

Return (Binary, 4)

The return code indicating the success or failure of the request. If the return code is 0 the request was successful.

CVTDecimal – Set decimal character options

Description

This function establishes the decimal character to be used in numeric fields. The default is to display the decimal character and to use the period (“.”). You can turn off the use of the decimal character or change it to another value with this API.

Call format

You can call the function in an ILE RPG program like this:

```
d TS DS
d DECCHR 1 2A
d SET 3 6B 0
d RETRB 7 10B 0
c*
c Eval DECCHR = '.' + x'00'
c Eval SET = 1
c*
c callb(d) 'CVTDecimal'
c parm SET
c parm DECCHR
c parm RETRB
c*
```

Parameters

Set value (Binary, 4)

This value is equal to 1 to set the decimal character to the following field, or 0 to indicate that no decimal character is to be used.

Decimal character (Character, 2)

This field contains the decimal character followed by a null (Hex 00). It can be any value. By default the value is a period.

Return (Binary, 4)

The return code indicating the success or failure of the request. If the return code is 0 the request was successful.

CVTNewLine – Set line termination options

Description

This function establishes the use of line termination characters for the termination of the data record in the output file. Normally the termination characters are a Carriage Return and Line Feed. But you can set these to any value you want, or turn off line termination altogether.

Call format

You can call the function in an ILE RPG program like this:

```
d TS DS
d TRMCHR 1 3A
d SET 4 7B 0
d RETRB 8 11B 0
c*
c Eval TRMCHR = x'0D2500'
c Eval SET = 1
c*
c callb(d) 'CVTNewLine'
c parm SET
c parm TRMCHR
c parm RETRB
c*
```

Parameters

Set value (Binary, 4)

This value is equal to 1 to set the line termination character(s) to the following field, or 0 to indicate that no line termination characters are to be used.

Termination character (Character, 3)

This field contains the line termination characters followed by a null (Hex 00). It can be any value. By default the value includes a Carriage Return and Line Feed character.

Return (Binary, 4)

The return code indicating the success or failure of the request. If the return code is 0 the request was successful.

CVTNumSign – Set numeric sign options

Description

This function establishes the numeric sign character to be used in negative numeric fields. The default is to display the negative sign character and to use the dash (“-”). You can turn off the use of the sign character or change it to another value with this API.

Call format

You can call the function in an ILE RPG program like this:

```
d TS DS
d NEGCHR 1 2A
d SET 3 6B 0
d RETRB 7 10B 0
c*
c Eval NEGCHR = '-' + x'00'
c Eval SET = 1
c*
c callb(d) 'CVTNumSign'
c parm SET
c parm NEGCHR
c parm RETRB
c*
```

Parameters

Set value (Binary, 4)

This value is equal to 1 to set the negative sign character to the following field, or 0 to indicate that no negative sign character is to be used.

Sign character (Character, 2)

This field contains the negative sign character followed by a null (Hex 00). It can be any value. By default the value is a dash (“-“).

Return (Binary, 4)

The return code indicating the success or failure of the request. If the return code is 0 the request was successful.

CVTTruncat – Truncate trailing blanks in character fields

Description

This function establishes how trailing blanks in character fields will be written to the output file. If you specify 0 for the option trailing blanks will not be removed from character fields (the default). If you specify 1 for the option trailing blanks will be removed from character fields.

Call format

You can call the function in an ILE RPG program like this:

```
d TS    DS
d SET   1 4B 0
d RETRB 5 8B 0
c*
c       Eval SET = 1
c*
c       callb(d) 'CVTTruncat'
c       parm SET
c       parm  RETRB
c*
```

Parameters

Set value (Binary, 4)

This value is equal to 1 to truncate trailing blanks. The value is equal to 0 to retain trailing blanks in character fields.

Return (Binary, 4)

The return code indicating the success or failure of the request. If the return code is 0 the request was successful.

CVTTruncNr – Truncate leading blanks or zeros in numeric fields

Description

This function establishes how leading blanks or zeros in numeric fields will be written to the output file. If you specify 0 for this option leading blanks or zeros will not be removed from numeric fields (the default). If you specify 1 for the option leading blanks or zeros will be removed from numeric fields.

Note that how this API works is dependent on the CVTZeroDsp API. If you've specified that zero value numeric fields should be blanks, this API will suppress leading blanks in a numeric field. If you specified that zero value numeric fields should be all zeros, this API will suppress leading zeros. If you use this API to suppress leading blanks and a numeric field has a zero

value, there will be one blank character in the output file for the field. If you use this API to suppress leading zeros and a numeric field has a zero value, there will be one zero character in the output file for the field.

Call format

You can call the function in an ILE RPG program like this:

```
d TS     DS
d SET    1 4B 0
d RETRB  5 8B 0
c*
c        Eval SET = 1
c*
c        callb(d) 'CVTTruncNr'
c        parm SET
c        parm RETRB
c*
```

Parameters

Set value (Binary, 4)

This value is equal to 1 to truncate leading blanks. The value is equal to 0 to retain leading blanks or zeros in character fields (the default). The value is equal to 2 to suppress zeros in numeric fields and blanks in character fields. Set this value to 3 to suppress zeros in numeric fields, but leave a single “0” character for zero value fields. Set this value to 4 to allow the numeric sign to float to the first position before a number if the value is negative.

Return (Binary, 4)

The return code indicating the success or failure of the request. If the return code is 0 the request was successful.

CVTZeroDsp – Set zero value numeric field display attributes

Description

This function establishes how zero value numeric fields will be written to the output file. You can specify that the field is filled with zeros (the default) or that all blanks should be written to the output file. It is common in ASCII fixed field length files that zero value numeric fields are blanks.

Call format

You can call the function in an ILE RPG program like this:

```
d TS    DS
d SET   1 4B 0
d RETRB 5 8B 0
c*
c       Eval SET = 1
c*
c       callb(d) 'CVTZeroDsp'
c       parm SET
c       parm RETRB
c*
```

Parameters

Set value (Binary, 4)

This value is equal to 1 to display the output as all zeros. The value is equal to 0 to display the output as all blanks.

Return (Binary, 4)

The return code indicating the success or failure of the request. If the return code is 0 the request was successful. s

CVTPk2Stor – Expand packed field output by one byte

Description

This function allows specifying that a packed field should be expanded by one byte during conversion. The default is not to expand the output. By expanding the output you can be compatible with the data conversion characteristics of certain PC based transfer applications.

Call format

You can call the function in an ILE RPG program like this:

```
d TS DS
d OPTION 1 4B 0
d RETRB 5 8B 0
c*
c Eval OPTION = 1
c*
c callb(d) 'CVTPk2Stor'
c parm OPTION
c parm RETRB
c*
```

Parameters Option (Binary, 4)

Set this value to 1 to expand packed numeric fields by one byte. Set this value to 0 to not expand numeric fields (the default).

Return (Binary, 4)

The return code indicating the success or failure of the request. If the return code is 0 the request was successful.

CVTQuote – Set character field quote options

Description

This function establishes the quote character to be used around character and numeric fields. The default is to use the quote character and to use the double quote (“). You can turn off the use of the quote character or change it to another value with this API.

Call format

You can call the function in an ILE RPG program like this:

```
d TS DS
d QOTCHR 1 2A
d SET 3 6B 0
d RETRB 7 10B 0
c*
c Eval QOTCHR = '"' + X'00'
c Eval SET = 1
c*
c callb(d) 'CVTQuote'
c parm SET
c parm QOTCHR
c parm RETRB
c*
```

Parameters

Set value (Binary, 4)

This value is equal to 1 to the quote character to the following field, or 0 to indicate that no quote character is to be used.

Quote character (Character, 2)

This field contains the quote character followed by a null (Hex 00). It can be any value. By default the value is a double quote (“).

Return (Binary, 4)

The return code indicating the success or failure of the request. If the return code is 0 the request was successful.

CVTTable – Set EBCDIC to ASCII translation table options

Description

This function establishes the name of the EBCDIC to ASCII table to be used when you convert data to the ASCII character set and whether the data will be translated. The default is to use the QASCII table supplied by IBM and to not perform translation.

Call format

You can call the function in an ILE RPG program like this:

```
d TS DS
d SET 1 4B 0
d RETRB 5 8B 0
d TRANNM 9 19A
d TRANLB 20 30A
c*
c Eval TRANNM = 'MYTABLE ' + x'00'
c Eval TRANLB = 'MYLIBR ' + x'00'
c Eval SET = 1
c*
c callb(d) 'CVTTable'
c parm SET
c parm TRANLB
c parm TRANNM
c parm RETRB
c*
```

Parameters

Set value (Binary, 4)

This value is equal to 1 to set EBCDIC to ASCII translation on, or 0 to indicate that no EBCDIC to ASCII translation is to be done.

Translation table name (Character, 11)

This field contains the name of the translation table followed by the null character (hex 00). The default is not to use translation and to use the QASCII table in library QSYS.

Translation table library (Character, 11)

This field contains the name of the library for the translation table followed by the null character (hex 00). The default is library QSYS.

Return (Binary, 4)

The return code indicating the success or failure of the request. If the return code is 0 the request was successful.

CVTTitles – Set CSV column title options

Description

This function establishes the names of the column titles in a CSV output file. By default CrossData/400 will not use column titles. If you want to use column titles you can choose to use the field names for the titles, or you can use the column text values in the DDS for titles, or you can specify your own titles for the columns. If you want to define your own titles you can specify up to three rows for the output file. The function is also used to turn column titles on or off.

The maximum length of a column title is 20 bytes. You must specify the entire twenty bytes for each field. You can use leading or trailing blanks for each title. If you want to supply titles for three fields you must provide 60 bytes of title characters. Probably the easiest way to do this is by using one or more arrays to hold the title text (see below.)

You can use one, two, or three rows of titles. If you only want to use one row of titles you can use this function to enable titles for that row, and then call the function with the set parameter of 0 to turn off the other two rows of titles.

Call format

You can call the function in an ILE RPG program like this:

```
d TS DS
d SET 1 4B 0
d RETRB 5 8B 0
d TITLES 9 29A
d OPTION 30 33B 0
c*
c Eval TITLES = 'SALES  ' + x'00'
c Eval SET = 1
c Eval OPTION = 4
c*
c callb(d) 'CVTTitles'
c parm SET
c parm OPTION
c parm TITLES
c parm RETRB
c*
```

Parameters

Set value (Binary, 4)

This value is equal to 1 to enable the use of titles with the option specified. Specify the value of 0 to turn off the use of titles with the option specified.

Option (Binary, 4)

This parameter sets the type of titles you want to use. Specify the following values for the option:

1 - Do not use titles

2 - Use field names for the titles

3 - Use Column Headers for the titles

4 - Use the text provided in the following parameter for the first row of titles

5 - Use the text provided in the following parameter for the second row of titles

6 - Use the text provided in the following parameter for the third row of titles

Title text (Character, 20 to nnn)

This field contains the titles you want to use for column headings. It is only used for options 4, 5, and 6 above. If you specify option 1, 2, or 3 this field is ignored. If you are not going to use this field you should pass a single character with a null value (hex 00).

Return (Binary, 4)

The return code indicating the success or failure of the request. If the return code is 0 the request was successful.

CVTConvert – Convert data

Description

This function performs the actual conversion of the data using the information you have supplied on previous calls to the APIs. You must have called the following functions before calling CVTConvert:

- CVTInit Initialize
- CVTInFile Define input file
- CVTOutFile Define output file

If the database conversion is successful the converted data will be placed in the output file.

Call format

You can call the function in an ILE RPG program like this:

```
d TS    DS
d RETRB 1 04B 0
c*
c       callb(d) 'CVTConvert'
c       parm RETRB
c*
```

Parameters Return (Binary, 4)

The return code indicating the success or failure of the request. If the return code is 0 the request was successful.

CVTTo400 – Convert data to AS/400 format

Description

This function performs the conversion of data in PC format (CSV, tab-delimited, etc) to AS/400 format. This is done on a record basis and does not work on a file. If you want to convert an entire file you can use one of the high level APIs to do this. If you want to write your own file conversion routine you can use this API to convert each record in the file. You will need to read each record, create the format string, and call this API.

Call format

You can call the function in an ILE RPG program like this:

```
d In                s                1a              dim(32766)
d InLength          s                10u 0
d Format@           s          32767a
d FmtLength         s                10u 0
d Out               s                1a              dim(32766)
d OutLength         s                10u 0
d RtnLength         s                10u 0
d APIreturn         s                10u 0

c       callb(d)  'CVTTo400'
c       parm              In
c       parm              InLength
c       parm              Format@
c       parm              FmtLength
c       parm              Out
c       parm              OutLength
c       parm              RtnLength
c       parm              APIreturn
```

Parameters

In (Character 32766)

The data to be converted to AS/400 format. The characteristics of this data should be described by other APIs (such as CVTFldSep, etc) before calling this API to perform the conversion.

InLength (Binary, 4.0)

The length of the data to be converted.

Format (Character, 32767)

The format string for the data. You can create the format string automatically by using the CVCRTFMT API, or you can specify the data format in a variable. See the specifications above on how to specify the conversion format.

FmtLength (Binary, 4.0)

The length of the format data.

Out (Character 32766)

The converted data returned by the API.

OutLength (Binary, 4.0)

The length of the variable Out.

RtnLength (Binary, 4.0)

The length of the data converted by the API. This is the actual length of the converted data.

Return (Binary, 4)

The return code indicating the success or failure of the request. If the return code is 0 the request was successful.

CVTFldSep – Inbound data field separator

Description

This function defines the field separator that is used for converting data from PC format to AS/400 format. The field separator would be a comma (“,”) for CSV files, etc. Be sure to define the field separator in the source character set. For example, you can define it as “,” for EBCDIC data, but you would define it as hex 22 for ASCII data.

Call format

You can call the function in an ILE RPG program like this:

```
d APIreturn             s       10u 0
d Namval                s       11a
d Libval                s       11a
d CharVal               s       3a

c       eval            CharVal = ',' + x'00'
c       callb(d)        'CVTFldSep'
c       parm            CharVal
c       parm            APIreturn
```

Parameters

CharVal (Character, 2)

The character that is used as the field separator. This can be any value and must be null terminated. For ASCII input files you can specify a hex value for the field separator.

Return (Binary, 4)

The return code indicating the success or failure of the request. If the return code is 0 the request was successful.

CVTHist – Create history record

Description

This function lets the programmer create a history record in the AFHIST file. It is useful when you have used the Programmer's Toolkit functions to perform a conversion.

Call format

You can call the function in an ILE RPG program like this:

```
d TS DS
d STATUS 1 1A
d RETURN 2 5A
d TEXT 6 85A
d FRMLIB 86 95A
d FRMFIL 96 105A
d FRMMBR 106 115A
d TOLIB 116 125A
d TOFIL 126 135A
d TOMBR 136 145A
c*
c call  'CVTHist'
c parm STATUS
c parm RETURN
c parm TEXT
c parm FRMLIB
c parm FRMFIL
c parm FRMMBR
c parm TOLIB
c parm TOFIL
c parm TOMBR
c*
```

Parameters

Status (Character, 1)

Provide a value of "1" to indicate the request was successful. Provide a value of "2" to indicate the request failed.

Return code (Character, 4)

Provide the error return code in this field. If you want to use your own error codes you should chose a number in the range of “8001” to “8999”.

Text (Character, 80)

Provide descriptive text for the return code.

From library (Character, 10)

Provide the name of the library in which the file to be converted resides.

From file (Character, 10)

Provide the name of the file you want to convert.

From member (Character, 10)

Provide the name of the member in the from file to be converted.

To library (Character, 10)

Provide the name of the library that will receive the file that is converted.

To file (Character, 10)

Provide the name of the file that is the output file for the conversion.

To member (Character, 10)

Provide the name of the member to receive the converted data.

Sample programs

Sample programs are provided in the QCLSRC, QRPGLESRC, and QRPGSRC files in the product library. You may use these source files as a template or guide for your own programs. Be sure to store your own programs in a library other than the CrossData/400 product library. This library will be refreshed periodically and you will lose any source you store in these source members.

Getting Support

Free support

Free support for bug fixes is available for the first 90 days after you install Alliance. You can receive free fixes to problems and support in the use of the product via e-mail or fax. At our website, www.townsendsecurity.com there is a link to ‘Support’, click on ‘Start a support incident’ and post your questions, attach logs, copy screen shots there. We attempt to answer all questions within 24 hours of receipt.

Alternatively you can send questions via fax to:

(360) 357-9047

Please be sure to include your name, return fax number, company name and address, and phone number. We try to answer all questions within 24 hours of receipt.

Paid support

A maintenance contract is available that provides you with telephone support, fixes to problems as they are available, new releases and enhancements on a regular basis, and other services. Contact your software vendor for information about the maintenance agreement.

Problem determination

Overview

Most problems you will encounter with Alliance will probably have to do with configuring the many options that are available for converting data. However, you may encounter actual program problems that require support. Please consider using the following procedures to analyze a problem before reporting the problem to your software vendor.

Using the history log

Alliance records the result of every conversion that is made via CrossData/400. Use the Display History Log option on the main menu to view the results of a conversion. The history log will provide a short description of any request that failed.

Using the application logs

For a more detailed description of the actions Alliance is taking during data conversion processing, you can activate the application log. Use the Configure CrossData/400 option to activate logging. After activating application logging, replicate the problem and view the log with the Display Application Log option on the main menu. The log will have a detailed list of functions performed during logon or request validation. You may be able to determine the cause of the problem by reviewing the log.

QSYSOPR messages

The Alliance applications rarely place messages in the QSYSOPR message queue, but you should check there if you are having difficulties. Use the DSPMSG command to view this message queue:

```
dspmsg qsysopr
```

Print any error messages from Alliance that may be there.

Further action

If you are unable to determine the cause of a problem, please contact your software vendor for support. Be prepared to provide a copy of the history log, application log, source code, and any QSYSOPR messages to your software vendor.

Appendix A – Configuring SMTP

Introduction

This document is designed to assist you in configuring SMTP and POP3 on your AS/400 for use with NetMail*400 from Net400. It takes you through a step-by-step process for configuring SMTP and POP3. It assumes you are configuring SMTP for the first time. If you already have SMTP and POP3 configured, you can skip this guide and proceed directly to configuring NetMail*400 for using SMTP.

AS/400 PTF requirements

IBM has released several PTFs related to SMTP mail in the last few months. You should be sure your system is current for cumulative PTFs before proceeding. After applying the most recent cumulative PTF package be sure the following PTFs are installed. You can use the Display PTF (DSPPTF) command to view PTFs.

SF40120 (Support for non-deliverable mail) SF40855 (Handling extended margins – V3R2) SF47957 (Handling extended margins – V3R7) SF44790 (V3R7) SF46876 (V3R7) SF45145 (V3R7)

SF45139 (SendMail API V4R1) SF45414 (Fix routing error in SNADS V3R7) SF36758 (SendMail API V3R7) SF36668 (SendMail API V3R7)

Configuring TCP/IP

Overview

Before you start configuring SMTP mail on the AS/400 you must have a valid TCP/IP configuration. If you have not yet configured TCP/IP you should stop and do this now. See the AS/400 TCP/IP reference manual for instructions on how to configure TCP/IP. You should make sure that you can use Telnet or FTP from a PC or other node on your network to connect to the AS/400 to verify the proper functioning of TCP/IP on the AS/400.

After configuring TCP/IP you should be sure you can Ping your mail server before continuing. If you have an ISP you can get the IP address of the mail server from them. If you are using an internal mail server, your LAN administrator can give you the IP address of the mail server. Be sure to use the IP address on the Ping request, not the domain name. If you cannot Ping your mail server contact your AS/400 support organization for assistance in configuring TCP/IP before continuing with SMTP configuration.

Automatically starting TCP/IP

You should consider automatically starting TCP/IP services on the AS/400 if you are not already doing so. You can retrieve the source for CL program QSTRUP from library QSYS and modify it to automatically start TCP/IP. The command to start TCP/IP is STRTCP. You should start TCP/IP before starting the SNADS subsystem and before starting subsystem TMAIL. If you configure SMTP to start automatically, which is the recommendation below, you do not need to do anything special to start SMTP services.

Starting QSNADS subsystem

IBM provides the QSNADS subsystem which must be started before you can exchange mail using SMTP. You can use the Start Subsystem (STRSBS) command to start it or you can modify the QSTRUP program to start it during IPL processing. This latter method is the preferred method.

Local domain and host name

As a part of your TCP/IP configuration you should enter a local domain name and local host name. If your company already has a registered domain name you should use this as your local domain name on the AS/400. Use the Configure TCP/IP (CFGTCP) command and use the option to Change Local Domain And Host Names. You can use any name for the AS/400 you wish that is a valid host name. The host name does not have to be the System Name of the AS/400. The host name and domain name are concatenated into an Internet name for purposes of SMTP mail. If you enter a domain name of “acme.com” and a host name of “as400” the SMTP Internet name for the AS/400 will be “as400.acme.com”.

Host table entries

After entering a local domain and host name you need to add the AS/400 name to the host table. Use the Configure TCP/IP (CFGTCP) command and use the option to Work With TCP/IP Host Table Entries. Add the Internet name for your AS/400 to the table. Use the IP address for the AS/400 and the Local Domain and Host names. Following the above example, you would use “as400.acme.com”. You must add this entry for SMTP mail to function properly.

Registering your AS/400 domain name

If you are going to send mail to Internet recipients you will probably want to register your AS/400 on the Internet. This is because some Internet Service Providers will not accept mail from an unregistered host. ISPs often enforce this requirement in order to reduce “spam,” or unsolicited mail, and to protect themselves from denial of service attacks. Your ISP can assist you in registering your AS/400 on the Internet.

If you are going to receive mail to your AS/400 POP3 server you definitely need to register the AS/400 on the Internet. If you’ve set up your mail service at the ISP as “acme.com” but want to receive mail to the AS/400 that is addresses to “as400.acme.com” you should ask your ISP to enter an MX entry for this domain name with the real IP address of the AS/400. The MX entry should have a higher priority than their mail server. Mail for the AS/400 will be queued by the ISP until your AS/400 is connected. Mail exchange will take place whenever the AS/400 is connected.

Configuring the Remote Name Server

Server address

Enter the IP address of the ISP domain name server (DNS). If you do not have an ISP (intranet only) and you do have an internal DNS server, enter the IP address of your internal DNS server. If you do not have either an ISP domain name server or an internal domain name server, leave this blank and set the Searched First parameter below to *LOCAL. If you have no connection to a domain name server you will have to enter the domain name of all recipients in the AS/400 TCP/IP host table.

Searched first

The default is *REMOTE which means that all attempts to resolve a domain name to an IP address will use the server address you entered above. This is the normal preferred setting.

If you do not have an ISP you should change this to *LOCAL. This will cause AS/400 SMTP to search the TCP/IP host table first to resolve domain names. In this case you must define all domain names to the AS/400 hosts table using CFGTCP option 10. SMTP must always resolve a domain name to an IP address and if you try to send mail to a user on another system it will fail unless there is a host entry.

If you do have an ISP and have defined a Server Address for the DNS above, but also send mail to SMTP mail systems within your local network, you should change this to *LOCAL. You will want SMTP to search domain names locally before trying to search the Internet DNS in case there are duplicate names in use.

Testing the connection

After configuring the IP address for the domain name server you should make sure the AS/400 can connect to it. Go to an AS/400 command line and use the PING command to ping the DNS server's IP address. For example:

```
ping rmtsys(*intnetadr) intnetadr('192.32.14.8')
```

You should observe several successful connections to the DNS server. If the ping command reports that the host is not found you may need to create a TCP/IP route using the Configure TCP/IP (CFGTCP) command. Consult your network administrator for the entries you need to make for the TCP/IP route.

Verify the connection to your mail server

After configuring TCP/IP and the Remote Name Server (DNS) you should verify that you can connect to the mail server you will be using. You can use the Ping command to do this. If your ISP is hosting your mail server you should obtain the domain name used to connect to the mail

server. If, for example, your ISP is located at “isp.com” the name of the mail server is probably “mail.isp.com”. However, you will need to get this information from your specific ISP. Once you have the name of the mail server you should verify the connection using the PING command:

```
ping mail.isp.com
```

You should observe several successful connections to the mail server. If the connection is not successful there can be more than one cause. The domain name server (DNS) may not have an entry for the mail server. Ask your ISP if this is the case. If you are using an internal domain name server you may need to add information for the ISP’s mail server address. If the DNS does have an entry for the mail server, perhaps you need to add a TCP/IP route for the IP address of the mail server. Ask the ISP for the actual IP address of the mail server. Use the Configure TCP/IP command to add the route. You may also need to discuss this with your network administrator. If your company uses an internal router or a router provided by the ISP you may need to add TCP/IP route information for the router.

If you are using an internal mail server such as Microsoft Exchange or Seattle Labs SLMAIL you will probably have to add name information for this server. If you are using an internal domain name server ask your network administrator to add an entry for the mail server. If you do not have an internal domain server, add an entry to the AS/400 host table for the mail server. You will need the name of the mail server and the TCP/IP IP address of the mail server to do this.

Verify subsystem QSYSWRK routing entry for SMTP

The subsystem QSYSWRK will contain all of the jobs associated with the TCP/IP SMTP application. You should verify that the proper routing entries exist in this subsystem for SMTP. Use the Display Subsystem Description (DSPSBSD) command to display subsystem QSYSWRK. Use option 7 to view routing entries. You should observer a routing entry with the following information:

```
Program: QCMD
Library: QSYS
Compare value: SMTPROUT
Start position: 1
```

If you do not see this entry you should use the Add Routing Entry (ADDRTGE) command to add it.

Verify subsystem QSYSWRK routing entry for POP3

The subsystem QSYSWRK will contain all of the jobs associated with the TCP/IP POP3 application. You should verify that the proper routing entries exist in this subsystem for POP3. Use the Display Subsystem Description (DSPSBSD) command to display subsystem QSYSWRK. Use option 7 to view routing entries. You should observer a routing entry with the following information:

```
Program: QTMMSRVR
Library: QTCO
Compare value: POPSERVE
Start position: 1
```

If you do not see this entry you should use the Add Routing Entry (ADDRTGE) command to add it.

Configuring Distribution Services

SMTP mail services requires a SNADS distribution queue and related configuration. You should verify that these configurations are in place before proceeding.

SNADS distribution queue

Use the Configure Distribution Services (CFGDSTSRV) command to display a list of configuration options. Select option 1 for Distribution Queues. You should observer a distribution queue named QSMTPQ. Use option 5 to display the distribution queue. You should observe the following values:

```
Queue: QSMTPQ
Queue type: *RPDS
Remote location name: TCPIPLOC
Mode: *NETATR
Remote net ID: *LOC
Local location name: *LOC
Normal priority: (all blanks)
High priority: (all blanks)
Number of retries: 3
Number of minutes: 5
```

If you do not see this entry use F6 from the list of queues to add the distribution queue.

SNADS routing entry

Use the Configure Distribution Services (CFGDSTSRV) command to display a list of configuration options. Select option 2 for Routing Table. You should observer a routing entry named TCPIP. Use option 5 to display the routing entry. You should observe the following values:

```
Destination system name/group: TCPIP
Description: TCP/IP routing
Service level:
        Fast: QSMTPQ / *DFT
        Status: QSMTPQ / *DFT
        Data high: QSMTPQ / *DFT
        Data low: QSMTPQ / *DFT
```

If you do not see this entry use F6 from the list of routing entries to add the routing entry.

Configuring the System Distribution Directory (SNADS)

SMTP mail services on the AS/400 require the configure of entries in the SNADS directory. Use the following procedures to review or add the directory entries.

Default SMTP entry

IBM recommends that you add a default SMTP name entry into the SNADS directory. Use the WRKDIRE command to maintain the directory. Use option 1 (Add) to add the following new directory entry:

```
User ID: QSMTPDMY
Address: QSMTPSYS
Description: QSMTP user
System name/Group:  TCPIP
Network user ID: QSMTPDMY QSMTPSYS
```

If you do not have this entry use F6 to add it.

Configuring SMTP Attributes

You must configure the SMTP attributes before you start using SMTP mail services. Use the Change SMTP attributes (CHGSMTPA) command to change these values.

Autostart server

You should change this value to *YES in order to automatically start SMTP when you start TCP/IP.

Retries by minute (number of retries, time interval)

The default is three retries at 30-minute intervals. This means that an error addressing a mail message will not be reported for at least 90 minutes after sending a message. You can lower the time interval to 5 or 10 minutes in order to get error reports more frequently.

Retries by day (number of retries, time interval)

The default number of retries is 0 and the default time interval in days is 0. You should leave these values at their defaults in order to avoid long delays in notification of undeliverable mail.

Retry remote name server

You should leave the default value of *NO for this parameter. If you change this value to *YES the SMTP application will continue to retry a DNS server when it cannot resolve a domain name. This can hide errors in mail addresses.

Automatic registration

You should specify *YES for this value if your AS/400 mail users will receive mail from the Internet. AS/400 SMTP will automatically enroll users in the SNADS directory so that the AS/400 recipient can reply to the message.

User ID prefix

QSM is the default for this value. We recommend you leave this as the default value so that you can identify which entries in the SDADS directory were automatically added.

Address

The default is QSMRMTAD. You should leave this as the default as it is used in the initial SNADS configuration for SMTP.

System name

The default system name is TCPIP. You should leave this as the default as this is used in the configuration of the SNADS distribution services routing entry.

Alias table type

IBM strongly recommends that you use the SMTP system alias table rather than personal alias tables. You should leave this value *SYSTEM.

User ID delimiter

The default is the question mark (“?”). It is more common to use a period to for the user ID delimiter. If you are configuring SMTP for the first time you should consider changing this to a period (“.”). If you are already using SMTP mail you should not change this parameter in order to avoid causing errors in mail delivery.

Configuring the POSTMASTER User

SMTP mail systems can generate mail to the POSTMASTER account. This is a normal convention for such systems. You should add support for this user. To do this you must create a user profile named POSTMASTER, add a SNADS directory entry with the user of POST, the address of MASTER, the user profile of POSTMASTER, and the SMTP alias of postmaster@your-domain. Change “your-domain” to the full host and domain name of your AS/400.

You should plan to review messages received to this address on a regular basis. You can set up a NetMail account for this, or you can forward mail from this account to an existing SNADS or NetMail account using the Forward From setting on the directory.

Undeliverable Mail

If mail is received on the AS/400 for a user that is unknown to the AS/400 SMTP mail system it is returned to the sender as undeliverable. The message sent back to the sender will indicate that the recipient is not known to the AS/400 system. If you want you can disable undeliverable

mail rejections and receive all undeliverable mail to a specific user on the AS/400 you can do this by creating a special SNADS entry and SMTP alias. Before you do this you must install certain PTFs from IBM. If you are on V3R7 you must install PTF SF40120. If you are on V4R1 you must install SF40090.

First create a user profile that can be used for undeliverable mail. You might name it UNDELIV or something similar. The password can be *NONE and initial menu can be *SIGNOFF to prevent anyone from actually using this profile. Then add a SNADS directory entry for this user. You might add an entry named “UNDELIV S1015973” for example. Be sure to specify the local AS/400 system name as the SNADS address and system name, and specify the UNDELIV user profile. After creating the SNADS entry, add an SMTP alias for this user with the SMTP user ID of “nondelivery” and an SMTP Domain of “your-domain” where your-domain is the AS/400’s SMTP domain.

You should plan to review messages received to this address on a regular basis. You can set up a NetMail account for this, or you can forward mail from this account to an existing SNADS or NetMail account using the Forward From setting on the directory.

Message splitting

It can occur that the AS/400 will split a large message into parts before sending to Internet or local SMTP recipients. This generally happens when there is a large attachment on a message. If this occurs you should take the following actions.

Change the POP attributes (command CHGPOPA) to increase the Message Split size to the maximum. At the time this document was written that size was 2 megabytes. Start the restart the SMTP server support using the ENDTCPSVR and STRTCPSVR commands.

Create a data area in the QUSRSYS library named QTMSNOSPLT. The data area should be 1 byte in size. You can use this command to create the data area:

```
Crtdtaara dtaara(qusrsys/qtmsnosplt) type(*char) size(1)
```

Creating this data area should inhibit any splitting of messages by AS/400 SMTP support.

Testing SMTP Mail

If you are using NetMail*400, OfficeVision, or other SNADS mail systems be sure to avoid using the QSECOFR user profile (or any IBM provided user profile) for testing SMTP or Internet mail. These user profiles can cause problems in delivery of mail. Instead, use a normal user profile that you have created using the Create User Profile (CRTUSRPRF) command.

Troubleshooting SMTP Mail

It can be difficult to determine the cause of problems with SMTP mail delivery. There are no readily-available interfaces provided by IBM to seeing what is happening with SMTP mail and why it may not be getting delivered or received properly. Use these procedures for determining

why mail might not be getting sent from the AS/400 correctly. These procedures assume that you have performed the above install procedures and can ping your DNS server and mail server.

Place the SNADS distribution queue QSMTPQ on hold. You can use the WRKDSTQ command to do this. Then send your message to the SMTP recipient using the SNDDST command. You should see the distribution in the queue after using SNDDST. If you do not see the distribution in the queue check the results of the SNDDST command and make sure you see the message “Distribution sent to 1 users”, or something similar. If you do not see this message you have not established the SNADS directory entries correctly. Fix these before proceeding. If the message is sent you should now see it in the distribution queue. If it is not in the queue you probably have not started the QSNADS subsystem. Start the subsystem and check again. Release the QSMTPQ.

If your SMTP message is getting in the QSMTPQ distribution queue but is not getting delivered, you should check the SMTP flight recorder. There can sometimes be helpful information in this area. The flight recorder is a set of entries that the IBM SMTP mail system places in a user queue. You have to dump the user queue and print the dump to read the entries. This queue automatically wraps, so you will also need to check this when you can be sure that there is not much activity in the SMTP mail system. Send the message and then use these commands:

```
dmpobj obj(qusrsys/qtmsflrcq1) objtype(*usrq)
dmpobj obj(qusrsys/qtmsflrcq2) objtype(*usrq)
dmpobj obj(qusrsys/qtmsflrcq3) objtype(*usrq)
dmpobj obj(qusrsys/qtmsflrcq4) objtype(*usrq)
```

Review the output to determine the cause of the problem.

If the above information is inadequate to solve the problem you can get a more detailed trace by creating the QTMSTRCNTL data area. When this data area exists the SMTP support will write detailed trace information to trace files. Use this command to create the data area:

```
Crtdtaara dtaara(qusrsys/qtmstrcntl) type(*char) len(8) value('1111
```

Stop the SMTP server, wait a few minutes, and then start it again. You can use the STRTCPSRV command to do this. After the SMTP server is started send the message. Wait a few minutes to be sure the message has been processed by AS/400 SMTP support. Then stop the SMTP server again. View the trace detail with these commands:

```
dsppfm file(qusrsys/qtmsflrcf1)
dsppfm file(qusrsys/qtmsflrcf2)
dsppfm file(qusrsys/qtmsflrcf3)
dsppfm file(qusrsys/qtmsflrcf4)
```

Review the contents of the trace file to determine the cause of the problem. Delete the data area and start SMTP support again.

If the above steps do not indicate the cause of the problem you can use a communications trace to view the conversation between AS/400 SMTP support and your Internet or Intranet mail server. First determine the IP Address and Line description used for network connections. You can use the CFGTCP command to do this. Then use the STRSST command to display the service tools. Select option 1 from the first menu, and then option 3 from the next menu to start a trace. Use function key F6 to add a trace. Enter the line description name, a text description, and “6” for the buffer size to make the buffer size as large as possible. After the trace is started send your mail message and wait a few minutes. Use the STRSST command again to view the active trace and use option “2” to stop the trace. After the trace is stopped use option “6” to print it. Be sure to enter the line description name, *ASCII for the data type, and select TCP/IP only. To minimize the size of the trace be sure to turn off SNA, broadcast data, and all other options. On the next display you will be prompted for the IP address of the source and destination systems. Enter the AS/400 IP address in the first field and print the trace. Review the trace to determine any error messages that may be sent by the remote mail system or the AS/400. You should be able to scan for the characters “HELO” to determine where the SMTP mail protocol starts.

Review any error messages and resolve the problem. Use STRSST to delete the trace.

Internal SMTP Message Cleanup

It can sometimes happen that IBM SMTP support gets into a loop and continuously processes mail. This can happen if the SMTP configuration you have created inadvertently causes a looping condition, or other conditions at the DNS server or local host table create a looping condition. The following procedure can be used to remove any SMTP messages being processed from the system. You should normally only use this procedure if you notice that the SMTP server jobs in subsystem QWRKSYS seem to be looping and using a large amount of CPU and I/O.

First stop the SMTP server using the ENDTCPSVR command. After the server is stopped create the following data area:

```
Crtdtaara dtaara(qusrsys/qtmsclean) type(*char) len(1) value('c') a
```

Note that the value of this data area must be a lower case “c”. Now restart the SMTP server using the STRTCPSVR command. The internal messages will be removed and the data area deleted.